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Scalable Quantum Machine Learning: Trainability, Expressivity and Efficiency

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Designing scalable parameterized quantum circuits for machine learning has remained elusive due to three fundamental obstacles: barren plateaus that prevent gradient-based training, the absence of provable guarantees that the learned function class is classically hard, and prohibitive circuit evaluations per gradient step. We propose a solution based on the *unitary brick-wall* circuit: a k-particle fermionic architecture for nearest-neighbor hardware, combining Reconfigurable Beam Splitter gates with interleaved single-qubit phase gates and a non-Gaussian magic-state encoding, where the particle number k is a tunable resource dial that trades classical simulation hardness against training cost.

Trainable. The brick-wall has dynamical Lie algebra $\mathfrak{u}(n)$ and is surjective onto $U(n)$ via Givens rotations, enabling Haar initialization. Two-body correlator readouts achieve gradient variance $\Theta(k^3/n^5)$, which remains polynomial in n throughout the regime $n - 2k = \Omega(n)$.

Expressive. The classical hardness of the pipeline is controlled by the particle number k: best-known classical algorithms for sampling run in time $2^{\Theta(k)}\,\mathrm{poly}(n)$ [1], worst-case #P-hardness holds from $k = n^{\epsilon}$, and the full average-case machinery of Fermion Sampling [2] applies at $k = \Theta(n)$. At our operating point $k = 60$, best-known classical simulation exceeds 10^{24} operations at every n.

Efficient. A multi-layer parallel parameter-shift rule computes all $O(n^2)$ gradients from $4kn$ distinct circuit evaluations per gradient step — a factor n/k reduction over the $4n^2$ evaluations required by the standard parameter-shift rule, growing linearly with n at fixed k.

The *unitary butterfly* variant targets all-to-all hardware, with depth $2\log n$ and $\frac{3}{2}n\log n$ parameters. It achieves similar expressivity and hardness guarantees to the brick-wall, and a gradient step needs $4k\log n$ circuit evaluations, the same factor-n/k reduction over the $4n\log n$ naive PSR cost. Its trainability is established at two levels: the absence of exponential barren plateaus is *unconditional*, whereas the sharp $\Theta(k^3/n^5)$ rate holds under a two-particle approximate-2-design conjecture.

I. INTRODUCTION

Quantum machine learning has developed along two largely distinct trajectories—statistical learning theory and kernel methods on one hand, and deep learning and neural networks on the other—corresponding to two different visions of how quantum computation can augment artificial intelligence.

The first draws on quantum algorithms for linear systems, matrix operations, and singular value transformations [3–5], covering a broad range of tasks including among others nearest-neighbor search [6], principal component analysis [7], clustering [8], convolutional networks [9], and recommendation systems [4]. The hardware requirements of these methods are fault-tolerant quantum computers with QRAM-encoded data and circuits of polynomial depth, and they achieve important theoretical speedups over the classical algorithms that are used in practice and operate on explicit matrix representations. The precise extent of their computational advantage was significantly revised by the dequantization work of Tang and collaborators [10–12], which showed that when classical algorithms are given analogous ℓ^2-norm sampling access to the input data, the quantum speedup is polynomial for a broad class of linear algebraic tasks. The conclusion is doubly nuanced. First, the dequantization results do not eliminate quantum advantage: they show that under ℓ^2-norm sampling access, the quantum speedup reduces from exponential to polynomial, and in some cases large polynomial. Second, the dequantized classical algorithms, while polynomially competitive with quantum under sampling access assumptions, appear in practice to be slower than the fastest classical algorithms that operate directly on explicit matrix representations—suggesting that the sampling access model, though theoretically interesting, may not correspond to practical classical computation. The net result is that quantum linear algebra may still offer genuine advantages over practical classical methods, but establishing this rigorously awaits large-scale, fast, fault-tolerant quantum hardware and efficient QRAM implementations. In fact, very recently, Zhao et al. [13] proved unconditional exponential quantum advantage in classification and dimension reduction via quantum oracle sketching, establishing machine learning on classical data as a concrete domain of quantum advantage in the linear algebra paradigm. This paradigm represents the quantum analogue of traditional, kernel-based machine learning.

The second area, which we call *quantum deep learning* (QDL), draws on parameterized quantum circuits (PQCs) trained by gradient descent [14–17], with the

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circuit playing the role of a neural network layer or feature extractor [18–20]. These methods are aligned with the deep learning revolution that has vastly enlarged the applicability and impact of machine learning: instead of hand-crafted features or explicit kernel methods, the circuit learns representations directly from data. This approach has been explored in supervised learning [21–23] and as policy and value function approximators in reinforcement learning agents [24–26]. Their hardware requirements are far more modest than the first paradigm—shallow circuits of linear or even logarithmic depth can suffice, the optimization loop runs classically, and the quantum device is called only for circuit evaluation—making QDL a natural target for near-term and early fault-tolerant quantum hardware [27]. Analogously to classical deep learning, where the value of a given architecture or training method is established empirically rather than by theoretical proofs of superiority, the value of QDL will ultimately be established on real hardware and real tasks: the key question is whether quantum representations can access function classes that lie outside the reach of classical models of comparable parameter count, enriching model capacity in ways that matter in practice. This is the paradigm this paper addresses.

A. The State of Quantum Deep Learning

The past decade has produced a rich body of work on quantum deep learning, advancing both the design of trainable architectures and our theoretical understanding of their limitations.

a. Quantum neural network architectures. Early proposals for quantum neural networks sought direct quantum analogues of classical layers: quantum perceptrons, circuits inspired by classical convolutional architectures, and variational models [16, 20, 28]. A key insight that emerged was that PQCs can be interpreted as nonlinear feature maps into quantum Hilbert space [15, 21], connecting quantum neural networks to the classical theory of kernel methods and opening a systematic framework for understanding their expressive power.

A central challenge identified early was the issue of *barren plateaus*: for generic, randomly initialized circuits, gradients vanish exponentially with system size, rendering training infeasible [29], with the severity depending on the locality of the cost function [30]. This motivated a sustained search for architectures with provable trainability. Quantum convolutional neural networks (QCNNs) [31], which interleave convolutional and pooling layers in a hierarchical structure, were shown to be free of exponential barren plateaus [32]. Hierarchical tensor-network classifiers [33] and equivariant quantum neural networks [34, 35] extended these ideas to symmetry-constrained architectures. The unifying theoretical framework—that gradient variance is controlled by the dimension of the dynamical Lie algebra (DLA)—was established by Fontana et al. [36] and Larocca et al. [37], providing a precise, quantitative account of why structured circuits train. A troubling corollary, however, is that circuits with small DLA—which avoid barren plateaus—can often be efficiently classically simulated [38]. This has been confirmed concretely for the QCNNs of [31]: Bermejo et al. [39] showed that they are effectively classically simulable because, when randomly initialized, they can only access information encoded in low-bodyness observables, and their prior empirical success is explained by benchmarking on datasets that are "locally easy" in precisely this sense. Similar simulability results hold for certain permutation-equivariant architectures [40], though the general equivariant case remains only partially understood. These results sharpen the fundamental tension between trainability and genuine quantum advantage: the same structural constraints that prevent barren plateaus tend to limit the circuit to a classically accessible function class.

b. Hamming-weight-preserving quantum neural networks. A parallel line of work introduced *orthogonal quantum neural networks* based on Hamming-weight-preserving fermionic gates, which implement orthogonal matrix multiplications as natural quantum operations [23, 41]. Depending on the hardware connectivity, different topologies were proposed: brick-wall and pyramid circuits for nearest-neighbor connectivity with linear depth and quadratic number of parameters, and butterfly circuits arranged in the pattern of the Cooley-Tukey FFT for all-to-all connectivity, operating in $O(\log n)$ depth with $O(n \log n)$ parameters [41]. These circuits appeared in subsequent work spanning quantum vision transformers [42], quantum deep hedging [26], financial forecasting [43], and clinical data imputation [44], and were tested on real quantum hardware for up to 32 qubits [45, 46]. The underlying subspace structure has also been studied as a primitive in its own right [47]. It is worth noting that in the special case of Hamming weight $k = 1$, these circuits are equivalent to linear optical networks and are efficiently classically simulable [48]. Far from being a limitation, this simulability was productive: it revealed that the RBS gate provides a natural and complete parametrization of the orthogonal group via Givens rotations, yielding a fully classical algorithm for training orthogonal neural networks with exact orthogonality preservation and $O(1)$ gradient cost per parameter—a combination not achieved by prior classical methods based on Cayley transforms, Householder reflections, or Stiefel manifold optimization [41].

The interesting regime for quantum advantage requires larger values of Hamming weight k. The RBS circuit itself is always a passive FLO transformation and remains efficiently classically simulable regardless of k [48]. Classical hardness arises not from the circuit but from the *input state*: when the k-particle input is non-Gaussian, it can be provably hard to simulate classically when combined with a passive FLO circuit [2], comparable to Bo-

son Sampling hardness. The theoretical properties of these circuits were subsequently analyzed from two complementary angles. Monbroussou et al. [49] showed that the gradient variance of the Hamming-weight preserving circuits (DLA $\mathfrak{so}(n)$) scales as $\Theta(1/\binom{n}{k})$. For constant k this is polynomial in n, but for values of k that grow with n (for example $k = n/2$) this variance is exponentially small, constituting an actual barren plateau. Independently, the adjoint representation framework of Fontana et al. [36] established that for a Lie Algebra Supported Ansatz (LASA)—where the measurement observable iM lies in the DLA—gradient variance scales as $\Theta(1/\dim(\mathfrak{g}))$ for any particle number k. For DLA $\mathfrak{u}(n)$ this gives $\Theta(1/n^2)$, a polynomial bound independent of k. Note that the occupation-number observable $n_j = c_j^\dagger c_j$ (measuring whether mode j is occupied) is the natural readout of a computational basis measurement; in_j lies in $\mathfrak{u}(n)$ but *not* in $\mathfrak{so}(n)$. This is the LASA condition for generic input states. For our specific magic-state encoding, however, every mode of the spread input carries equal occupation k/n (Section II), and losses built from statistics of fixed body number are classically computable via r-RDM propagation (Proposition 8); the trainable readout of interest is the two-body correlator $n_i n_j$, for which the relevant gradient variance is $\Theta(k^3/n^5)$ (Section IV).

c. The classical-simulation boundary for non-Gaussian fermionic inputs. The complexity of classically simulating non-Gaussian fermionic inputs evolved under passive FLO has been refined considerably in recent work, and the resulting picture directly shapes our architectural choices. Sampling-based hardness (multiplicative precision on individual probabilities) for *fully-packed* magic state inputs, i.e. at extensive particle number $k = \Theta(n)$, follows from the Fermion Sampling argument of Oszmaniec et al. [2]; whether hardness survives with fewer magic blocks was posed there as an open problem, a question that directly shapes the choice of k in our construction (Section II D). Dias and König [50] and Reardon-Smith et al. [1] gave classical algorithms for arbitrary non-Gaussian inputs whose cost scales as $O(\chi^2 \cdot \mathrm{poly}(n))$, where χ is the FLO extent of the input. More recently, Oh, Oszmaniec, Reardon-Smith, and Zimborás [51] exhibited a structurally tighter result for the specific class of *paired* non-Gaussian inputs (form-rank-2 states such as $1/\sqrt{2}(|0011\rangle + |1100\rangle)$): an additive-error classical estimator for transition amplitudes, overlaps, and arbitrary-weight number correlators, with sample complexity $O(\epsilon^{-2}\log\delta^{-1})$ matching the operational shot complexity of quantum hardware. Their algebraic core is a *mixed-Pfaffian* reduction that compresses an exponentially large sum of determinants into a single coefficient of a multivariate Pfaffian polynomial. This places paired non-Gaussian inputs in the additive-error-tractable regime, where the FLO extent provides only an upper bound that the Pfaffian shortcut beats.

d. Training quantum neural networks. The standard training algorithm for PQCs is the parameter-shift rule [14, 52], which computes the exact gradient of each parameter using two circuit evaluations per parameter. For a circuit with T trainable parameters, this requires $2T$ circuit evaluations per gradient step—growing linearly with the number of parameters and becoming prohibitive for large circuits. Stochastic approximations can reduce the per-step cost to $O(1)$ circuit evaluations but introduce bias and increased variance [53, 54]. Recent stochastic block coordinate descent [55] selectively updates parameter subsets but does not reduce the fundamental per-parameter cost. A complementary line of work casts stochastic approximations, random coordinate descent, and the parameter-shift rule as special cases of a single forward-gradient estimator parametrized by the number of random directions probed per step, with an adaptive optimiser choosing this number from the measurement statistics themselves [56]. An alternative approach is to reduce the effective number of parameters by mixing over ensembles of circuits—the density QML framework [57] showed that this can improve both trainability and per-step cost, but at the price of reduced expressivity, trading the capacity to represent complex functions for easier optimization. Note as well that when there is a single trainable layer where all gates act on disjoint qubit pairs, their generators commute, and in principle all gradients within that layer could be estimated simultaneously from a single set of circuit evaluations rather than sequentially [46]. From a practical point of view, for circuits with $T = O(n^2)$ parameters, namely a linear depth parametrized quantum circuit, the naive PSR requires $O(n^2)$ circuit evaluations per gradient step, making on-hardware training at scale impractical even for 100 qubits, regardless of which of these strategies is employed.

e. Data loading. A key distinction from the quantum linear algebra paradigm is that quantum neural networks process one data point at a time: there is no requirement to load an entire dataset into a quantum state or to maintain a QRAM over the full data matrix. The data loading problem here is therefore a per-sample question—how to efficiently encode a single classical vector $\mathbf{x} \in \mathbb{R}^n$ into a quantum state that is expressive enough to support learning, without incurring circuit depth that negates the advantage of the shallow trainable circuit that follows. The simplest approach, *angle encoding*, uses one rotation gate per feature and requires $O(1)$ depth, but encodes data in rotation angles rather than amplitudes and does not preserve inner product structure [15]. *Amplitude encoding* represents n features as the amplitudes of the unary basis of an n-qubit state, preserving ℓ_2 distances, and can be implemented with $n-1$ RBS gates in a ladder formation on nearest-neighbor architectures or with a $O(\log n)$ depth RBS gate tree on all-to-all connectivity architectures [41]. For k-Hamming weight states, Bärtschi and Eidenbenz [58] gave short-depth circuits for Dicke states and Farias et al. [59] gave optimal RBS-based encoders for k-particle states. The right encoding strategy for QDL remains an important

question, since the encoding choice simultaneously impacts the trainability of the circuit and the expressivity of the learned function class [60].

B. The Potential of Quantum Deep Learning

The progress surveyed above has significantly advanced our understanding of what quantum neural networks can and cannot do: we know when they train, when they do not, when they are classically simulable, and what structural properties govern each. Yet despite this theoretical clarity, the fundamental question—what is the genuine potential of quantum deep learning, and what should we concretely expect from it?—remains wide open. Answering it requires more than additional technical results. It requires first being precise about what we mean by potential, and then identifying the specific challenges that must be overcome to realize it.

We argue that the right question is not whether quantum circuits can replace classical neural networks, or whether they offer provable worst-case speedups on standard benchmark tasks. Analogously to classical deep learning—where the value of convolutional networks or transformers was not established by theoretical proofs of superiority but by empirical demonstration of expressivity on real tasks—the right question is whether quantum circuits can represent useful functions or policies that lie outside the reach of classical models of comparable parameter count, enriching the model class in ways that matter in practice.

Two conditions must therefore be met for quantum deep learning to realize its potential. The first is *expressivity*: the quantum model must access function classes that are provably beyond the reach of classical models of comparable size. This is a question of quantum complexity theory, and for the right architecture, precise lower bounds on the classical simulation cost can be established. The second condition is *practical utility*: that the functions accessible to the quantum model are not merely hard for classical computers but actually useful—that they correspond to inductive biases or representations that help on real tasks. This second condition cannot be established by complexity-theoretic arguments alone; it requires running quantum models on real data and real hardware. And this is precisely where scalability becomes the binding constraint: there is no point in demonstrating quantum expressivity on toy problems if the architecture cannot be trained efficiently at scale. The path to understanding the practical potential of quantum neural networks therefore runs directly through the problem of building scalable, trainable, and expressive architectures—which is the problem this paper addresses.

We identify five concrete challenges that any scalable quantum framework must address simultaneously, and we show that two complementary architectural choices—the unitary brick-wall for nearest-neighbor hardware and the unitary butterfly for all-to-all connectivity—and an efficient algorithm for computing the gradients—the multi-layer parallel parameter-shift rule—resolve all five. The challenges are: efficient data loading (C1); efficient parametrized quantum circuits as quantum neural layers (C2), where the optimal depth depends on hardware connectivity; provable avoidance of exponential barren plateaus (C3); efficient gradient computation with sub-linear circuit evaluations per step (C4); and genuine classical hardness of the deployed model output (C5). The key insight unifying the solution is that particle-number preservation—the defining property of RBS gates—simultaneously enables efficient gradient computation, prevents exponential barren plateaus, and places the circuit in the Fermion Sampling framework for classical hardness with the correct data loading procedure.

A sixth challenge—whether the function classes accessible to quantum models are not merely hard for classical computers but genuinely useful on real tasks—cannot be resolved by theory alone. It requires running scalable quantum models on real data and real hardware, at sizes where the quantum expressivity advantage is meaningful. Closing C1–C5 is the prerequisite for even posing this question seriously; the structure of the framework does, however, narrow the field of candidate applications considerably, and we identify them in Section VIII.

This paper builds on a long series of work on Hamming-weight-preserving quantum neural networks. The prior program established the architecture, tested it on hardware, and demonstrated its utility across applications in finance and healthcare—but left two gaps open: the barren plateau at large particle number k, and the prohibitive gradient computation cost at scale. We close both simultaneously by augmenting the RBS brick-wall and butterfly circuits with single-qubit R_z gates (which lifts the DLA from $\mathfrak{so}(n)$ to $\mathfrak{u}(n)$ and resolves the barren plateau issue) and by proving the multi-layer parallel parameter-shift rule (which achieves a factor-$O(n/k)$ reduction in circuit evaluations per gradient step for each architecture: from $O(n^2)$ to $O(kn)$ for the brick-wall, and from $O(n \log n)$ to $O(k \log n)$ for the butterfly). The first contribution ensures that the lifted architecture has polynomial gradient variance throughout the regime $n - 2k = \Omega(n)$, including at extensive particle number $k = cn$ with $c < 1/2$, where the sampling-hardness guarantees of Section VII apply—so challenges C3 (trainability) and C5 (classical hardness) are met simultaneously, the former unconditionally for the brick-wall. (All trainability statements are for the regime $n - 2k = \Omega(n)$; at exactly half filling the one-body signal vanishes identically and only an $O(1/n^4)$ upper bound on the rate is proved.)

The second contribution makes training cheap in the practically relevant regime $k \ll n$. In addition, we organize the classical-hardness guarantees of the framework into an explicit ladder in the particle number k, calibrated against the best-known classical simulation algorithms [1, 51], and identify the concrete operating point

at which quantum advantage is possible while training remains efficient.

It is worth stating at the outset how challenges C3 and C5 coexist, since they are met on *different objects* of the same circuit. Trainability is established for the gradient channel: losses that depend on the measurement samples through two-body statistics, for which the gradient variance is $\Theta(k^3/n^5)$. Classical hardness is established for the deployed output: the sampling task, whose best-known classical cost is $2^{\Omega(k)}\mathrm{poly}(n)$. These are genuinely distinct objects—knowledge of the two-body moments does not permit sampling, since exponentially many distributions on Hamming-weight-k strings share them—and the separation is what allows both conditions to hold at once for a single architecture, with the two-body correlator serving as the trainable anchor and the sampler as the deployed model. Section VI C develops this into a taxonomy of pipelines, including which gradients may be computed classically and which require the quantum device inside the training loop.

Our main technical contribution is the **multi-layer parallel parameter-shift rule** (Theorem 24): all gradients of the QNN can be computed simultaneously from a number of circuit evaluations that is a factor n/k smaller than the naive parameter-shift rule, using random sign vectors and the commuting structure of the gates within each layer. The estimator is unbiased (exact in expectation), requires no assumption on the observable beyond particle-number preservation, and applies identically to both architectures. The naive PSR costs 4 circuit evaluations per RBS parameter and 2 per R_z parameter; the parallel PSR computes all gradients in a single layer from $2k$ evaluations for the RBS parameters and $2k$ for the R_z parameters, independently of n. The resulting per-gradient-step costs and reductions are:

- **Brick-wall** ($T = 3n^2/2$ parameters, n columns, each a full-width R_z layer followed by that column's $n/2$ RBS gates): naive PSR requires $4n^2$ circuit evaluations; parallel PSR requires $4kn$—a factor of n/k reduction counted in distinct circuit configurations.

- **Butterfly** ($T = \frac{3}{2}n\log n$ parameters, $\log n$ layers): naive PSR requires $4n\log n$ circuit evaluations; parallel PSR requires $4k\log n$—a factor of n/k reduction.

For $n = 1024$ at the sampling operating point $k = 60$ (Section II D), the butterfly requires 2,400 circuit evaluations per gradient step (vs. 40,960 for the naive PSR on the butterfly, a 17.1× reduction), and the brick-wall requires 245,760 evaluations (vs. 4,194,304 for the naive PSR on the brick-wall, also 17.1×). Crucially, the reduction factor n/k grows linearly with n at fixed hardness level: it reaches 68.3× at $n = 4096$, and the butterfly's absolute cost, $4k\log n$, is nearly independent of n. Throughout, the naive baseline is computed for the *same* circuit, and we quote the absolute per-step cost alongside the ratio (Corollary 27). As n grows the training cost scales only as $O(k\log n)$ for the butterfly and $O(kn)$ for the brick-wall—polylogarithmic and linear respectively—while the Hilbert space dimension grows as $\binom{n}{k}$. The extra factor of n in the brick-wall reflects the price of nearest-neighbor connectivity: $O(n)$ depth instead of $O(\log n)$, but with no long-range gates required. In both cases the training cost scales far below the expressivity of the model.

a. Paper structure. Section II presents the fermionic magic-state data encoding. Section III introduces the unitary brick-wall and unitary butterfly architectures and their mathematical properties, including the depth-connectivity tradeoff for each. Section IV establishes the trainability and absence-of-barren-plateaus result. Section V proves the multi-layer parallel parameter-shift rule. Section VI presents the complete pipeline: a single sample-based readout interface shared by all task families, from distributional losses to classical heads—linear, multilayer, or transformer—consuming the sampled Fock strings, together with an account of which gradients admit a variance guarantee and which require the quantum device in the training loop. Section VII proves classical hardness of the learned function class for sampling-based readouts. Section VIII discusses implications and open problems.

II. DATA LOADING: MAGIC-STATE ENCODING

The data loading happens in three steps:

A. Magic State Preparation: we prepare a non-Gaussian magic state on a set of active blocks;

B. Spreading Operation: we spread the particles across all n modes using a fixed Hadamard spreading circuit;

C. Data Encoding: we encode classical data through a layer of single-qubit phase gates.

The encoding pipeline is the same for both architectures; the particle number k is calibrated against classical simulation cost in Section II D, while the only connectivity-dependent element is the spreading circuit in step B, which has depth $O(\log n)$ on all-to-all hardware (Hadamard butterfly) and $O(n)$ on nearest-neighbor hardware (Hadamard brick-wall).

We now detail each step.

A. Magic State Preparation

The magic state is the minimal non-Gaussian construction adequate for the Fermion Sampling hardness structure of Section VII. Divide n qubits into $k/2$ active blocks

of 4 qubits and $n-2k$ idle qubits initialized to $|0\rangle$. Prepare:

$$|\Psi_{\text{in}}^{(4)}\rangle = |\Psi_4\rangle^{\otimes k/2} \otimes |0\rangle^{\otimes(n-2k)}, \qquad |\Psi_4\rangle = \frac{|1100\rangle + |0011\rangle}{\sqrt{2}}. \tag{1}$$

Each block is prepared independently using a circuit of depth 4 and 3 entangling gates [2]; all $k/2$ blocks run in parallel, so the total depth is 4. The state has exactly k particles and is non-Gaussian: a superposition of two distinct Fock states differing in all modes, which cannot be written as a single Slater determinant [2]. Each block is a 2-form (sum of two pair-creation operators), so the overall input lies in the form-rank-2 manifold. Two complexity-theoretic consequences follow, developed in Section VII: the mixed-discriminant structure of its output amplitudes underlies the Fermion Sampling hardness argument at sufficiently large particle number [2], while its mixed-Pfaffian structure makes few-body expectation values classically estimable to additive error in polynomial time at *any* k [51].

Remark 1 (Sparse inputs: hardness scales with the number of magic blocks)**.** *The original Fermion Sampling hardness argument of Oszmaniec et al. [2] considers fully-packed magic inputs* $|\Psi_4\rangle^{\otimes n/4}$ *with* $k = n/2$ *particles across all* n *modes. Our construction uses only* $B = k/2$ *magic blocks and* $n - 2k$ *idle modes initialized to vacuum. The mixed-discriminant* structure *carries through directly to the sparse case: the idle modes simply do not appear in the rows and columns of the single-particle unitary indexed by the support of the magic state, and, exactly as in the fully-packed case [2, Lemma 11], the output amplitude is a sum of* 2^B *determinants of* $k \times k$ *submatrices of the single-particle unitary* W*,*

$$\langle x'|V|\Psi_{\text{in}}^{(4)}\rangle = 2^{-B/2} \sum_{S\in\mathcal{S}} \det\big(W_{x',S}\big), \qquad |\mathcal{S}| = 2^B. \tag{2}$$

What does not *carry through automatically is hardness, because Eq.* (2) *is simultaneously a classical algorithm of cost* $O(2^B \cdot k^3)$ *per amplitude. The hard-instance size of the Ivanov–Gurvits reduction to the permanent [61, 62] is the block count* B*, so worst-case #P-hardness requires* $B = n^{\Omega(1)}$*. Moreover, the FLO extent of the input is* $\chi = 2^B$*, and all best-known simulation methods—exact amplitudes, few-body marginals, and, via the chain rule over marginals, exact sampling—run in time* $2^{\Theta(B)}\,\text{poly}(n)$ *[1]. In particular, for* $B = O(\log n)$ *the entire pipeline,* including sampling*, is classically simulable in polynomial time. Whether hardness survives with fewer magic blocks than the fully-packed input was posed as an open problem in [2, Sec. IV]; the extent-based algorithms of [1] resolve its hardness half negatively for logarithmically many blocks, while the regime of block number between* $\omega(\log n)$ *and* $o(n)$ *remains open at the level of average-case hardness. Section VII organizes our hardness claims along this ladder.*

B. Spreading Operation

The remaining encoding stages are the Hadamard spreading and the data-dependent phase layer. Throughout, we write $|\Psi_{\text{in}}\rangle$ for the magic state $|\Psi_{\text{in}}^{(4)}\rangle$.

For the spreading operation, we apply U^{had}—a passive FLO circuit with all RBS angles fixed at $\pi/4$, implementing the Walsh–Hadamard transform in the single-particle sector. On all-to-all connectivity this is the Hadamard butterfly of depth $\log n$ (defined formally in Section III); on nearest-neighbor connectivity it is the Hadamard brick-wall of depth $n-1$ (defined formally in Section III). In both cases the circuit is non-trainable and fixed once and for all. This delocalizes the k particles across all n modes:

$$|\phi_0\rangle = U^{\text{had}}\,|\Psi_{\text{in}}\rangle = \sum_{S\subseteq[n],\,|S|=k} \alpha_S\,|S\rangle\,, \tag{3}$$

where every qubit j has single-particle occupation $\rho_1^{\text{out}}[j,j] = k/n$ (Lemma 2), and particle number is preserved exactly. Non-Gaussianity is preserved: U^{had} is a passive FLO transformation, and passive FLO maps non-Gaussian states to non-Gaussian states [63]. Crucially, passive FLO also preserves form-rank, so the form-rank-2 structure of the input is preserved by spreading.

Lemma 2 (Hadamard spreading equalizes single-particle occupations)**.** *Let* $|\Psi_{\text{in}}\rangle = |\Psi_{\text{in}}^{(4)}\rangle$ *be the magic state of Section II A, with* k *particles distributed across* $2k$ *active modes (the remaining* $n-2k$ *modes initialized to* $|0\rangle$*). After applying the Hadamard spreading circuit* U^{had} *on all* n *modes, the single-particle reduced density matrix of* $|\phi_0\rangle = U^{\text{had}}\,|\Psi_{\text{in}}\rangle$ *satisfies*

$$\rho_1^{\text{out}} \;=\; \tfrac{1}{2}\,\Pi, \qquad \rho_1^{\text{out}}[j,j] \;=\; \frac{k}{n} \;>\; 0 \quad \textit{for every } j, \tag{4}$$

where Π *is an orthogonal projector of rank* $2k$*. In particular, no qubit is a vacuum mode after spreading: every mode carries single-particle weight* k/n*, so the data-encoding phase layer imprints a nontrivial phase on every feature. The same holds for* $\rho_1(\mathbf{x})$ *after the diagonal data layer, with* Π *replaced by a unitary conjugate of itself.*

Proof. The $n-2k$ idle qubits, initialised to $|0\rangle$, carry no particles before the spreading circuit. Let $c_j^\dagger$ and c_j denote the fermionic creation and annihilation operators for mode j (under the Jordan–Wigner correspondence, $c_j^\dagger$ maps the vacuum on qubit j to an occupied mode), so that $n_j = c_j^\dagger c_j$ is the occupation-number operator. For any k-particle passive FLO unitary with single-particle matrix W, the single-particle reduced density matrix transforms as $\rho_1^{\text{out}} = W\,\rho_1^{\text{in}}\,W^\dagger$, where $\rho_1^{\text{in}}[j,j'] = \langle\Psi_{\text{in}}|\,c_j^\dagger c_{j'}\,|\Psi_{\text{in}}\rangle$ is the single-particle reduced density matrix of the input.

Diagonal entries. Within each block the two Fock-state contributions partition the block's modes into

two disjoint groups of equal amplitude: block g occupies modes $\{4g, 4g{+}1, 4g{+}2, 4g{+}3\}$ and superposes $\{4g, 4g{+}1\}$ with $\{4g{+}2, 4g{+}3\}$ at equal amplitude $1/\sqrt{2}$. Each active mode $j \in \{0, \ldots, 2k-1\}$ belongs to exactly one block and appears in exactly one of that block's two Fock-state choices, so it is occupied in exactly half of the input Fock states, giving $\rho_1^{\text{in}}[j,j] = \frac{1}{2}$ for $j \in \{0, \ldots, 2k-1\}$ and 0 otherwise, with $\operatorname{tr}(\rho_1^{\text{in}}) = (2k) \cdot \frac{1}{2} = k$.

Off-diagonal entries vanish. For $j \neq j'$, $\rho_1^{\text{in}}[j,j'] = \langle \Psi_{\text{in}} | \, c_j^\dagger c_{j'} \, | \Psi_{\text{in}} \rangle$. If j and j' lie in different blocks, $c_j^\dagger c_{j'}$ changes the particle number within each of the two blocks; since $|\Psi_{\text{in}}\rangle$ is a tensor product of blocks each carrying a definite particle number (2 per block), the result is orthogonal to $|\Psi_{\text{in}}\rangle$ and the overlap is zero. If j and j' lie in the same block, $c_j^\dagger c_{j'}$ moves a single particle within that block; acting on either Fock component (e.g. $|1100\rangle$ or $|0011\rangle$ for a paired block) it produces a basis state of the same Hamming weight orthogonal to both components—converting one component into the other would require moving *two* particles—so the overlap is again zero. Hence $\rho_1^{\text{in}} = \frac{1}{2}\Pi_A$ with Π_A the projector onto the $2k$ active modes, and $\rho_1^{\text{out}} = \frac{1}{2} W \Pi_A W^\dagger$ is $\frac{1}{2}$ times a rank-$2k$ projector.

Walsh–Hadamard transform. Both the Hadamard butterfly and the Hadamard brick-wall implement the normalised Walsh–Hadamard transform on n modes in the single-particle sector, $H[i,j] = (-1)^{\langle i,j \rangle}/\sqrt{n}$, where $\langle i,j \rangle = \bigoplus_b i_b j_b$ is the bitwise inner product of the binary expansions of i and j, so $|H[i,j]|^2 = 1/n$ for every pair i, j.

Output occupations. Because ρ_1^{in} is diagonal, $\rho_1^{\text{out}}[i,i] = [H \rho_1^{\text{in}} H^\dagger]_{ii} = \sum_j |H[i,j]|^2 \, \rho_1^{\text{in}}[j,j]$. Using $|H[i,j]|^2 = 1/n$ for every pair (i,j),

$$\rho_1^{\text{out}}[i,i] = \frac{1}{n} \sum_{j=0}^{2k-1} \rho_1^{\text{in}}[j,j] = \frac{1}{n} \operatorname{tr}(\rho_1^{\text{in}}) = \frac{k}{n} \tag{5}$$

for every output mode i, as claimed. The data layer $D(\boldsymbol{\varphi}(\mathbf{x}))$ is a diagonal unitary, so $\rho_1(\mathbf{x}) = D \rho_1^{\text{out}} D^\dagger$ is again $\frac{1}{2}$ times a rank-$2k$ projector with the same diagonal. □

C. Data Encoding Layer

Apply $D(\boldsymbol{\varphi}) = \bigotimes_{i=1}^n R_z(\varphi_i)$ where $\varphi_i = f(x_i)$, e.g. $\varphi_i = \pi x_i / \max_i |x_i|$. Since $R_z(\varphi) \, |1\rangle = e^{-i\varphi/2} \, |1\rangle$ and $R_z(\varphi) \, |0\rangle = e^{+i\varphi/2} \, |0\rangle$:

$$|\psi(\mathbf{x})\rangle = D(\boldsymbol{\varphi}(\mathbf{x})) \, |\phi_0\rangle = \sum_{|S|=k} \alpha_S \, e^{-i \sum_{j \in S} \varphi_j} \, |S\rangle \, . \tag{6}$$

All n features enter through independent phases. R_z is diagonal in the computational basis—it preserves Hamming weight, non-Gaussianity, and form-rank. Note that this differs from standard angle encoding, where R_z gates are applied directly to $|0\rangle^{\otimes n}$ to produce a product state; here the R_z layer imprints data as phases on the pre-spread entangled state $|\phi_0\rangle$, so the data interacts with all $\binom{n}{k}$ superposition components simultaneously rather than independently qubit by qubit. Unlike amplitude encoding, this scheme does not preserve ℓ_2 inner products between data points: the encoded state defines an implicit kernel that is translation-invariant in feature-phase differences. This is by design—the goal of this encoding is expressivity and classical hardness, not metric preservation.

The total encoding depth is $\log_2 n + 5$ on all-to-all connectivity (depth-4 magic preparation + $\log n$ Hadamard butterfly + 1 data layer), and $n + 4$ on nearest-neighbor connectivity. All magic blocks are prepared in parallel in Step A, so the magic preparation depth does not scale with k.

D. The Right Choice of k

The particle number k is a free parameter of the construction. A single quantity controls the classical simulation cost of the full pipeline: the number of non-Gaussian blocks, $B = k/2$, through the FLO extent $\chi = 2^B$ (Section VII). Exact amplitudes cost $O(2^B \operatorname{poly}(n))$ classically; sampling costs $O(2^{2B} \operatorname{poly}(n))$ under best-known algorithms [1]. Few-body expectation values are additively tractable at every k for this encoding [51], so k calibrates the hardness of *sampling* specifically. The training cost of the multi-layer parallel parameter-shift rule of Section V scales in the opposite direction: $4kn$ circuit evaluations per gradient step for the brick-wall and $4k \log n$ for the butterfly, a factor n/k below the naive parameter-shift rule for both. Larger k therefore buys stronger classical hardness at linearly growing training cost. These tradeoffs apply to both the unitary brick-wall and the unitary butterfly, which share the same encoding pipeline and differ only in the trainable and spreading circuits that follow. We first state the asymptotic regimes that organize the hardness results of Section VII, and then fix the concrete operating values used throughout the paper.

a. $k = O(\log n)$*: classically simulable.* Here $\chi = 2^B = \operatorname{poly}(n)$, and the full pipeline—amplitudes, few-body expectation values, and sampling—is classically simulable in polynomial time (Remark 1). No quantum advantage is possible in this regime. It is nonetheless operationally valuable: the same architecture, run at small k, can be simulated exactly, which permits end-to-end classical validation of the training pipeline and of hardware behavior before scaling k up. Training is cheapest here ($O(kn)$ evaluations per step for the brick-wall, $O(k \log n)$ for the butterfly).

b. $k = \omega(\log n)$*: superpolynomial hardness under best-known algorithms.* Once the block count exceeds logarithmic, χ is superpolynomial and best-known classical algorithms cost $2^{\Omega(k)} \operatorname{poly}(n)$. At the representative

point $k = \log^2 n$ this is $n^{\Omega(\log n)}$, quasi-polynomial hardness for sampling (the additive-error estimator of [51] does not enable sampling, which requires relative precision on exponentially small marginals). No polynomial-hierarchy-based statement is available in this regime—the worst case itself is computable in quasi-polynomial time—but training remains genuinely cheap.

c. $k = n^\epsilon$: worst-case #P-hardness. For any constant $\epsilon > 0$, the Ivanov–Gurvits reduction embeds permanents of $n^{\Omega(1)}$-size matrices, and there exist parameter settings of the brick-wall (which is surjective onto $U(n)$, Section III) whose output probabilities are #P-hard to compute exactly (Section VII; for the butterfly, whose fixed-depth parametrization covers only a submanifold of $U(n)$, see the reachability remark there). Average-case hardness and anticoncentration for such sparse inputs are open—this is precisely the "fermion sampling with less magic" problem of [2]—and we state them as a conjecture. Best-known simulation cost is $2^{\Omega(n^\epsilon)}$, and training costs $O(kn) = O(n^{1+\epsilon})$ (brick-wall) or $O(n^\epsilon \log n)$ (butterfly) evaluations per step.

d. $k = \Theta(n)$: the extensive limit. At $k = cn$ with a constant filling fraction c the Hilbert space dimension $\binom{n}{k}$ is exponential in n, and at $c = 1/2$ the construction matches the original Fermion Sampling setup of [2]; the full machinery—anticoncentration, worst-to-average-case reduction, and conditional hardness of approximate sampling—applies to the Haar-initialized brick-wall (Section VII). Trainability is at its strongest here as well: for $c < 1/2$ the gradient variance is $\Theta(1/n^2)$, the largest order it attains anywhere in the range (Section IV). What is lost is only *cheap* training: at $k = \Theta(n)$ both the naive and the parallel parameter-shift rule cost $\Theta(n^2)$ circuit evaluations per gradient step, the standard cost for a circuit with $\Theta(n^2)$ parameters. This regime is thus the strongest-hardness theoretical limit; the operating point below trades a calibrated amount of that hardness for training cost that falls with n.

e. Operating point: thirty blocks. Asymptotic labels hide constants, so for concrete deployments we calibrate k directly against classical compute. We require that best-known classical simulation of sampling from the pipeline exceed $\sim 10^{24}$ elementary operations—about two weeks on a 10^{18}-flop/s machine—which, with 2^{2B} cross terms each costing at least $\mathrm{poly}(n) \gtrsim 10^6$ operations (each cross term requires cubic-cost linear algebra on $n \times n$ matrices and we take $n \geq 120$), is reached at $\chi^2 = 2^{2B} \approx 10^{18}$, i.e. at

$$B = 30 \text{ blocks:} \quad k = 60, \ |\Psi_4\rangle^{\otimes 30}. \tag{7}$$

The operating point sits at hardness level $\chi^2 = 2^{60}$. It requires $n \geq 120$ modes, and the training advantage of the parallel parameter-shift rule, n/k, *grows linearly with n at fixed hardness*: it reaches 17.1× at $n = 1024$, and 68.3× at $n = 4096$. Two caveats about this calibration before moving on. First, it is pinned to *best-known* algorithms [1, 51]; any future algorithmic improvement is absorbed by increasing B, at training cost linear in k. Second, individual amplitudes at $B = 30$ cost only $2^{30}\,\mathrm{poly}(n)$ operations classically; the sampling barrier rests on the 2^{2B} cost of marginal-based samplers, so a hypothetical sampler operating at amplitude-level cost—no fermionic analogue is currently known—would halve the exponent and be absorbed by doubling B.

Throughout the remainder of the paper we carry k as a free parameter in all statements, instantiate asymptotic claims on the ladder above, and use $B = 30$ (Eq. (7)) for all concrete numbers. Complete parameter values at representative problem sizes are collected in Section VIII A.

Remark 3 (Alternative data loading via Dicke states)**.** *The block-product magic states are not the only route to a classically hard, non-Gaussian input. A natural alternative is the Dicke state $|D_k^n\rangle$—the uniform superposition of all $\binom{n}{k}$ Hamming-weight-k Fock states—which has form-rank exactly k. Its 1-RDM is $(k/n)I_n$ by permutation symmetry, so the uniform-occupation property of Lemma 2 holds* without *a separate spreading step. The price is preparation depth: $O(k \log(n/k))$ on all-to-all connectivity [58, 64] (polylogarithmic, but deeper than the constant-per-block magic-state preparation) and $O(n)$ on nearest-neighbor connectivity. The block-product construction is therefore preferable when depth is the binding constraint. We note that the trainability analysis of Section IV does not transfer to this input: it is driven by $\tau = \|[g, \rho_1]\|_F^2$, which vanishes identically when $\rho_1 \propto I$, so both the disconnected and the contracted contributions of Proposition 10 are zero and the gradient variance is determined entirely by the connected two-body correlations of the Dicke state, which we do not analyze here.*

III. THE UNITARY BRICK-WALL AND BUTTERFLY ARCHITECTURES

A. Reconfigurable Beam Splitter Gates

The basic gate primitive is the Reconfigurable Beam Splitter (RBS) gate, introduced in the context of quantum machine learning in [23, 41] as the natural building block for implementing orthogonal transformations on quantum hardware. Acting on qubits i and j, it is defined as:

$$\mathrm{RBS}(\theta) = \exp(-i\theta G_{ij}), \quad G_{ij} = i(|01\rangle\langle 10| - |10\rangle\langle 01|)_{ij}$$

with matrix representation in the $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ basis:

$$\mathrm{RBS}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{8}$$

The generator G_{ij} has eigenvalues $\{-1, 0, 0, +1\}$ and acts only on the $\{|01\rangle, |10\rangle\}$ subspace. Crucially, RBS gates

are *particle-number preserving*: they map $|01\rangle \leftrightarrow |10\rangle$ and fix $|00\rangle$ and $|11\rangle$.

This gate is a member of several well-studied families that have appeared across quantum information and condensed matter physics under different names. It is a *matchgate* in the sense of Valiant [65] and Terhal–DiVincenzo [66], who showed that nearest-neighbor circuits of such gates are efficiently classically simulable via their connection to noninteracting fermions through the Jordan–Wigner transformation [48, 67]. In the language of fermionic linear optics (FLO), the RBS gate is a passive FLO operation: it implements a beam splitter transformation on two fermionic modes, mixing them without creating or destroying particles. This connection places RBS circuits in the Fermion Sampling framework [2].

Augmenting each RBS gate with a single-qubit R_z phase gate on *each* qubit of the pair gives the three-parameter gate primitive that we use throughout both architectures,

$$\mathcal{G}(\theta, \phi_a, \phi_b) \;=\; \mathrm{RBS}(\theta) \cdot \big(R_z(\phi_a) \otimes R_z(\phi_b)\big). \tag{9}$$

This primitive contains the passive FLO two-qubit gate $D_{\mathrm{pas}}(\alpha_1, \alpha_2)$ of Oszmaniec et al. [2] (their Eq. (20)):

$$D_{\mathrm{pas}}(\alpha_1, \alpha_2) = \big(e^{-i\alpha_1 Z_1/2}\, e^{i\alpha_1 Z_2/2}\big)\, e^{i\alpha_2 (X_1X_2 + Y_1Y_2)/2}, \tag{10}$$

where the second factor is the RBS gate with $\theta = \alpha_2$, and the first factor is a *correlated* R_z pair $(R_z(-\alpha_1) \otimes R_z(\alpha_1))$ up to global phase. Our primitive $\mathcal{G}$ carries two independent phases rather than a correlated pair, so it strictly contains D_{pas}: setting $\phi_a = -\phi_b = -\alpha_1$ and $\theta = \alpha_2$ recovers it exactly, up to global phase. The extra phase direction is a single-qubit Z rotation already present in the Cartan subalgebra, so both parametrizations generate the same dynamical Lie algebra; its role here is not algebraic but algorithmic—a phase layer covering all n modes is what the R_z half of the parallel parameter-shift rule requires (Section V). The unitary brick-wall and butterfly circuits introduced below are therefore direct realizations of the passive FLO circuit families of [2] with the DLA lifted from $\mathfrak{so}(n)$ to $\mathfrak{u}(n)$ by the addition of R_z gates.

B. The Unitary Brick-Wall Circuit

The unitary brick-wall is the natural architecture for nearest-neighbor hardware. It is surjective onto $U(n)$ in the single-particle sector, containing a rectangular mesh of Givens rotations as a submanifold. The brick-wall consists of n columns of nearest-neighbor RBS gates on a ring of n modes (a chain with periodic boundary), in an alternating pattern—odd columns act on the $n/2$ pairs $(1,2), (3,4), \ldots, (n-1, n)$, even columns on the $n/2$ pairs $(2,3), (4,5), \ldots, (n-2, n-1), (n, 1)$, for $n^2/2$ RBS gates in total—each column preceded by a full-width layer of n single-qubit R_z gates. Every column therefore covers all n modes with disjoint pairs, a property the parallel parameter-shift rule of Section V exploits; the open-boundary (chain) variant is discussed in Remark 6. Formally, the ℓ-th column applies a phase layer on all n modes followed by the RBS gates of that column:

$$U^{(\ell)} = \Big[\bigotimes_{j \in P_\ell} \mathrm{RBS}(\theta_\ell^{(j)})\Big] \cdot \bigotimes_{i=1}^{n} R_z(\phi_\ell^{(i)}), \tag{11}$$

where P_ℓ is the set of $n/2$ active nearest-neighbor pairs of column ℓ; equivalently, each active pair carries the primitive $\mathcal{G}$ of Eq. (9). The full trainable circuit is the ordered product of the n columns,

$$U(\boldsymbol{\theta}, \boldsymbol{\phi}) = U^{(n)} \cdots U^{(1)}. \tag{12}$$

Within each column all active pairs act on disjoint qubits, so generators from different pairs commute—the structure exploited by the multi-layer parallel parameter-shift rule.

We call this circuit the *unitary brick-wall*, to distinguish it from the *orthogonal brick-wall* of prior work [41], which implements $SO(n)$ in the single-particle sector. The R_z gates are precisely what promote the DLA from $\mathfrak{so}(n)$ to $\mathfrak{u}(n)$.

Proposition 4 (Properties of the unitary brick-wall circuit)**.** *The unitary brick-wall $U(\boldsymbol{\theta}, \boldsymbol{\phi})$ satisfies:*

1. ***Depth:*** $2n$
2. ***Parameters:*** $T = n^2/2 + n^2 = 3n^2/2$
3. ***Universality:*** $\{U(\boldsymbol{\theta}, \boldsymbol{\phi})\}$ *is* surjective onto $U(n)$ *in the single-particle sector, and contains the n^2-dimensional Givens mesh as the submanifold obtained by zeroing the boundary-pair angles and the redundant phases*
4. ***DLA:*** *The dynamical Lie algebra is $\mathfrak{u}(n)$, with* $\dim(\mathfrak{u}(n)) = n^2$.

Proof. Items 1 and 2. The brick-wall consists of n columns of nearest-neighbor RBS gates on the ring, with $n/2$ disjoint active pairs in every column, giving $n \cdot \frac{n}{2} = n^2/2$ RBS gates in total. Each column is preceded by a full-width layer of n single-qubit R_z gates, contributing n parameters per column and n^2 in total; each column therefore has depth 2 (one parallel R_z layer followed by one parallel RBS layer), giving total depth $2n$. Total parameter count: $T = n^2/2 + n^2 = 3n^2/2$. *Item 3 (surjectivity).* Restrict the parameters by setting the boundary-pair angles of the even columns to zero—reducing each even column to its $n/2-1$ chain pairs—and, in every column $\ell \geq 2$, the phase on the second qubit of each remaining active pair and the phases on the modes not covered by those pairs to zero. What remains is exactly one phase per Givens rotation, $n(n-1)/2$ of them, together with the n free phases of the first column. In the single-particle sector each $\mathrm{RBS}(\theta_\ell^{(j)})$ preceded by its single phase acts as

a two-parameter complex Givens rotation on the corresponding pair of modes, and the rectangular mesh decomposition of $U(n)$ states that any $W \in U(n)$ factors as a product of $n(n-1)/2$ nearest-neighbor Givens rotations in this pattern preceded by a diagonal phase matrix. The restricted map from $\mathbb{R}^{n(n-1)+n} = \mathbb{R}^{n^2}$ to $U(n)$ is therefore surjective, with the inverse given by the constructive Givens (QR-type) decomposition of any $W \in U(n)$: the Givens rotations zero out the off-diagonal entries one at a time, and the residual diagonal unitary is absorbed by the first column's phase layer. Since $n(n-1)+n = n^2 = \dim_{\mathbb{R}} U(n)$, this submanifold is a direct parametrization; the full circuit, with $3n^2/2$ parameters, is surjective but over-complete (Remark 5).

Item 4 (DLA). The trainable RBS gates of the brick-wall act on nearest-neighbor pairs of the ring, so their generators $\{G_{i,i+1}\}_{i=1}^{n-1} \cup \{G_{n,1}\}$ do not by themselves span $\mathfrak{so}(n)$; rather, since the chain $1-2-\cdots-n$ is connected, iterated commutation $[G_{i,i+1}, G_{i+1,i+2}] \propto G_{i,i+2}$ reaches G_{ik} for every pair (i,k), so the RBS generators generate $\mathfrak{so}(n)$ by Lie closure (exactly as in Proposition 7). The R_z generators span the full Cartan subalgebra $i\mathfrak{h} \subset \mathfrak{u}(n)$ of imaginary diagonal matrices. The Lie closure of $\mathfrak{so}(n)$ and $i\mathfrak{h}$ is $\mathfrak{u}(n)$ by the same commutator argument as in Proposition 7 (commutators of the G_{ij} with the diagonal R_z generators supply the symmetric off-diagonal generators absent from $\mathfrak{so}(n)$) [36, 37]. This Lie-closure statement for the *algebra* is logically separate from the mesh parametrization of the *group* $U(n)$ in Item 3: the layout reaches every $W \in U(n)$ as an ordered product of nearest-neighbor Givens rotations, even though the nearest-neighbor generators span $\mathfrak{so}(n)$ only after Lie closure. □

Remark 5 (Over-completeness of the full-width phase layers)**.** *The full-width phase layers and the boundary pairs exceed what the Givens mesh requires: the brick-wall carries $3n^2/2$ parameters—$(n^2-n)/2$ redundant phases and $n/2$ boundary-pair angles beyond the n^2 that already saturate $\dim_{\mathbb{R}} U(n)$—and in the single-particle sector the extra parameters add no expressivity. We adopt them for two reasons. First, they make every column's layer primitive identical to the butterfly's ($n/2$ disjoint pairs covering all n modes, each carrying $\mathcal{G}$), so that the two architectures share one gate, one Lie-algebra argument, and one parameter-shift accounting. Second, and operationally, a phase layer covering all n modes and an RBS layer whose pairs cover all n modes are precisely the hypotheses under which the parallel parameter-shift rule costs $2k$ circuit evaluations per column per gate family (Theorems 24 and 26, Lemma 22 and Remark 23).*

Remark 6 (Open-boundary (chain) variant)**.** *On hardware with open boundary—a chain lacking the link between modes n and 1—the $n/2$ boundary gates of the even columns are simply omitted, recovering the standard rectangular Givens mesh with $n(n-1)/2$ RBS gates and $T = (3n^2-n)/2$ parameters. Surjectivity and Haar initialization (Remark 15), the trainability results of Section IV, and the hardness results of Section VII are unaffected: the chain circuit is the submanifold of the ring circuit with the boundary angles frozen at zero, and every such statement is either proved on that submanifold or depends only on the realized single-particle unitary. The only change is to the parallel parameter-shift rule of Section V: an even column then leaves modes 1 and n unpaired, the covering hypothesis of Lemma 22 fails for its RBS gradients, and those columns use instead the $2k$-node quadrature exact on* all *polynomials of degree at most $2k-1$ (the system $\sum_q a_q \sin r_q \cos^{m-1} r_q = \frac{1}{2}$, $m = 1, \ldots, 2k$), at $4k$ circuit evaluations per even column. The per-gradient-step total becomes $\frac{n}{2} \cdot 2k + \frac{n}{2} \cdot 4k + 2kn = 5kn$ in place of $4kn$, and the reduction factor $(4n-2)/(5k) \approx 0.8\, n/k$ in place of n/k; all asymptotic statements are unchanged.*

The key property of the brick-wall that distinguishes it from the butterfly is that it is *surjective onto* $U(n)$: every element of $U(n)$ is reachable, and when the parameters $(\boldsymbol{\theta}, \boldsymbol{\phi})$ are drawn from the appropriate distribution, the brick-wall realizes the Haar measure on $U(n)$. This is a strictly stronger statement than the butterfly's approximate 2-design property established below. In particular, the anticoncentration result of Oszmaniec et al. [2] (their Theorem 1) applies directly to the brick-wall in the fully-packed regime: for Haar-random passive FLO circuits on $U(n)$—which the brick-wall realizes—initialized in fully-packed magic inputs ($k = n/2$), the output distribution anticoncentrates, and the full average-case sampling hardness chain (anticoncentration + worst-to-average-case reduction via Cayley path) goes through without modification. Whether it extends to sparse magic inputs is open. We state this formally, together with its regime of validity in k, in Section VII.

C. The Unitary Butterfly Circuit

An n-qubit unitary butterfly circuit of depth $K = \log_2 n$ consists of K layers. Layer $\ell \in \{1, \ldots, K\}$ applies a full-width layer of n single-qubit phase gates followed by RBS gates on each of the $n/2$ disjoint pairs with stride $2^{\ell-1}$:

$$U(\boldsymbol{\theta}, \boldsymbol{\phi}) = U^{(K)} \cdots U^{(1)}, \tag{13}$$

$$U^{(\ell)} = \Big[\bigotimes_{j=1}^{n/2} \mathrm{RBS}(\theta_\ell^{(j)})\Big] \cdot \bigotimes_{i=1}^{n} R_z(\phi_\ell^{(i)}). \tag{14}$$

Equivalently, each pair carries the three-parameter primitive $\mathcal{G}$ of Eq. (9), one R_z on each of its two qubits. The R_z gate acts as $R_z(\phi)\,|0\rangle = e^{i\phi/2}\,|0\rangle$, $R_z(\phi)\,|1\rangle = e^{-i\phi/2}\,|1\rangle$: a diagonal one-body unitary, hence passive FLO and particle-number preserving. Within each layer, all $n/2$ gate pairs act on disjoint qubits, so generators from different pairs commute—the key structure exploited by the multi-layer parallel parameter-shift rule.

We call this circuit the *unitary butterfly*, to distinguish it from the *orthogonal butterfly* of prior work [23, 41],

which implements $SO(n)$ in the single-particle sector. The R_z gates promote the DLA from $\mathfrak{so}(n)$ to $\mathfrak{u}(n)$.

Proposition 7 (Properties of the unitary butterfly circuit)**.** *The unitary butterfly $U(\boldsymbol{\theta}, \boldsymbol{\phi})$ satisfies:*

1. ***Depth:*** $2 \log n$
2. ***Parameters:*** $T = \frac{3}{2} n \log n$.
3. ***Universality:*** *the subgroup of $U(n)$ generated by $\{U(\boldsymbol{\theta}, \boldsymbol{\phi})\}$ is dense in $U(n)$ in the single-particle sector: products of finitely many butterfly circuits approximate any $W \in U(n)$. A* single *depth-$2 \log n$ butterfly, by contrast, parametrizes a compact submanifold of $U(n)$ of dimension at most $\frac{3}{2} n \log n < n^2$, and is therefore* not *dense in $U(n)$.*
4. ***DLA:*** *The dynamical Lie algebra is $\mathfrak{u}(n)$, with $\dim(\mathfrak{u}(n)) = n^2$.*

Proof. Items 1 and 2. Each of the $K = \log_2 n$ layers acts on $n/2$ disjoint pairs in parallel; the full-width R_z layer (depth 1) is followed by the RBS layer (depth 1), giving per-layer depth 2 and total depth $2 \log n$. Each layer contributes $n/2$ angles $\theta_\ell^{(j)}$ and n angles $\phi_\ell^{(i)}$, for $3n/2$ parameters per layer and $T = \frac{3}{2} n \log n$ total.

Items 3 and 4. The RBS generators G_{ij} are purely imaginary antisymmetric matrices. In the single-particle sector they span $\mathfrak{so}(n)$ in the sense of Lie closure: the butterfly uses only $n \log n/2$ distinct qubit pairs across its $\log n$ layers, but the stride-doubling layer pattern makes these pairs form a connected graph on n modes with diameter $\log n$. Since $[G_{ij}, G_{jk}] \propto G_{ik}$, iterated commutation along connecting paths reaches G_{ik} for arbitrary (i, k), spanning all of $\mathfrak{so}(n)$. The R_z generators span the full Cartan subalgebra $i\mathfrak{h} \subset \mathfrak{u}(n)$ with dimension n. Commutators $[G_{ij}, iH_{kk}]$ produce the symmetric off-diagonal generators $i(e_{ij} + e_{ji})/2$ absent from $\mathfrak{so}(n)$, and together $\mathfrak{so}(n)$, $i\mathfrak{h}$, and these commutators span all of $\mathfrak{u}(n)$ [36, 37], giving DLA $= \mathfrak{u}(n)$ of dimension n^2. Density of the *generated subgroup* in $U(n)$ follows from the density theorem for Lie groups; the dimension count $\frac{3}{2} n \log n < n^2$ shows that the image of a single fixed-depth butterfly, a compact set, cannot itself be dense. The induced action on the k-particle antisymmetric subspace is the same exterior-power representation of $U(n)$ described at the start of Section IV. □

While the butterfly and brick-wall have the same DLA $\mathfrak{u}(n)$, they differ in an important way: the brick-wall is *surjective onto* $U(n)$, while a fixed-depth butterfly covers only a submanifold of $U(n)$ of dimension at most $\frac{3}{2} n \log n$, achieving the full algebra $\mathfrak{u}(n)$ only through Lie closure. As a consequence, a uniformly random butterfly circuit does not realize the Haar measure on $U(n)$; instead it induces a structured distribution whose second-moment operator, restricted to the antisymmetric (fermionic two-mode) sector that carries all observables used in this paper, has spectral gap $1 - 1/n$, establishing the butterfly as an ε-approximate unitary 2-design on that sector with $\varepsilon = O(1/n)$. This spectral gap is established in Appendix B and underlies the barren-plateau analysis of Section IV. The hardness results for the two architectures are correspondingly different: the brick-wall supports the full average-case sampling-hardness machinery of the Oszmaniec et al. framework (in the fully-packed regime) and worst-case #P-hardness at $k = n^\epsilon$, while for the butterfly the unconditional statement is the extent-based best-known-algorithm cost, with worst-case #P-hardness holding under a reachability assumption; we return to this in Section VII.

D. Connection to the FFT and Walsh-Hadamard Transform

At $\theta = \pi/4$ and $\phi = 0$, each gate pair of the butterfly reduces to a beam splitter performing $|01\rangle \mapsto (|01\rangle + |10\rangle)/\sqrt{2}$. The butterfly at all-$\pi/4$ angles and zero phases implements the Walsh–Hadamard transform in the single-particle sector [41]:

$$U^{\mathrm{had}} |e_i\rangle = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} (-1)^{\langle i,j \rangle} |e_j\rangle , \tag{15}$$

where $\langle i, j \rangle$ is the bitwise inner product. The circuit structure follows exactly the radix-2 butterfly decomposition of Cooley and Tukey [68]: at each of the $\log n$ layers, pairs of modes are mixed by a two-mode gate with the pairing stride doubling at each stage. More generally, by setting the phase gates to the appropriate twiddle factors $\omega^{jk/2^\ell}$ with $\omega = e^{2\pi i/n}$, following the Cooley–Tukey decomposition, the circuit implements the full discrete Fourier transform in the single-particle sector [69]. The fact that both the Walsh–Hadamard transform and the discrete Fourier transform are reachable within the parametrized family highlights the architectural richness of the butterfly: canonical passive-FLO operations central to quantum algorithmic primitives are specific points in the parameter space, and the trainable circuit can interpolate between and beyond them. In this paper we use U^{had} (all phases zero) as the fixed Hadamard spreading circuit of Section II for the all-to-all connectivity regime, and the fully parametrized butterfly (free $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$) as the trainable circuit.

The analogous fixed-angle brick-wall (all chain-pair RBS angles set to $\pi/4$, the boundary-pair angles and all R_z angles set to zero) also implements the Walsh–Hadamard transform in the single-particle sector, at depth $n - 1$. This is the Hadamard brick-wall spreading circuit used in the nearest-neighbor encoding of Section II. The trainable brick-wall then follows as a fully parametrized circuit with free $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$.

IV. BARREN PLATEAUS: PROVABLE TRAINABILITY

Both the unitary brick-wall and the unitary butterfly share the same dynamical Lie algebra. As established in Propositions 4 and 7, the Lie closure of the RBS generators (spanning $\mathfrak{so}(n)$) and the R_z generators (spanning the Cartan subalgebra $i\mathfrak{h}$) gives:

$$\mathfrak{g} = \mathfrak{u}(n), \quad \dim(\mathfrak{g}) = n^2, \tag{16}$$

for both architectures. In both cases the induced action on the k-particle antisymmetric subspace is the k-fold antisymmetric tensor (exterior power) representation of $U(n)$, whose image is an n^2-dimensional Lie subgroup of the full unitary group on the $\binom{n}{k}$-dimensional k-particle space. The trainability results of this section apply to the magic-state encoding of Section II, in particular at the operating point of Eq. (7). The gradient-variance analysis depends only on the structure of the trainable circuit and the single-particle occupation structure of the encoded input (Lemma 2).

Despite sharing the same DLA, the two architectures differ in what can be established unconditionally about their gradient variance:

- **Brick-wall**: the brick-wall is surjective onto $U(n)$ and realizes the Haar measure on $U(n)$ for the appropriate parameter distribution. This allows the Weingarten formula to be applied exactly, yielding a *closed expression* for the gradient variance of two-body readouts (Theorem 9) which evaluates to $\Theta(k^3/n^5)$ throughout $n - 2k = \Omega(n)$. The statement is parametrization-independent: it is proved for the invariant directional derivative at a Haar point, hence exactly for the input-layer parameters (Corollary 11), and for every gate of the circuit in the independent-halves ensemble (Theorem 13).
- **Butterfly**: the butterfly is an ε-approximate unitary 2-design with $\varepsilon = O(1/n)$ on the antisymmetric two-mode sector of the single-particle tensor square—the sector that carries all fermionic observables used in this paper (Theorem 16). This gives an unconditional polynomial lower bound $\Omega(k^4/n^6)$ on the gradient variance, and the sharp $\Theta(k^3/n^5)$ rate conditional on Conjecture 20 (the butterfly is an approximate 2-design on $\Lambda^2\mathbb{C}^n$).

In both cases the gradient variance is inverse polynomial in n and grows polynomially in the particle number k, ruling out exponential barren plateaus throughout $n - 2k = \Omega(n)$. The contrast with the orthogonal architectures (DLA $\mathfrak{so}(n)$) is sharpest here: for the orthogonal brick-wall or butterfly, the gradient variance of the two-body readout $n_i n_j$ scales as $\Theta(1/\binom{n}{k})$ [49], which is $\Theta(n^{-60})$ already at the sampling operating point $k = 60$ and superpolynomially small at $k = \omega(\log n)$. The R_z gates that lift the DLA from $\mathfrak{so}(n)$ to $\mathfrak{u}(n)$ are therefore essential for trainability at the particle numbers required for classical hardness.

A. Few-Body Readouts Are Classically Tractable

We first observe that expectation values of observables of fixed body number, while free of any exponential barren plateau, are classically simulable and therefore cannot be the locus of quantum advantage under either architecture. This is a consequence solely of the fact that such expectation values depend on the encoded input through a reduced density matrix of the same body number, which propagates through the corresponding exterior power of the single-particle unitary.

Proposition 8 (Fixed-body readouts depend only on the r-RDM)**.** *Fix $r \geq 1$. For the data-encoded input $|\psi(\mathbf{x})\rangle$ of Section II and any passive FLO unitary W with single-particle matrix $W^{(1)} \in U(n)$ (in particular, either the unitary brick-wall or the unitary butterfly), the expectation value of any r-body observable M is*

$$\langle\psi(\mathbf{x})|\, W^\dagger M W\, |\psi(\mathbf{x})\rangle = \mathrm{tr}\big(M^{(r)}\, \Lambda^r(W^{(1)})\, \rho_r(\mathbf{x})\, \Lambda^r(W^{(1)})^\dagger\big), \tag{17}$$

a function of the r-particle data $\big(M^{(r)}, W^{(1)}, \rho_r(\mathbf{x})\big)$ alone, where $M^{(r)}$ and $\rho_r(\mathbf{x})$ are the r-body matrix of M and the r-particle reduced density matrix of $|\psi(\mathbf{x})\rangle$, both of dimension $\binom{n}{r}$. For the block-product inputs of Section II the matrix $\rho_r(\mathbf{x})$ is available in closed form, so (17) *is computable classically in* $\mathrm{poly}(n)$ *time for every fixed r, at any parameter setting and independently of the particle number k, the FLO extent, and the form-rank of the input. In particular one-body readouts cost $O(n^2)$, with*

$$\langle n_j\rangle(\mathbf{x}) = \big[W^{(1)}\rho_1(\mathbf{x})(W^{(1)})^\dagger\big]_{jj}, \qquad \sum_j \langle n_j\rangle = k, \tag{18}$$

and mean occupation k/n per mode (Lemma 2). Consequently few-body readouts admit efficient classical simulation regardless of the input's non-Gaussian structure and cannot witness quantum advantage.

Proof. For an r-body observable M and passive FLO unitary W, conjugation replaces each mode operator by a linear combination of mode operators, $W^\dagger a_i W = \sum_p W^{(1)}_{ip} a_p$; antisymmetrizing a product of r such operators produces the $r \times r$ minors of $W^{(1)}$, that is the exterior power $\Lambda^r(W^{(1)})$. The Heisenberg evolution therefore stays within the r-body algebra and the expectation value contracts to the r-particle sector, giving (17); this is the standard fixed-body moment-closure property of passive fermionic linear optics [70]. Both $M^{(r)}$ and $\rho_r(\mathbf{x})$ have dimension $\binom{n}{r} = O(n^r)$, so evaluating (17) costs $\mathrm{poly}(n)$ for fixed r; the data layer acts diagonally, $\rho_r(\mathbf{x}) = \Lambda^r(D(\boldsymbol{\varphi}(\mathbf{x})))\, \rho_r^{\mathrm{out}}\, \Lambda^r(D(\boldsymbol{\varphi}(\mathbf{x})))^\dagger$, and ρ_r^{out} is supported on Λ^r of the $2k$-dimensional active subspace and computed blockwise from the magic-block structure. For $r = 1$ the matrices are $n \times n$ and the cost is $O(n^2)$ per entry; Lemma 2 fixes the diagonal of ρ_1^{out} (and hence of $\rho_1(\mathbf{x})$, since D is diagonal) at k/n, giving $\sum_j \langle n_j\rangle = \mathrm{tr}(\rho_1) = k$. □

Few-body readouts are therefore classical baselines: no advantage claim attaches to them, and the hardness results of Section VII are stated at the level of sampling. Their role in this paper is as the *trainable* readout, and among them the two-body correlator $\langle n_i n_j \rangle$ is the family for which we establish polynomial gradient variance for both architectures.

B. Trainability of the Unitary Brick-Wall

The unitary brick-wall is surjective onto $U(n)$ via the Givens rotation decomposition: every element of $U(n)$ is reachable, and the map from circuit parameters $(\boldsymbol{\theta}, \boldsymbol{\phi})$ to single-particle unitary $W \in U(n)$ is surjective. This surjectivity enables a natural and efficient initialization strategy: sample W from the Haar measure on $U(n)$, then compute the corresponding RBS angles $\boldsymbol{\theta}$ and R_z angles $\boldsymbol{\phi}$ via the Givens decomposition (equivalently, via QR decomposition of the sampled matrix). This decomposition is numerically stable and requires $O(n^2)$ operations. Under this initialization, the brick-wall realizes the Haar measure on $U(n)$ exactly.

The computation rests on five structural lemmas, which we state and prove after the theorem so as not to interrupt the argument: one bounds the effect of a *local* gate generator on a Hadamard-delocalized two-particle state, two are exact identities on $\Lambda^2\mathbb{C}^n$ that together evaluate the Weingarten sum in closed form, and two more supply the Haar-averaged versions of the first needed for the interior-gate statements of Theorem 13.

Throughout we use the following notation. We write $d = \binom{n}{2}$ and identify an operator M on $\Lambda^2\mathbb{C}^n$ with its kernel M^{ij}_{kl}, antisymmetric in the upper and in the lower index pair, so that $\mathrm{tr}M = \frac{1}{2}M^{ij}_{ij}$ and $\mathrm{tr}(MN) = \frac{1}{4}M^{ij}_{kl}N^{kl}_{ij}$ (repeated indices summed). The *one-particle contraction* of M is the $n \times n$ matrix

$$\Gamma(M)^i{}_k \;:=\; \sum_j M^{ij}_{kj}, \qquad \Gamma(I_{\Lambda^2}) = (n-1)\, I_n. \tag{19}$$

For a k-particle state, $\Gamma(\rho_2) = (k-1)\rho_1$: contracting one index of the two-particle reduced density matrix against itself removes one particle and counts the $k-1$ ways of choosing the partner. Finally, $d\Lambda^2(g)$ denotes the derivation lift of a single-particle operator g to $\Lambda^2\mathbb{C}^n$, $d\Lambda^2(g)(u\wedge v) = (gu)\wedge v + u\wedge(gv)$; we write $G^{(2)} = d\Lambda^2(g)$ for the lift of the single-particle generator g of a gate.

Throughout this section θ ranges over the trainable parameters whose gate lies in the backward causal cone of the readout observable: a gate acting on modes disjoint from the support of M propagated back through the layers that follow it leaves $\mathcal{L}$ invariant, so its gradient vanishes identically. For the trained readout $M_{\mathrm{eff}} = \sum_{i<j} w_{ij}\, n_i n_j$ of Section VI B, with all w_{ij} nonzero, the cone is the whole circuit.

Theorem 9 (Closed-form gradient variance for the brick-wall, invariant form)**.** *Let $M = n_i n_j$ be a two-body correlator observable ($i \neq j$) and let $|\psi(\mathbf{x})\rangle$ be the data-encoded input of Section II. Write $\mathcal{L}(W;\mathbf{x})$ for the loss of the unitary brick-wall viewed as a function of the single-particle unitary $W \in U(n)$ realized by the trainable circuit, so that $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\phi};\mathbf{x}) = \mathcal{L}\big(W(\boldsymbol{\theta},\boldsymbol{\phi});\mathbf{x}\big)$, and let $W \sim \mathrm{Haar}(U(n))$ as in Remark 15. For a fixed direction $g \in \mathfrak{u}(n)$ define the* invariant *(input-side, right-invariant) directional derivative*

$$D_g\mathcal{L} \;:=\; \frac{d}{ds}\,\mathcal{L}\big(We^{-isg};\mathbf{x}\big)\Big|_{s=0}, \tag{20}$$

and set

$$B \;:=\; \big[\, G^{(2)},\, \rho_{2,\mathfrak{g}_2}(\mathbf{x})\,\big], \qquad G^{(2)} = d\Lambda^2(g), \tag{21}$$

where $\rho_{2,\mathfrak{g}_2}$ is the traceless part of the encoded 2-RDM. Then $\mathbb{E}[D_g\mathcal{L}] = 0$ and, exactly and for every $\mathbf{x}$,

$$\mathrm{Var}[D_g\mathcal{L}] \;=\; \frac{4\,\|B\|_F^2 \;+\; 2\,\|\Gamma(B)\|_F^2}{n^2(n^2-1)}, \tag{22}$$

$$\Gamma(B) \;=\; (k-1)\,[\,g,\,\rho_1(\mathbf{x})\,]. \tag{23}$$

Writing $\rho_1(\mathbf{x}) = \frac{1}{2}\Pi$ with Π a rank-$2k$ projector (Lemma 2) and $\tau := \|[g,\Pi]\|_F^2$, one has

$$\|\Gamma(B)\|_F^2 = \frac{(k-1)^2}{4}\,\tau, \;\; \|B\|_F^2 = \frac{(2k-1)\,\tau}{16} + O\!\left(\frac{k^{3/2}}{n}\right), \tag{24}$$

and, for the encoded state and either generator family, $\tau = \Theta\big(k(n-2k)/n^2\big)$. Consequently, whenever $n - 2k = \Omega(n)$—in particular at the operating point $k = 60$ of Eq. (7), and at any constant filling fraction $k = cn$ with $c < 1/2$—

$$\mathrm{Var}[D_g\mathcal{L}] \;=\; \Theta\!\left(\frac{k^3}{n^5}\right), \tag{25}$$

polynomial in n and cubic in the particle number k, for every $k \geq 2$. The variance therefore decays at most polynomially in n and there is no exponential barren plateau.

Proof sketch; full proof in Appendix A. Right invariance of the Haar measure makes We^{-isg} Haar-distributed for every s, so $\mathbb{E}[D_g\mathcal{L}] = 0$. We do *not* split W into independent factors—the partial products of Givens rotations before and after a fixed gate are not individually Haar-distributed—and instead use that the Haar measure is an exact unitary 2-design on every polynomial irreducible representation, applying the second-moment identity directly on $\Lambda^2\mathbb{C}^n$. Three steps then give the result.

Step 1. As a $U(n)$-representation, $\Lambda^2\mathbb{C}^n \otimes \Lambda^2\mathbb{C}^n = V_{(2,2)} \oplus V_{(2,1,1)} \oplus V_{(1,1,1,1)}$, all three of dimension $\Theta(n^4)$, so Schur's lemma turns the variance into the exact irreducible sum (A19) pairing the weights of $M^{(2)\otimes 2}$ against those of $B^{\otimes 2}$, the generator entering *only* through B.

Step 2. Because $M^{(2)} = |\{i,j\}\rangle\langle\{i,j\}|$ is a rank-one projector, $(\mathrm{tr}M^{(2)})^2 = \mathrm{tr}(M^{(2)2})$ and its weights on $V_{(2,1,1)}$ and on $\Lambda^4\mathbb{C}^n$ both vanish; the sum collapses to the single $V_{(2,2)}$ channel. It is this projector property, and

not any exchange symmetry, that annihilates the other two channels—the general two-body readout of Proposition 12 does not have it.

Step 3. Evaluating the $(2,2)$ weight of $B^{\otimes 2}$ for anti-Hermitian traceless B gives $-\frac{1}{3}\|B\|_F^2 - \frac{1}{6}\|\Gamma(B)\|_F^2$, which upon division by $\dim V_{(2,2)} = n^2(n^2-1)/12$ is exactly (22).

The remaining content of the theorem is the evaluation of the two norms for the encoded state, stated as Proposition 10 below; the supporting lemmas and all computations are deferred to Appendix A. □

Proposition 10 (Evaluation of the two norms)**.** *For the data-encoded input of Section II, write $\rho_1(\mathbf{x}) = \frac{1}{2}\Pi$ with Π a rank-$2k$ projector (Lemma 2) and $\tau := \|[g,\Pi]\|_F^2$. Then, exactly and for every $\mathbf{x}$,*

$$\Gamma(B) = (k-1)\,[\,g,\,\rho_1(\mathbf{x})\,], \qquad \|\Gamma(B)\|_F^2 = \frac{(k-1)^2}{4}\,\tau. \tag{26}$$

Moreover $\|B^{\mathrm{disc}}\|_F^2 = (2k-1)\tau/16$ exactly, and $\|B\|_F^2 = (2k-1)\tau/16 + O(k^{3/2}/n)$, uniformly in $\mathbf{x}$. For the brick-wall, $\tau = 8k(n-2k)/n^2$ holds exactly and unconditionally for every trainable RBS gate, and $\tau = 4k(n-2k)/n^2$ holds exactly and unconditionally for every trainable phase gate. Substituting into (22) *gives* (25)*.*

The one-body term is the larger of the two and is what produces the cubic dependence on k; both terms carry the factor $(n-2k)$, the supply of idle modes, which is why the sharp rate is claimed throughout $n-2k=\Omega(n)$ and not at exactly half filling.

Corollary 11 (Front phase-layer parameters)**.** *Let U_P be the unitary brick-wall initialized by sampling $W \sim \mathrm{Haar}(U(n))$ and computing angles via the Givens decomposition (Remark 15), and let ϕ_j be one of the n phases of the front R_z layer—the first gates the circuit applies. Then $\partial_{\phi_j}\mathcal{L} = D_{\hat{n}_j}\mathcal{L}$ for the occupation-number generator $\hat{n}_j$ of that gate, and Theorem 9 applies verbatim, giving $\mathrm{Var}[\partial_{\phi_j}\mathcal{L}] = \Theta(k^3/n^5)$ exactly and unconditionally, for every $\mathbf{x}$.*

Proof. Write $W = W' D_1$ with $D_1 = \bigotimes_i R_z(\phi_i)$ the front phase layer and W' the product of all later gates. Since $R_z(\phi_j) = e^{-i\phi_j \hat{n}_j}$ up to a global phase and $\hat{n}_j$ is diagonal, it commutes with D_1, so

$$\partial_{\phi_j} W = W'(-i\hat{n}_j)D_1 = -i\,W' D_1\left(D_1^\dagger \hat{n}_j D_1\right) = -i\,W\,\hat{n}_j, \tag{27}$$

which is exactly the right translation of (20) in the fixed, non-random direction $g = \hat{n}_j$. Theorem 9 therefore applies, with $\tau = 4k(n-2k)/n^2$ from Step 4. The front placement of the residual diagonal (Remark 15), adopted because a terminal phase layer would carry identically zero gradient, is what puts this whole layer in the rigorous set. □

Proposition 12 (Arbitrary two-body readout and trained heads)**.** *Let $n \geq 4$ and let M be* any *two-body observable, with image $M^{(2)}$ on $\Lambda^2\mathbb{C}^n$; for a trained head $M_{\mathrm{eff}} = \sum_{i<j} w_{ij}\, n_i n_j$ one has $M^{(2)}_{\mathrm{eff}} = \sum_{i<j} w_{ij}\,|\{i,j\}\rangle\langle\{i,j\}|$ and $\Gamma(M^{(2)}_{\mathrm{eff}}) = \mathrm{diag}(r)$ with $r_i = \sum_{j\neq i} w_{ij}$. Write $\tilde{M} := M^{(2)} - \frac{\mathrm{tr}M^{(2)}}{\binom{n}{2}}\, I_{\Lambda^2}$ for the traceless part and set $p := \|\tilde{M}\|_F^2$, $q := \|\Gamma(\tilde{M})\|_F^2$. Then, in the setting of Theorem 9, exactly and for every $\mathbf{x}$,*

$$\mathrm{Var}[D_g\mathcal{L}] \;=\; c_2(M)\,\|B\|_F^2 \;+\; c_1(M)\,\|\Gamma(B)\|_F^2, \tag{28}$$

with the two head-dependent, k-independent scalars

$$c_1(M) = \frac{(n^2-n+2)\,q \;-\; 4(n-1)\,p}{n^2(n^2-1)(n-2)(n-3)},$$
$$c_2(M) = \frac{4\big[(n-2)\,p \;-\; q\big]}{n^2(n+1)(n-2)(n-3)}. \tag{29}$$

Three consequences.

(i) (consistency) *For $M = n_i n_j$ one has $p = 1 - 2/(n(n-1))$ and $q = 2 - 4/n$, giving $c_2 = 4/(n^2(n^2-1))$ and $c_1 = 2/(n^2(n^2-1))$, i.e. exactly* (22)*. For $M^{(2)} \propto I_{\Lambda^2}$ one has $p = q = 0$ and $\mathrm{Var} = 0$, as it must be.*

(ii) (the head is not a $\Theta(1)$ prefactor) *Combining* (28) *with Step 4 and* (A31)*,*

$$\mathrm{Var}[D_g\mathcal{L}] \;=\; \frac{\tau}{16}\Big[(2k-1)\,c_2(M) + 4(k-1)^2\,c_1(M)\Big] + c_2(M)\cdot O\Big(\tfrac{k^{3/2}}{n}\Big). \tag{30}$$

Since c_1 and c_2 may differ in order and in sign, the rate itself, and not merely its constant, depends on the head. For the single-pair readout $c_1 = \Theta(n^{-4})$ dominates and the rate is $\Theta(k^3/n^5)$; for the uniform nearest-neighbour head $w_{i,i+1} = 1$ one computes $p = (n-1)(n-2)/n$, $q = 2(n-2)/n$, hence $c_2 = 4/(n^2(n+1))$ and $c_1 = -2/(n^2(n^2-1))$, and the rate is $\Theta(k^2/n^4)$—larger than $\Theta(k^3/n^5)$ by the factor n/k.

(iii) (no exponential barren plateau, for every head) *For $k \geq 2$ and $n \geq 2k+2$ the bracket in* (30) *is positive and bounded below by its value at $q = 0$:*

$$(2k-1)c_2 + 4(k-1)^2 c_1 \;\geq\; \frac{4\,p\,(n-1)\big[(2k-1)(n-2) - 4(k-1)^2\big]}{n^2(n^2-1)(n-2)(n-3)} \;>\; 0. \tag{31}$$

Consequently, for $n-2k=\Omega(n)$ and k above a constant, $\mathrm{Var}[D_g\mathcal{L}] = \Omega\big(p\,k^2/n^5\big)$: the gradient variance of any *two-body readout with $\tilde{M} \neq 0$ is at most polynomially small.*

Theorem 13 (Every gate, independent-halves ensemble)**.** *Let θ be any trainable parameter of the brick-wall, with generator g, and write the trainable circuit as $W = W_2\,U_\theta\,W_1$, where W_1 is the product of the gates preceding θ and W_2 the product of those following it. Assume the*

independent-halves ensemble: *W_1 and W_2 are independent and Haar-distributed on $U(n)$. Then, exactly and for every $\mathbf{x}$,*

$$\mathrm{Var}\left[\frac{\partial \mathcal{L}}{\partial \theta}\right] = \frac{4\,\mathbb{E}_{W_1}\|B_{\tilde{g}}\|_F^2 \;+\; 2\,\mathbb{E}_{W_1}\|\Gamma(B_{\tilde{g}})\|_F^2}{n^2(n^2-1)},\ \tilde{g} = W_1^\dagger g W_1, \tag{32}$$

with $B_{\tilde{g}} = [\, d\Lambda^2(\tilde{g}),\, \rho_{2,\mathfrak{g}_2}(\mathbf{x})\,]$, and, for either generator family,

$$\mathrm{Var}\left[\frac{\partial \mathcal{L}}{\partial \theta}\right] \;=\; \Theta\!\left(\frac{k^3}{n^5}\right) \ \textit{whenever } k \geq 2 \textit{ and } n-2k = \Omega(n), \tag{33}$$

unconditionally, with the same constants as in Theorem 9. The independent-halves ensemble is the standard setting for interior-gate statements in the dynamical-Lie-algebra barren-plateau literature [36, 37].

Proposition 14 (Interior gates of the literal single mesh)**.** *Let W be the brick-wall at the Haar initialization of Remark 15, realized through the Hurwitz factorization of the Haar measure into Givens rotations, in which the phase attached to each rotation is uniform on $[0, 2\pi)$ and independent of all rotation angles [71, 72]. Let θ be a trainable parameter in column ℓ, let W_1 be the product of the columns preceding it, and set $\Pi^{(\ell)} := W_1 \Pi W_1^\dagger$. Then:*

(a) (trivial prefix) *for a parameter of the front R_z layer, $W_1 = I$ and Corollary 11 gives the exact rate $\Theta(k^3/n^5)$;*

(b) (Haar prefix) *if W_1 is Haar and independent of the suffix, Theorem 13 gives the same rate, with the same constants;*

(c) (first moment at every depth) *for every ℓ and every j, $\mathbb{E}\big[\Pi_{jj}^{(\ell)}\big] = 2k/n$; the uniform diagonal produced by the Walsh–Hadamard spreading is preserved in expectation at all depths.*

Parts (a) and (b) pin the interior case at both endpoints of the prefix, with matching constants, and (c) is the corresponding first-moment statement at every intermediate depth.

Remark 15 (Haar initialization via Givens decomposition)**.** *The Haar initialization procedure is both natural and practical for the brick-wall. Given a Haar-random matrix $W \sim \mathrm{Haar}(U(n))$, its QR decomposition yields $n(n-1)/2$ complex Givens rotations—each decomposing into one RBS angle θ and one R_z angle ϕ—and n residual diagonal phases, absorbed by the first column's phase layer, producing n^2 circuit angles; this is the standard rectangular mesh decomposition [73, 74] (with the residual diagonal placed at the input rather than the output of the mesh), and the induced law on the angles is the Hurwitz factorization of the Haar measure [71, 72], in which each Givens phase is uniform on $[0, 2\pi)$ and independent of the rotation angles; the remaining $(n^2-n)/2$ redundant phases of the full-width layers and the $n/2$ boundary-pair angles of the even columns are set to zero (Remark 5), which places the initialization on the Givens submanifold and makes W exactly Haar-distributed. The decomposition is numerically stable and requires $O(n^2)$ operations. This contrasts with the butterfly, for which no analogous inversion is possible: the butterfly does not fully parametrize $U(n)$, so there is no efficient procedure to find butterfly angles corresponding to a given $W \in U(n)$.*

C. Trainability of the Unitary Butterfly

The unitary butterfly does not fully parametrize $U(n)$: with only $O(n \log n)$ parameters it cannot reach every element of $U(n)$, and there is no efficient procedure to find butterfly angles corresponding to a given $W \in U(n)$. Consequently, Haar initialization is not available for the butterfly, and the gradient variance analysis must proceed differently. Instead, we initialize all RBS angles θ and all phase angles ϕ independently and uniformly on $[0, 2\pi)$, and rely on the approximate 2-design property of the butterfly established in Theorem 16. This gives two results: an unconditional polynomial lower bound on gradient variance, and a sharp $\Theta(k^3/n^5)$ rate conditional on Conjecture 20.

1. *The Butterfly as an Approximate 2-Design*

The Fontana et al. [36] framework for barren plateaus requires that the parametrized circuit's second-moment operator has the correct fixed-point structure and a sufficiently large spectral gap. Density of the generated subgroup in $U(n)$ is not sufficient for this; we need to establish the spectral gap of the second-moment operator directly. Let $\Phi_2^{W_n}$ denote the second-moment operator of the butterfly on $\mathrm{End}(\mathbb{C}^n \otimes \mathbb{C}^n)$:

$$\Phi_2^{W_n}[Y] = \mathbb{E}_{W_n}\left[(W_n \otimes W_n)\, Y\, (W_n^\dagger \otimes W_n^\dagger)\right], \tag{34}$$

where the expectation is over the parameters of the butterfly, sampled uniformly on their domain. Since $W_n \otimes W_n$ commutes with SWAP, the antisymmetric sector $\mathrm{End}(\Lambda^2\mathbb{C}^n) = \{P_A Y P_A\}$, with $P_A = (I_{n^2} - \mathrm{SWAP})/2$ the projector onto $\Lambda^2\mathbb{C}^n \subset \mathbb{C}^n \otimes \mathbb{C}^n$, is $\Phi_2^{W_n}$-invariant; we write $\Phi_2^{W_n, A}$ for the restriction. This is the sector that carries every quantity appearing in this paper: k-particle states and particle-number-preserving observables evolve in exterior powers of the single-particle action, and the variance computations below engage the second moment only through antisymmetric-sector operators.

Theorem 16 (Butterfly is an approximate 2-design on the antisymmetric sector)**.** *The restriction $\Phi_2^{W_n, A}$ of the second-moment operator of the unitary butterfly at uniformly sampled parameters has fixed-point space $\mathrm{span}\{P_A\}$ (eigenvalue 1), and every other eigenvalue lies in $\{0, 1/n\}$; its spectral gap is $1 - 1/n$. Consequently,*

the butterfly is an ε-approximate unitary 2-design on the antisymmetric sector,

$$\varepsilon_{2\mathrm{d}}(n) \;:=\; \|\Phi_2^{W_n,A} - \Phi_2^{\mathrm{Haar},A}\|_{\mathrm{op}} \;=\; O(1/n). \qquad (35)$$

The proof is given in Appendix B. It proceeds via a parity-decoupling recursion that tracks how the second-moment operator acts on the antisymmetric projector basis $\{P_{ij}\}_{i<j}$ through the butterfly's stride-doubling layer structure, showing that the associated transfer matrix is doubly stochastic and that its only non-zero, non-Perron eigenvalue is $1/n$ (with multiplicity $K-1$). The general case $n = 2^K$ is established in closed form by diagonalizing the second-moment operator using the $(\mathbb{Z}_2)^K$ translation (Fourier) symmetry of the butterfly: every character $y \neq 0$ is annihilated layer by layer, and the surviving $y = 0$ block is diagonalized by an explicit induction on K. Theorem 16 therefore holds unconditionally for all $n = 2^K$.

Two features of this result deserve emphasis.

- The spectral gap is established on the antisymmetric sector of the tensor square of the *single-particle* representation. Extending it to the antisymmetric 2-particle representation $\Lambda^2\mathbb{C}^n$ governs two-body correlator readouts and is the content of Conjecture 20 below.
- The butterfly achieves spectral gap $1 - 1/n$ at depth $2\log n$. This is notable because generic local circuits typically require polynomial depth to reach a comparable spectral gap [75, 76]. The stride-doubling structure of the butterfly is what enables this near-optimal spectral gap at logarithmic depth.

We stress that $\Lambda^2\mathbb{C}^n$ is the same space in both statements; what differs is the number of copies of the Λ^2 action. Theorem 16 controls $\mathbb{E}[\Lambda^2(W_n) \cdot \Lambda^2(W_n)^\dagger]$—a single copy, quadratic in the entries of W_n, obtained by restricting the second-moment operator of the fundamental representation—whereas Conjecture 20 controls $\mathbb{E}[\Lambda^2(W_n)^{\otimes 2} \cdot \Lambda^2(W_n)^{\dagger\otimes 2}]$, which is quartic in those entries. This is why the transfer-matrix recursion of Appendix B settles the former and not the latter.

2. Gradient Variance Bounds

We now establish the gradient variance results for the butterfly. We separate them into an unconditional polynomial lower bound and a sharp conditional rate.

Theorem 17 (Absence of exponential barren plateau, unconditional)**.** *Let $n = 2^K$ and let the particle number satisfy $4k \le n$. Let U_B be the unitary butterfly at uniformly random initialization and $|\psi(\mathbf{x})\rangle$ the data-encoded input of Section II. Then there exist a two-body correlator $M = n_i n_j$ and a trainable RBS parameter θ of the butterfly such that, with $\mathcal{L} = \langle\psi(\mathbf{x})| U_B^\dagger M U_B |\psi(\mathbf{x})\rangle$,*

$$\mathrm{Var}\left[\frac{\partial\mathcal{L}}{\partial\theta}\right] \;\ge\; \frac{k^2(k-1)^2}{2\,n^6}. \qquad (36)$$

The bound uses no design property of the butterfly. In particular the gradient of the readout decays at most polynomially in n: there is no exponential barren plateau.

Remark 18 (On the hypothesis $4k \le n$)**.** *The condition $4k \le n$ is used only in Step 4 below, to guarantee that the stride Σ of the last butterfly layer maps every occupied pre-spread mode to an unoccupied one. It is a mild constraint at the operating point of Eq. (7): it requires $n \ge 240$ at $k = 60$, far below the sizes at which the hardness calibration of Section II D becomes relevant.*

Theorem 19 (Sharp $\Theta(k^3/n^5)$ gradient variance, conditional)**.** *Let $M = n_i n_j$ be a two-body correlator observable ($i \neq j$) and let $\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}) = \langle\psi(\mathbf{x})| U_B^\dagger M U_B |\psi(\mathbf{x})\rangle$ be the associated loss with U_B the unitary butterfly at uniformly random initialization and $|\psi(\mathbf{x})\rangle$ the data-encoded input of Section II. Assume Conjecture 20 below. Then, for every $\mathbf{x}$ and every $2 \le k \le \gamma_0 n$, for a small enough constant $\gamma_0 > 0$ determined by the implied constant of Conjecture 20, and for every phase parameter θ of the butterfly's* first R_z layer*—as well as, in the independent-halves ensemble of Theorem 13, for every trainable parameter in the causal cone of M, where the statement is in fact* unconditional*: the ensemble makes the composite Haar and Theorem 13 applies without Conjecture 20, which is needed only for the literal fixed-depth butterfly—*

$$\mathrm{Var}\left[\frac{\partial\mathcal{L}}{\partial\theta}\right] \;=\; \Theta\!\left(\frac{k^3}{n^5}\right), \qquad (37)$$

polynomial in n and cubic in k, matching the unconditional brick-wall rate of Theorem 9. Whether the same statement holds pointwise for an interior parameter of the literal single-mesh butterfly is open, as is its brick-wall counterpart (Section VIII).

Conjecture 20 (Butterfly is a relative-error 2-design on Λ^2)**.** *The unitary butterfly W_n at uniformly random parameters is an ε-approximate unitary 2-design on $U(n)$ in the antisymmetric 2-particle representation $\Lambda^2\mathbb{C}^n$, in the* relative *(multiplicative) sense, with $\varepsilon = O(1/n)$: writing*

$$\Phi_2^{\Lambda^2,W_n}[Y] := \mathbb{E}_{W_n}\left[\Lambda^2(W_n)^{\otimes 2}\, Y\, \Lambda^2(W_n)^{\dagger\otimes 2}\right], \qquad (38)$$

for $Y \in \mathrm{End}(\Lambda^2\mathbb{C}^n \otimes \Lambda^2\mathbb{C}^n)$, and $\Phi_2^{\Lambda^2,\mathrm{Haar}}$ for the corresponding Haar second moment,

$$(1-\varepsilon)\,\Phi_2^{\Lambda^2,\mathrm{Haar}} \;\preceq\; \Phi_2^{\Lambda^2,W_n} \;\preceq\; (1+\varepsilon)\,\Phi_2^{\Lambda^2,\mathrm{Haar}} \qquad (39)$$

as completely positive maps. In particular $\Phi_2^{\Lambda^2,W_n}$ has fixed-point space of dimension 3, matching the Haar decomposition $\Lambda^2 \otimes \Lambda^2 = V_{(2,2)} \oplus V_{(2,1,1)} \oplus V_{(1,1,1,1)}$ into three irreducible $U(n)$-representations.

The relative form is what makes the conjecture usable here. The additive form—$\|\Phi_2^{\Lambda^2,W_n} - \Phi_2^{\Lambda^2,\mathrm{Haar}}\| \le \varepsilon$, which is what a spectral gap such as Theorem 16 supplies, and which is strictly weaker—does not suffice at any $\varepsilon = \Omega(n^{-4})$: it permits an additive error $O(\varepsilon\|B\|_F^2)$ in Eq. (40) below, whereas the quantity being estimated is itself of order $\|B\|_F^2/n^4$. Conversely, Conjecture 20 is stronger than what is used below: Claim 1 consumes only the single pairing of $\Phi_2^{\Lambda^2,W_n}[M^{(2)\otimes 2}]$ against $B^{\otimes 2}$ for the readouts of Section VI B, so any statement reproducing that scalar up to constants suffices, and Theorem 19 survives correspondingly weaker hypotheses. Should the conjecture fail as stated, a further degree of freedom is the initialization—or a slight modification of the architecture—which may be chosen to bring the second moment closer to the Haar one; we do not pursue this here.

The proof of Theorem 19 follows the brick-wall computation with the exact Haar second moment replaced by the approximate one. We record the two ingredients.

Claim 1 (Weingarten variance formula on Λ^2). *Assume Conjecture 20, fix a trainable parameter θ with generator g, and write $W = V_2 U_\theta V_1$ and $\tilde{g} = V_1^\dagger g V_1$. Assume moreover that the prefix V_1 is independent of the twirl variable—automatic for a parameter of the first R_z layer, where $V_1 = I$ and $\tilde{g} = g = \hat{n}_j$ is deterministic, and imposed by construction in the independent-halves ensemble. Then*

$$\begin{aligned} \mathrm{Var}[\partial_\theta \mathcal{L}] &= \frac{4\|B\|_F^2 + 2\|\Gamma(B)\|_F^2}{n^2(n^2-1)}\Big(1 + O(k/n)\Big), \\ B &= [\tilde{G}^{(2)}, \rho_{2,\mathfrak{g}_2}(\mathbf{x})], \end{aligned} \tag{40}$$

where $\tilde{G}^{(2)} = d\Lambda^2(\tilde{g})$ and the relative $O(k/n)$ is the design error of Conjecture 20, amplified by the effective-rank factor $\|B\|_1^2/\big(\|B\|_F^2 + \frac{1}{2}\|\Gamma(B)\|_F^2\big) = O(k)$ as in the proof below; it is $o(1)$ throughout $k = o(n)$.

Proof. As in Theorem 9, M is two-body, so $\partial_\theta \mathcal{L} = -i\,\mathrm{tr}_{\Lambda^2}\big(\Lambda^2(W)^\dagger M^{(2)} \Lambda^2(W)\, B\big)$, with the commutator carried by the conjugated generator $\tilde{g}$; the independence hypothesis is what allows B to be held fixed while the twirl acts on $M^{(2)\otimes 2}$, and without it the twirl and the conjugation are correlated. Steps 1–3 of that proof, run with the twirl of Conjecture 20 in place of the exact Haar one, then give $4\|B\|_F^2 + 2\|\Gamma(B)\|_F^2$ over $\dim V_{(2,2)}$—the relevant invariant theory being that of $U(n)$ on Λ^2, with three irreducibles, and not that of the full group $U(\binom{n}{2})$ of the two-particle space.

It remains to size the error. Splitting the Hermitian $B^{\otimes 2}$ into positive and negative parts, Eq. (39) reproduces each pairing to relative accuracy $\varepsilon = O(1/n)$, so the total error is at most $\varepsilon\,\mathrm{tr}\big(\Phi_2^{\Lambda^2,\mathrm{Haar}}[M^{(2)\otimes 2}]\,|B^{\otimes 2}|\big)$. By Step 2 that twirl equals $P_{(2,2)}/D_{(2,2)}$, and $|B \otimes B| = |B| \otimes |B|$, so the ratio of error to signal is at most $\|B\|_1^2/\big(\|B\|_F^2 + \frac{1}{2}\|\Gamma(B)\|_F^2\big)$. The rank of B is $O(k^2)$: the trace part of $\rho_{2,\mathfrak{g}^2}$ drops from the commutator, and ρ_2 is supported on $\Lambda^2(\mathrm{ran}\,\Pi)$ with $\mathrm{rank}\,\Pi = 2k$, so $\mathrm{rank}(B) \le 2\,\mathrm{rank}(\rho_2) \le 2\binom{2k}{2}$. Hence, by $\|B\|_1^2 \le \mathrm{rank}(B)\,\|B\|_F^2$ and the norm evaluations of Claim 2 below—whose proof is independent of the present claim—

$$\frac{\|B\|_1^2}{\|B\|_F^2 + \frac{1}{2}\|\Gamma(B)\|_F^2} \;\le\; \frac{O(k^2)\cdot O(k^2/n)}{\Theta(k^3/n)} \;=\; O(k), \tag{41}$$

pointwise when $V_1 = I$; in the independent-halves ensemble the same bound holds with both numerator and denominator averaged over V_1, using $\mathbb{E}\|B\|_1^2 \le \mathrm{rank}(B)\,\mathbb{E}\|B\|_F^2$, which suffices since the unconditional variance is the V_1-average of the conditional one. The total relative error is therefore $\varepsilon \cdot O(k) = O(k/n)$, which is the relative $O(k/n)$ of (40). □

Claim 2 (Evaluation of the two norms). *Write $\tau_{\tilde{g}} := \|[\tilde{g}, \Pi]\|_F^2$. For the data-encoded input of Section II, $M = n_i n_j$, any trainable gate generator of the butterfly and every $\mathbf{x}$, one has exactly*

$$\|\Gamma(B)\|_F^2 = \tfrac{(k-1)^2}{4}\,\tau_{\tilde{g}}, \qquad \|B^{\mathrm{disc}}\|_F^2 = \tfrac{2k-1}{16}\,\tau_{\tilde{g}},$$

and, in either of the two settings of Theorem 19 and for every k with $n - 2k = \Omega(n)$, $\tau_{\tilde{g}} = \Theta\big(k(n-2k)/n^2\big)$, whence $\|\Gamma(B)\|_F^2 = \Theta(k^3/n)$ and $\|B\|_F^2 = O(k^2/n)$ (pointwise when $V_1 = I$, in expectation over V_1 in the independent-halves ensemble).

Proof. The two exact identities are Lemma 44 applied to $h = \tilde{g}$, which uses only that $\tilde{g}$ is Hermitian; in particular $\Gamma(B) = \frac{k-1}{2}[\tilde{g}, \Pi]$ by Lemma 41 and $\Gamma(\rho_2) = (k-1)\rho_1$, exactly as in Step 4 of Theorem 9. For a parameter of the first R_z layer, $\tilde{g} = g$ is coordinate-local, Step 4 of Theorem 9 evaluates $\tau_{\tilde{g}}$, and the connected contribution to $\|B\|_F^2$ is $O(k/n)$ by Lemma 39. For an independent Haar prefix, Lemma 43 gives $\mathbb{E}[\tau_{\tilde{g}}] = \Theta\big(k(n-2k)/n^2\big)$ with the same constants and Lemma 42 gives $\mathbb{E}\|B^{\mathrm{conn}}\|_F^2 = O(k/n)$; the cross term is $O(k^{3/2}/n)$ by Cauchy–Schwarz. Note that the pointwise evaluation of $\tau_{\tilde{g}}$ for a shallow butterfly prefix does *not* follow from either computation, which is why Theorem 19 is stated in the two settings above. □

Proof of Theorem 19. Claims 1 and 2 combine to give

$$\mathrm{Var}\left[\frac{\partial \mathcal{L}}{\partial \theta}\right] = \frac{\Theta(k^3/n) + O(k^2/n)}{n^2(n^2-1)}\big(1 + O(k/n)\big).$$

Writing Ck/n for the relative error of Claim 1 and setting $\gamma_0 := \min\{1/(2C),\, 1/4\}$, for every $k \le \gamma_0 n$ the last factor lies in $[\frac{1}{2}, \frac{3}{2}]$ and $n - 2k \ge n/2$, so the right-hand side is $\Theta(k^3/n^5)$. This grows polynomially in k. □

D. Summary and Comparison

Both architectures rule out exponential barren plateaus for two-body correlator readouts, and both achieve the same $\Theta(k^3/n^5)$ gradient variance rate

throughout $n - 2k = \Omega(n)$ — polynomial in n and cubic in the particle number k. They differ in what is provable unconditionally:

- **Brick-wall**: $\Theta(k^3/n^5)$ unconditionally and exactly, with no conjecture, under Haar initialization via the Givens decomposition. The closed form is proved for the invariant directional derivative at a Haar point (Theorem 9); it specializes to the n phases of the front R_z layer (Corollary 11) and, in the independent-halves ensemble, to every gate of the circuit (Theorem 13). Whether the same statement holds pointwise for an interior gate of the literal single mesh is open; Proposition 14 settles it at both endpoints of the prefix, and no claim above depends on it.

- **Butterfly**: $\Omega(k^4/n^6)$ unconditionally; $\Theta(k^3/n^5)$ conditional on Conjecture 20. Under uniform initialization.

The structure of the rate is transparent from the closed form (22). Two Frobenius norms enter, both divided by $\dim V_{(2,2)} = \Theta(n^4)$: the two-body norm $\|B\|_F^2$ of the commutator of the gate generator with the encoded 2-RDM, and the one-body norm $\|\Gamma(B)\|_F^2$ of its one-particle contraction. The one-body term is the larger of the two and is what produces the cubic dependence on k: by Lemma 41 it equals $(k-1)^2\|[g,\rho_1]\|_F^2$, and the encoded 1-RDM is $\frac{1}{2}$ times a rank-$2k$ projector, so $\|[g,\rho_1]\|_F^2 = \Theta(k(n-2k)/n^2)$. Both terms are proportional to the supply of idle modes $n - 2k$: it is precisely the failure of ρ_1 to be proportional to I_n (Lemma 2 equalizes only its diagonal) that carries the trainable signal, and at a constant filling fraction $k = cn$ with $c < 1/2$ the rate reaches its maximum $\Theta(1/n^2)$. At the fully-packed point $k = n/2$ the idle modes are exhausted, $\rho_1 = \frac{1}{2}I_n$ exactly, and (22) reduces to the connected-cumulant contribution alone, for which we prove only the upper bound $O(1/n^4)$; the sharp rate is therefore claimed throughout $n - 2k = \Omega(n)$ and not at exactly half filling. The contrast with the orthogonal architectures remains sharp: for both the orthogonal brick-wall and the orthogonal butterfly (DLA $\mathfrak{so}(n)$), the gradient variance of $n_i n_j$ scales as $\Theta(1/\binom{n}{k})$ [49]. The R_z gates that lift the DLA from $\mathfrak{so}(n)$ to $\mathfrak{u}(n)$ convert this catastrophic decay into the polynomial $\Theta(k^3/n^5)$—an overwhelming improvement in the trainable signal at every k that grows with n; they simultaneously make one-body readouts classically tractable while making two-body readouts the locus of trainable signal (Proposition 8).

Two consequences are worth stating explicitly. First, trainability is never the binding constraint in this framework: the gradient variance is polynomial in n at every particle number with $n - 2k = \Omega(n)$, from $k = O(1)$ through the operating point of Eq. (7) to extensive filling $k = cn$, $c < 1/2$, where it is $\Theta(1/n^2)$. The ladder of Section II D therefore concerns the training *cost* and the *type* of hardness statement available, not whether the model trains. Second, at extensive particle number $k = cn$ with $c < 1/2$ the architecture is simultaneously trainable, with the largest gradient variance it attains anywhere, and classically hard under the best-known-algorithm calibration of Section II D; worst-case #P-hardness of individual output probabilities is available at polynomial particle number $k = n^\epsilon$ (Theorem 34), and the full average-case Fermion Sampling machinery at exactly half filling (Theorem 37), where the sharp rate degrades to the connected-cumulant upper bound $O(1/n^4)$. This is precisely the combination that the tension between small dynamical Lie algebras and classical simulability [38] would seem to preclude, and that the orthogonal architectures do not achieve at any k that grows with n.

V. TRAINING ALGORITHM: MULTI-LAYER PARALLEL PARAMETER-SHIFT RULE

This section proves the multi-layer parallel parameter-shift rule. The rule applies identically to both the unitary brick-wall and the unitary butterfly: in both architectures, gates within each layer act on disjoint qubit pairs covering all n modes, with commuting generators, and the input state has particle number exactly k. These structural properties are all that the proof requires. The difference between the two architectures appears only in the total training cost: the brick-wall has $O(n)$ columns and $\Theta(n^2)$ parameters, while the butterfly has $\log n$ layers and $\Theta(n \log n)$ parameters. In both cases the parallel PSR achieves a factor-$\Theta(n/k)$ reduction in circuit evaluations over the naive parameter-shift rule:

- **brick-wall**: naive PSR costs $4n^2$ evaluations per gradient step (4 per RBS parameter, 2 per R_z parameter); parallel PSR costs $4kn$ — a factor of n/k reduction.

- **Butterfly**: naive PSR costs $4n \log n$ evaluations per gradient step; parallel PSR costs $4k \log n$ — a factor of n/k reduction.

Two remarks position the rule. First, hardware gradient estimation is the operative training mode for losses that are estimated from model samples—reinforcement-learning returns, adversarial and other likelihood-free objectives—where the per-sample function is an arbitrary diagonal observable and no classical route to the gradient is known short of the sampling task itself (Section VII); it is likewise the mode for training or fine-tuning the physical device against its own noise and calibration. Second, for losses anchored on classically estimable statistics (Section VI B), the same gradients may equally be computed classically; all comparisons in this section are therefore between quantum gradient estimators—the parallel rule against the naive parameter-shift rule—and make no claim against classical gradient computation. The verification regime $k = O(\log n)$ of Section II D already exploits this freedom. The algorithm is identical in both

cases. The key challenge is to estimate all gradients of a given layer simultaneously from a number of circuit evaluations that depends only on k, not on n. Two structural properties make this possible. First, within each layer ℓ, all $n/2$ RBS gates act on disjoint qubit pairs, so their generators commute: $[G_j^{(\ell)}, G_{j'}^{(\ell)}] = 0$ for $j \neq j'$; likewise the R_z gates within a layer act on disjoint single qubits. Second, the input state has particle number *exactly* k, which both limits the total degree of the loss as a trigonometric polynomial in the layer parameters (Lemma 21) and forces its Fourier support to have even ℓ_1 weight (Lemma 22, using that each layer's pairs and phases cover all n modes). Together these allow a random sign vector $\mathbf{s}$ to simultaneously probe all gradient directions in a single layer from only k shifts, with the k-particle Fourier structure ensuring the estimator is exactly unbiased.

A. Setup and Fourier Degree Bound

The loss restricted to layer ℓ is:

$$\mathcal{L}(\boldsymbol{\theta}_\ell, \boldsymbol{\phi}_\ell) = \langle\phi|\, U_\ell^\dagger(\boldsymbol{\theta}_\ell, \boldsymbol{\phi}_\ell)\, \tilde{M}\, U_\ell(\boldsymbol{\theta}_\ell, \boldsymbol{\phi}_\ell)\, |\phi\rangle \tag{42}$$

where $|\phi\rangle = U^{(\ell-1)} \cdots U^{(1)} |\psi(\mathbf{x})\rangle$ is the state entering layer ℓ, and $\tilde{M} = (U^{(K)} \cdots U^{(\ell+1)})^\dagger M\, U^{(K)} \cdots U^{(\ell+1)}$ is the effective observable for the downstream circuit. This formulation is identical for brick-wall and butterfly: $U^{(\ell)}$ is a layer of disjoint RBS-R_z pairs regardless of the architecture, and $K = O(n)$ for the brick-wall or $K = \log n$ for the butterfly.

Lemma 21 (Fourier degree bound)**.** *For a k-particle input state $|\phi\rangle$, the loss $\mathcal{L}(\boldsymbol{\theta}_\ell, \boldsymbol{\phi}_\ell)$ as a function of the RBS parameters $\boldsymbol{\theta}_\ell$ (with $\boldsymbol{\phi}_\ell$ held fixed) is a trigonometric polynomial of total degree at most $2k$: all nonzero Fourier coefficients $c_{\boldsymbol{\delta}}$ in*

$$\mathcal{L}(\boldsymbol{\theta}_\ell, \boldsymbol{\phi}_\ell) = \sum_{\boldsymbol{\delta}} c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\boldsymbol{\theta}_\ell}$$

satisfy $\|\boldsymbol{\delta}\|_1 \leq 2k$.

Proof. All gates in the downstream circuit are RBS and R_z, both particle-number preserving; since M is diagonal in the computational basis and therefore particle-number preserving, the effective observable $\tilde{M}$ is also particle-number preserving. For the k-particle input $|\phi\rangle$, the expectation value $\langle\phi|\, U_\ell^\dagger \tilde{M} U_\ell\, |\phi\rangle$ decomposes as a sum of matrix elements $\langle S|\, U_\ell\, |T\rangle$ with $|S| = |T| = k$.

Fix a basis state $|S\rangle$ with $|S| = k$. For each of the $n/2$ disjoint pairs of qubits in layer ℓ (nearest-neighbor for the brick-wall, stride-$2^{\ell-1}$ for the butterfly), the pair is *active* on $|S\rangle$ if exactly one qubit in the pair is occupied; inactive otherwise. Active pairs contribute a factor of $\cos\theta_j$ or $\sin\theta_j$ (each of trigonometric degree 1) to the matrix element; inactive pairs contribute a factor of 1. The number of active pairs is at most k (bounded by the particle number), so $\langle S|\, U_\ell\, |T\rangle$ is a trigonometric polynomial in $\boldsymbol{\theta}_\ell$ of total degree at most k. The same bound applies to $U_\ell^\dagger$, giving a bound of $2k$ on the product. □

The degree bound alone would call for $2k$ shifts. The following parity property of the k-particle Fourier support brings this down to k, and is what fixes the cost of the estimator at $2k$ circuit evaluations per gate family per layer.

Lemma 22 (Fourier parity)**.** *Let $|\phi\rangle$ have particle number* exactly *k and let $\tilde{M}$ be particle-number preserving. Then every frequency vector $\boldsymbol{\delta}$ occurring in the layer-ℓ Fourier expansion of $\mathcal{L}$ satisfies*

$$\|\boldsymbol{\delta}\|_1 \equiv 0 \pmod 2, \tag{43}$$

for the RBS angles $\boldsymbol{\theta}_\ell$ provided the pairs of layer ℓ cover all *n* modes*, and for the phases $\boldsymbol{\phi}_\ell$ provided the R_z layer acts on* all *n* modes*.*

Proof. RBS angles. As in the proof of Lemma 21, the matrix element $\langle S|\, U_\ell\, |T\rangle$ is nonzero only if S and T have the same occupancy on every pair of layer ℓ, and its $\boldsymbol{\theta}_\ell$-support is exactly the set of active pairs, each contributing a frequency ± 1. Writing $a(S,T)$ for the number of active pairs and using that the pairs of layer ℓ cover all n modes—so that every particle sits on exactly one pair—counting particles gives $k = a(S,T) + 2\,\#\{\text{pairs of occupancy 2}\}$, hence $a(S,T) \equiv k \pmod 2$. The loss is a sum of terms $\overline{\langle S'|\, U_\ell\, |T'\rangle}\, \tilde{M}_{S'S}\, \langle S|\, U_\ell\, |T\rangle$, whose frequency vector is the difference of two vectors of ℓ_1 weights $a(S,T)$ and $a(S',T')$; since $|x-y| \equiv x+y \pmod 2$ for integers, $\|\boldsymbol{\delta}\|_1 \equiv a(S,T) + a(S',T') \equiv 2k \equiv 0$.

R_z angles, full-width layer. With R_z gates on all n modes, $D(\boldsymbol{\phi}_\ell)\, |T\rangle = e^{-i\sum_{j\in T}\phi_\ell^{(j)}}\, |T\rangle$ up to a global phase, so the term indexed by (T,T') carries $\delta_j = \mathbb{1}[j \in T'] - \mathbb{1}[j \in T] \in \{0, \pm 1\}$, supported exactly on $T \triangle T'$. Since $|T| = |T'| = k$ we have $|T \setminus T'| = |T' \setminus T|$, so $\|\boldsymbol{\delta}\|_1 = |T \triangle T'|$ is even (and, in addition, $\sum_j \delta_j = 0$). □

Remark 23 (The full-width hypothesis is necessary)**.** *If the R_z layer covers only a proper subset $A \subsetneq [n]$ of the modes—for instance one phase attached to each pair—the frequency support is $(T \triangle T') \cap A$, which carries no parity constraint: taking $T \setminus T' = \{j\}$ with $j \in A$ and $T' \setminus T = \{j'\}$ with $j' \notin A$ gives $\|\boldsymbol{\delta}\|_1 = 1$. The argument below then fails and $2k$ shifts are genuinely required. This is why both architectures of Section III carry a phase layer on all n modes. The RBS half of the argument carries the analogous hypothesis: the layer's pairs must cover all n modes. Every butterfly layer does, and so does every brick-wall column, the even columns owing this to the boundary pair $(n,1)$; a layer that leaves modes uncovered—an even column on an open chain, for instance—admits odd-weight frequencies by the same mechanism (a particle sitting on an uncovered mode in the bra and on a covered pair in the ket), and its RBS gradients require the $2k$-shift rule (Remark 6).*

B. Main Theorem: Parallel PSR for RBS Parameters

Theorem 24 (Multi-layer parallel parameter-shift rule, RBS gradients)**.** *Let the input state have particle number exactly k, let the disjoint pairs of layer ℓ cover all n modes, and let $a_1, \dots, a_k$ and $r_1, \dots, r_k$ solve the linear system:*

$$\sum_{q=1}^{k} a_q \sin(r_q) \cos(r_q)^{2m-1} = \frac{1}{2}, \quad m = 1, \dots, k. \tag{44}$$

Let $\mathbf{s} = (s_1, \dots, s_{n/2}) \sim \mathrm{Uniform}(\{\pm 1\}^{n/2})$ and define the estimator

$$\hat{g}_j(\mathbf{s}) = s_j \sum_{q=1}^{k} a_q \left[\mathcal{L}(\boldsymbol{\theta}_\ell + r_q \mathbf{s}) - \mathcal{L}(\boldsymbol{\theta}_\ell - r_q \mathbf{s}) \right]. \tag{45}$$

Then for all $j \in \{1, \dots, n/2\}$ simultaneously,

$$\mathbb{E}_{\mathbf{s}}[\hat{g}_j(\mathbf{s})] = \frac{\partial \mathcal{L}}{\partial \theta_\ell^{(j)}}, \tag{46}$$

using $2k$ circuit evaluations in total—k shifts, each evaluated at $\pm r_q$—independently of n, for any particle-number-preserving observable M. The covering hypothesis holds for every layer of both architectures: every butterfly layer and every brick-wall column consists of $n/2$ disjoint pairs covering all n modes.

Remark 25 (Closed form and cost of the shift coefficients)**.** *The shifts $r_1, \dots, r_k \in (0, \pi/2)$ are free design parameters; once they are fixed (with distinct $\cos^2 r_q$), the coefficients are determined in closed form by Lagrange interpolation at $u = 1$. Writing $u_q = \cos^2 r_q$ and $c_q = a_q \sin r_q \cos r_q$, the system (44) reads $\sum_q c_q u_q^p = \frac{1}{2}$ for $p = 0, \dots, k-1$—the condition that the nodes $\{u_q\}$ reproduce the first k moments of the target measure $\frac{1}{2}\delta_1$—whose solution is*

$$a_q \;=\; \frac{1}{\sin 2r_q} \prod_{p \neq q} \frac{\sin^2 r_p}{\cos^2 r_q - \cos^2 r_p}, \qquad q = 1, \dots, k. \tag{47}$$

At $k = 1$ this is $a_1 = 1/\sin 2r_1$, and $r_1 = \pi/4$ gives $a_1 = 1$: the familiar $\pm\pi/4$ rule for a single frequency-2 parameter, which is what the parity of Lemma 22 leaves at $k = 1$. We recommend the nodes

$$u_q = \tfrac{1}{2}\Big(1 + \cos \tfrac{(2q-1)\pi}{2k}\Big), \; r_q = \arccos \sqrt{u_q}, \; q = 1, \dots, k, \tag{48}$$

i.e. Chebyshev–Gauss points in the variable $u = \cos^2 r$, for which the interpolation underlying (47) is well-conditioned and the weights are ℓ_1-optimal (Remark 28). Moreover k shifts are necessary*: exactness on odd polynomials of degree at most $2k-1$ is equivalent, in the variable u, to $\sum_q c_q Q(u_q) = \frac{1}{2}Q(1)$ for every Q of degree at most $k-1$, and with only $N \leq k-1$ nodes the choice $Q(u) = \prod_{q \leq N}(u - u_q)$ forces $Q(1) = 0$, i.e. $r_q = 0$ for some q, which is not a shift.*

The pair $\{a_q, r_q\}$ depends only on k: it is independent of the circuit parameters, the data, the layer index, the architecture and the system size n. The coefficients are therefore computed once*, before training, in $O(k^2)$ classical time, cached, and reused unchanged for every layer, every gradient step and every data point; this preprocessing is negligible against the per-step quantum cost.*

Proof. The proof proceeds in five steps.

Step 1: Reduction to the active layer. Freezing all layers except ℓ reduces $\mathcal{L}$ to a function of $\boldsymbol{\theta}_\ell$ alone, with the effective observable $\tilde{M}$ absorbing the downstream circuit. Since all downstream gates are RBS and R_z (both particle-number preserving) and M is a computational basis measurement (particle-number preserving), $\tilde{M}$ is particle-number preserving. No assumption on M beyond particle-number preservation is needed. This reduction applies identically to both architectures.

Step 2: Fourier structure of the loss. By Lemma 21, $\mathcal{L}(\boldsymbol{\theta}_\ell)$ is a trigonometric polynomial of total degree at most $2k$:

$$\mathcal{L}(\boldsymbol{\theta}_\ell) = \sum_{\boldsymbol{\delta}:\, \|\boldsymbol{\delta}\|_1 \leq 2k} c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\boldsymbol{\theta}_\ell}, \tag{49}$$

where each $\delta_j \in \{-2, -1, 0, +1, +2\}$ (the pairwise-difference frequencies from the eigenvalues $\{-1, 0, 0, +1\}$ of $G_j^{(\ell)}$) and at most $2k$ coordinates are nonzero simultaneously. By Lemma 22—whose covering hypothesis holds here by assumption—$\|\boldsymbol{\delta}\|_1$ is moreover *even* for every frequency vector occurring in this sum.

Step 3: Gradient in Fourier form.

$$\frac{\partial \mathcal{L}}{\partial \theta_\ell^{(j)}} = i \sum_{\boldsymbol{\delta}:\, \|\boldsymbol{\delta}\|_1 \leq 2k} \delta_j\, c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\boldsymbol{\theta}_\ell}. \tag{50}$$

Step 4: The multi-shift formula handles all cross-terms. We show that for every frequency vector $\boldsymbol{\delta}$ with $\|\boldsymbol{\delta}\|_1 \leq 2k$ even and $\delta_j \neq 0$,

$$2 \sum_{q=1}^{k} a_q \sin(\delta_j r_q) \prod_{j'' \neq j} \cos(\delta_{j''} r_q) = \delta_j. \tag{51}$$

To see why this is what we need, observe that for $s_j \in \{\pm 1\}$ uniform,

$$\mathbb{E}[s_j\, e^{i\alpha s_j}] = i \sin\alpha, \qquad \mathbb{E}[e^{i\alpha s_j}] = \cos\alpha, \tag{52}$$

so by independence of the $s_{j''}$,

$$\mathbb{E}_{\mathbf{s}}[s_j \sin(r_q \boldsymbol{\delta}\cdot\mathbf{s})] = \sin(r_q\delta_j) \prod_{j'' \neq j} \cos(r_q \delta_{j''}). \tag{53}$$

Condition (51) is therefore exactly what is required for the estimator to be unbiased.

To verify (51), we use the Chebyshev polynomial representations $\sin(mr) = \sin(r)\, U_{m-1}(\cos r)$ and $\cos(mr) =$

$T_m(\cos r)$, where U_{m-1} (second kind) and T_m (first kind) satisfy $U_{m-1}(1) = m$ and $T_m(1) = 1$. Writing

$$R(x) = U_{|\delta_j|-1}(x) \prod_{j'' \neq j} T_{|\delta_{j''}|}(x), \tag{54}$$

the left-hand side of (51) becomes

$$\operatorname{sign}(\delta_j) \cdot 2 \sum_{q=1}^{k} a_q \sin(r_q) R(\cos r_q). \tag{55}$$

The polynomial R has degree $(|\delta_j| - 1) + \sum_{j'' \neq j} |\delta_{j''}| = \|\boldsymbol{\delta}\|_1 - 1 \leq 2k - 1$. Crucially, T_m has parity $(-1)^m$ and U_{m-1} has parity $(-1)^{m-1}$, so R has parity $(-1)^{\|\boldsymbol{\delta}\|_1 - 1} = -1$ by Lemma 22: R is an *odd* polynomial of degree at most $2k - 1$, hence lies in the k-dimensional span $\{x, x^3, \ldots, x^{2k-1}\}$ rather than in the $2k$-dimensional space of all polynomials of that degree.

The linear system (44) is precisely the statement that the functional $P \mapsto \sum_q a_q \sin(r_q) P(\cos r_q)$ equals $P(1)/2$ on all *odd* polynomials of degree at most $2k - 1$: it imposes this on the monomials x^{2m-1}, $m = 1, \ldots, k$, and the general case follows by linearity. These are k conditions in the k unknowns a_q, and the system is nonsingular whenever the $\cos^2 r_q$ are distinct and nonzero, its matrix being $\operatorname{diag}(\cos r_q)$ times a Vandermonde matrix in $\cos^2 r_q$. Applying this quadrature to R gives

$$\operatorname{sign}(\delta_j) \cdot 2 \cdot \tfrac{R(1)}{2} = \operatorname{sign}(\delta_j) \cdot U_{|\delta_j|-1}(1) \prod_{j'' \neq j} T_{|\delta_{j''}|}(1) \tag{56}$$

$$= \operatorname{sign}(\delta_j) \cdot |\delta_j| \cdot 1 = \delta_j. \tag{57}$$

Step 5: Unbiasedness. Combining Steps 2–4:

$$\mathbb{E}_{\mathbf{s}}[\hat{g}_j(\mathbf{s})] = 2i \sum_{\boldsymbol{\delta}} c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\boldsymbol{\theta}_\ell} \sum_{q=1}^{k} a_q\, \mathbb{E}_{\mathbf{s}}[s_j \sin(r_q \boldsymbol{\delta}\cdot\mathbf{s})]$$

$$= 2i \sum_{\boldsymbol{\delta}} c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\boldsymbol{\theta}_\ell} \cdot \frac{\delta_j}{2} \tag{58}$$

$$= i \sum_{\boldsymbol{\delta}} \delta_j\, c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\boldsymbol{\theta}_\ell} = \frac{\partial \mathcal{L}}{\partial \theta_\ell^{(j)}}. \tag{59}$$

The total circuit cost is $2k$: two evaluations (shifted $\pm$) per shift r_q, for k shifts, independently of n and of the architecture. □

C. Parallel PSR for R_z Parameters

The R_z gate generators have spectrum $\{-\frac{1}{2}, +\frac{1}{2}\}$, so the Fourier degree of the loss in each $\phi_\ell^{(j)}$ individually is only 1, with per-gate difference frequencies $\delta_j \in \{-1, 0, +1\}$. The joint degree is nevertheless as large as in the RBS case: in the interference terms $\langle S| \cdots |S'\rangle$ of the loss the bra and ket Fock components can occupy *disjoint* sets of R_z-active qubits, so the frequency vector satisfies only $\|\boldsymbol{\delta}\|_1 \leq 2k$ (each of bra and ket contributes up to k occupied R_z qubits), and $\mathcal{L}(\boldsymbol{\phi}_\ell + r \cdot \mathbf{s}^\phi)$ has degree up to $2k$ in the scalar shift r. What rescues the shift count is therefore not the degree but the parity: when the phase layer covers all n modes, Lemma 22 forces $\|\boldsymbol{\delta}\|_1$ to be even, the polynomial R of Step 4 is odd, and k shifts suffice—exactly as for the RBS parameters. The full-width phase layers of Section III are chosen for exactly this reason; a partial layer breaks the argument (Remark 23). The R_z rule is then a direct corollary of the machinery of Theorem 24.

Theorem 26 (Multi-layer parallel parameter-shift rule, R_z gradients)**.** *Let the input state have particle number exactly k, let the R_z layer act on all n modes, and let $a_1, \ldots, a_k$ and $r_1, \ldots, r_k$ solve the linear system* (44)*. Let $\mathbf{s}^\phi \sim \text{Uniform}(\{\pm 1\}^n)$ and define*

$$\hat{h}_j(\mathbf{s}^\phi) = s_j^\phi \sum_{q=1}^{k} a_q \left[\mathcal{L}(\boldsymbol{\phi}_\ell + r_q \mathbf{s}^\phi) - \mathcal{L}(\boldsymbol{\phi}_\ell - r_q \mathbf{s}^\phi)\right]. \tag{60}$$

Then for all $j \in \{1, \ldots, n\}$ simultaneously,

$$\mathbb{E}_{\mathbf{s}^\phi}\left[\hat{h}_j(\mathbf{s}^\phi)\right] = \frac{\partial \mathcal{L}}{\partial \phi_\ell^{(j)}}, \tag{61}$$

using $2k$ circuit evaluations in total, independently of n and of the architecture.

The proof is identical to that of Theorem 24: the frequencies satisfy $\delta_j \in \{-1, 0, +1\}$ and, by Lemma 22, $\|\boldsymbol{\delta}\|_1 = 2m$ is even with $m \leq k$; the polynomial $R(x)$ of Step 4 is therefore the odd monomial $U_0(x) \prod_{j''} T_1(x) = x^{2m-1}$, and the quadrature (44) reproduces $P(1)/2$ on all odd polynomials of degree at most $2k - 1$. The same nodes and weights $\{a_q, r_q\}$ as for the RBS gradients are reused unchanged.

D. Total Training Cost

Corollary 27 (Full gradient cost and factor-n/k reduction)**.** *Applying Theorem 24 to each layer with independent sign vectors and Theorem 26 analogously, the full gradient of the loss with respect to all trainable parameters is estimated with the following costs:*

- ***Butterfly*** *($K = \log n$ layers, $T = \frac{3}{2} n \log n$ parameters): $2k \log n$ circuit evaluations for all RBS gradients and $2k \log n$ for all R_z gradients, giving $4k \log n$ total. The naive parameter-shift rule requires $4n \log n$ evaluations (4 per RBS parameter and 2 per R_z parameter); the parallel PSR achieves a reduction by a factor of n/k.*
- ***Brick-wall*** *($K = n$ columns, each a full-width R_z layer followed by that column's $n/2$ RBS gates, $T = 3n^2/2$ parameters): $2kn$ circuit evaluations for all*

RBS gradients and $2kn$ for all R_z gradients, giving $4kn$ total. The naive parameter-shift rule requires $4 \cdot n^2/2 + 2 \cdot n^2 = 4n^2$ evaluations; the parallel PSR achieves a reduction by a factor of exactly n/k, the same as the butterfly's.

In both cases the reduction factor is exactly n/k: $\Theta(n)$ at $k = O(1)$, growing linearly in n at any fixed k. At the operating point of Eq. (7), the butterfly requires $240 \log n$ circuit evaluations per gradient step—nearly independent of n—while the brick-wall requires $4kn = O(kn)$.

Per layer the naive rule costs $4 \cdot \frac{n}{2} + 2 \cdot n = 4n$ evaluations against the parallel rule's $2k + 2k = 4k$, so the parallel rule is cheaper precisely when $k < n$. Since the encoding requires $2k \leq n$, the parallel rule is never worse than the naive one anywhere in the admissible range. (Taking the estimator layer-wise as $\min(\text{naive}, \text{parallel})$ makes this monotonicity hold by definition, at no cost.)

For $n = 1024$: at the operating point $k = 60$, the butterfly requires 2,400 circuit evaluations per gradient step (vs. 40,960 for the naive PSR, a 17.1× reduction) and the brick-wall requires 245,760 (vs. 4,194,304, also 17.1×). At $n = 4096$ the reduction is 68.3×: the advantage of the parallel PSR grows linearly with n at fixed hardness level. The naive baseline here is computed for the *same* circuit. Since the parallel cost is independent of the parameter count while the naive cost is not, the ratio can be inflated by adding parameters, whereas the absolute per-step cost—$4k \log n$ and $4kn$—can only rise; it is the absolute cost that we report throughout. The resource counted is the number of distinct circuit configurations per gradient step: shots of a fixed configuration are repetitions of a single compiled and uploaded circuit, whereas each configuration must be compiled, uploaded and dispatched separately, so it is the configuration count, rather than the shot budget per configuration, that limits on-hardware training at scale (Remark 28).

Remark 28 (Configurations versus shots)**.** *At the nodes (48) the weights have the closed form $a_q = (-1)^{q-1} / \big(2k \sin^2(\theta_q/2)\big)$ with $\theta_q = (2q-1)\pi/(2k)$, whence $\sum_q |a_q| = k$ and $\sum_q a_q^2 = \frac{1}{3}(2k^2+1)$. The second sum sets the price of parallelism: with S shots per configuration and a loss evaluation of variance σ^2/S, the parallel estimator carries shot noise $\frac{4}{3}k^2\sigma^2/S\,(1+o(1))$ against $O(\sigma^2/S)$ for the naive rule. At a fixed total shot budget per gradient step this is largely bought back: the parallel rule spreads the budget over $\sim k/n$ as many configurations, so each receives $\sim n/k$ times more shots, leaving a net variance ratio $\Theta(k^3/n)$. Since the estimator is exactly unbiased at every shot count (Theorem 24), the residual noise enters stochastic optimization as variance rather than bias, alongside the mini-batch noise already present. Trading configurations for shots at comparable order is therefore favourable on hardware, where shots of a fixed configuration are repetitions of a single compiled and uploaded circuit, dispatched in one job, while every configuration must be compiled, uploaded and dispatched separately.*

E. Layer-wise Block Coordinate Descent

The multi-layer parallel PSR enables two optimization strategies for both architectures.

a. Standard gradient descent. Use Corollary 27 to compute all gradients in $4kK$ circuit evaluations ($K = \log n$ for butterfly, $K = n$ for brick-wall), then update all parameters simultaneously.

b. Layer-wise block coordinate descent. Alternatively, update layers one at a time in forward-backward sweeps, a strategy analogous to the sweep structure of DMRG in quantum many-body physics [77] and a special case of block coordinate descent [78]:

$$\begin{aligned} &\text{for } \ell \;=\; 1, \ldots, K, K-1, \ldots, 1: \\ &\quad \boldsymbol{\theta}_\ell \;\leftarrow\; \boldsymbol{\theta}_\ell - \eta\, \hat{\nabla}_{\boldsymbol{\theta}_\ell} \mathcal{L}, \quad \boldsymbol{\phi}_\ell \leftarrow \boldsymbol{\phi}_\ell - \eta\, \hat{\nabla}_{\boldsymbol{\phi}_\ell} \mathcal{L}, \end{aligned}$$

using $4k$ circuit evaluations per layer update ($2k$ for RBS, $2k$ for R_z), independently of the architecture. The cost per full forward-backward sweep is $4k \cdot 2K$: for the butterfly this is $O(k \log n)$, and for the brick-wall this is $O(kn)$—identical in order to full gradient descent in both cases. The potential advantage is that each layer's subproblem has reduced effective dimension: when training layer ℓ alone, the parameter space has dimension n (one RBS and one R_z parameter per pair) rather than T. Whether this translates to faster convergence in practice depends on the loss landscape and is an empirical question we leave for future work.

VI. THE QNN AS AN END-TO-END ML PIPELINE

We have built three ingredients: a magic-state encoder (Section II), unitary brick-wall and butterfly processors (Section III) that are provably trainable (Section IV) with efficient training cost (Section V). In this section we assemble these into complete machine-learning pipelines by specifying the readout (how the quantum state is turned into a classical output) and the loss function, and we discuss which tasks each pipeline variant is suited to. The two architectures share the same encoding, readout families, and task structure; they differ only in circuit depth, parameter count, and training cost. The classical simulation cost of each variant is analyzed in Section VII.

A. Pipeline Stages

The full inference pipeline consists of five stages, summarized in Table I. The first four stages build the quantum state $|\psi(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\phi})\rangle$ that encodes the classical input $\mathbf{x}$ under the trained parameters $(\boldsymbol{\theta}, \boldsymbol{\phi})$; the fifth stage extracts a classical output from this state via measurement.

All stages share the same two gate primitives—RBS and R_z—both particle-number preserving. Particle number k is therefore preserved exactly through the full circuit in both architectures, and the overall circuit $V =$

TABLE I. The five stages of the unitary brick-wall and butterfly QNN pipelines. All magic blocks are prepared in parallel in Stage 1, so the preparation depth does not scale with k. The total circuit depth is $3\log n + 6$ for the butterfly, and $3n + 5$ for the brick-wall.

#	Stage	Butterfly	Brick-wall
1	Magic state preparation	4	4
2	Fixed Hadamard spreading U^{had}	$\log n$	$n-1$
3	R_z data encoding $D(\boldsymbol{\varphi}(\mathbf{x}))$	1	1
4	Trainable circuit $U(\boldsymbol{\theta}, \boldsymbol{\phi})$	$2\log n$	$2n$
5	Measurement + readout	1	1

$U \circ D \circ U^{\mathrm{had}}$ is a passive FLO unitary acting on the non-Gaussian magic state $|\Psi_{\mathrm{in}}\rangle$. This structural coherence is what enables simultaneously:

- the multi-layer parallel parameter-shift rule (Lemma 21 requires particle-number preservation),
- the no-barren-plateau results: for the brick-wall, the closed-form gradient variance (22) under Haar initialization, evaluating to $\Theta(k^3/n^5)$ (Theorem 9); for the butterfly, unconditional $\Omega(k^4/n^6)$ and conditional $\Theta(k^3/n^5)$ under Conjecture 20 (Theorems 17 and 19); one-body readouts classically tractable for both (Proposition 8),
- the classical hardness results of Section VII, which require non-Gaussian input to a passive FLO circuit.

B. The Sampler as the Universal Readout

Stage 5 is the same for every task in this paper: the quantum device performs computational-basis measurements and returns m shots, each a bit string of Hamming weight k (by particle-number preservation). Every pipeline consumes these samples and nothing else; the QNN is always used as a sampler of $p(\mathbf{x}' \mid \mathbf{x}) = |\langle \mathbf{x}'| V |\Psi_{\mathrm{in}}\rangle|^2$, and the classically hard object of Section VII—the sampling task itself—therefore sits at the base of every task family. Tasks differ only in the loss applied to the samples and in the classical head that processes them; both apply identically to the brick-wall and the butterfly.

The ways of consuming the samples form a hierarchy of increasing generality. *Distributional losses* compare the multi-set of sampled strings directly to data through a divergence such as maximum mean discrepancy or KL, forming no summary statistics. *Few-body sample statistics* reduce the shots to a vector of empirical averages of observables of fixed body number, for example the two-body correlator vector $\mathbf{q} = (\langle n_i n_j \rangle)_{i<j}$, obtained by counting the fraction of shots in which modes i and j are both occupied; these are the simplest permutation-invariant functions of the sample set and the family for which trainability is proven (Section IV). *General set-valued heads* apply an arbitrary permutation-invariant classical network to the shots—a simple linear head on $\mathbf{q}$, a DeepSets aggregator or a set transformer whose tokens are the sampled strings—and are sensitive to sample structure beyond few-body statistics.

Two objects must be kept apart throughout what follows: the *loss*, which is a scalar functional used to steer the parameters during training, and the *deployed output*, which is what the trained model emits at inference. The hardness results of Section VII attach to the second, at the level of the sampling task. The two can therefore have different complexities, and in the pipelines below they generally do: by Proposition 8 every statistic of fixed body number is classically computable, so a model whose deployed output is a function of $\mathbf{q}$ alone is classically simulable end to end, whereas a model whose deployed output is the sample set itself is not—knowledge of $\mathbf{q}$ does not permit sampling, since exponentially many distributions on Hamming-weight-k strings share the same two-body moments.

C. Where the Computation Happens

Sections IV and V established a gradient rate and a gradient cost for the two-body correlator. Which pipelines those results actually cover is settled by two independent questions. Does a given loss inherit the variance guarantee? That is decided by the body number of the effective observable it induces. Can its gradient be computed classically? That is decided by whether the expectation defining it is taken over a distribution we have a model of. Keeping the two apart organizes the design space; the concrete task families are catalogued in Section VI D.

One preliminary applies to every case below. For a per-sample head h the model output is the expectation $\mathbb{E}_p[h]$ of the diagonal observable h; diagonal observables commute with the particle number, and the multi-layer parallel parameter-shift rule (Theorem 24) requires nothing of the observable beyond particle-number preservation. The gradient *cost* on hardware is therefore the same for every head: $4kn$ circuit evaluations per step for the brick-wall, $4k\log n$ for the butterfly. What varies is the gradient *variance*, and it is controlled by the body number of the effective observable that the loss induces.

a. Losses anchored on few-body statistics. The canonical distributional loss for an implicit generative model is moment matching: the model's two-body moments $q_{ij} = \langle n_i n_j \rangle$ are compared to the data's, and the squared discrepancy is minimized. Both of our questions have favourable answers for such a loss. By the chain rule its gradient is that of a single two-body effective observable, so the rate of Section IV transfers verbatim—provided the induced observable is not degenerate, which would make the guarantee vacuous. It is not, and degeneracy occurs only at convergence.

Proposition 29 (Moment-matching MMD induces a nondegenerate two-body head)**.** *Let $q_{ij}(W) = \langle n_i n_j \rangle$ be the model's two-body moments, μ_{ij} the corresponding data moments, both on Hamming-weight-k strings, and $\mathcal{L}_{\mathrm{MMD}} = \sum_{i<j}(q_{ij} - \mu_{ij})^2$. Then at every parameter point*

$$\partial_\theta \mathcal{L}_{\mathrm{MMD}} = \partial_\theta \langle M_{\mathrm{eff}} \rangle, \qquad M_{\mathrm{eff}} = 2 \sum_{i<j} \big(q_{ij} - \mu_{ij}\big)\, n_i n_j, \tag{62}$$

a two-body, particle-number-preserving observable, whose traceless part satisfies $\tilde{M}_{\mathrm{eff}} = 0$ if and only if $q = \mu$.

Proof. The identity is the chain rule together with linearity of each q_{ij} in the observable $n_i n_j$; the residual is evaluated at the current parameter point, and M_{eff} is a weighted two-body correlator, diagonal in the Fock basis and hence particle-number preserving, so Theorem 24 and Proposition 12 apply to it verbatim. For the degeneracy claim: $\tilde{M}_w = 0$ on $\Lambda^2\mathbb{C}^n$ if and only if all weights w_{ij} are equal. Since both model and data strings carry exactly k particles, $\sum_{i<j} q_{ij} = \sum_{i<j} \mu_{ij} = \binom{k}{2}$, so the residual sums to zero; equal residuals are therefore zero residuals, and $\tilde{M}_{\mathrm{eff}} = 0$ forces $q = \mu$. □

In applying the rate, two objects must be kept apart: the weights of M_{eff} depend on the current parameter point through the residual, whereas the variance statements of Section IV are for a *fixed* observable averaged over the Haar ensemble. The trainability statement for the MMD loss is therefore conditional on the point. Fix any parameter point, write $r := q - \mu$ for its residual, and let $\mathcal{L}_r$ denote the fixed-head loss with observable $M_r = 2\sum_{i<j} r_{ij}\, n_i n_j$; by Proposition 29, the gradient of $\mathcal{L}_{\mathrm{MMD}}$ at that point equals the gradient of $\mathcal{L}_r$ at that point. For the fixed head M_r, whenever $r \neq 0$—equivalently $\tilde{M}_r \neq 0$—Proposition 12(iii) gives, for $k \geq 2$ with $n - 2k = \Omega(n)$, $\mathrm{Var}[D_g \mathcal{L}_r] = \Omega\big(\|\tilde{M}_r\|_F^2\, k^2/n^5\big)$ over the Haar ensemble. The MMD landscape therefore carries no exponential barren plateau in the following precise sense: away from the matched point $q = \mu$, its gradient coincides pointwise with that of a two-body fixed-head loss whose ensemble gradient variance is at most polynomially small.

The class of losses covered is neither artificial nor exhaustive. Moment-matching objectives of this kind are the standard first choice for implicit generative models, and in the multivariate setting they amount to matching conditional pairwise co-occurrence structure—the quantity that downstream portfolio and risk applications consume. What falls outside the class is equally concrete: kernels whose Mercer features run to unbounded order, such as the Gaussian kernel commonly used with MMD, adversarial objectives evaluated by a trained discriminator, and generic reinforcement-learning returns. These are the subject of the next two paragraphs.

Turning to the second question: since the encoding and the circuit are known, the required statistics are obtained by propagating the corresponding reduced density matrix (Proposition 8), so gradients may be computed classically and exactly rather than estimated from shots. Training of this kind needs no quantum device at all; the device is required only at inference, to draw samples from the trained distribution. Whether classical propagation or hardware estimation is faster in practice is an engineering comparison—the former costs poly(n) per gradient evaluation with a sizeable exponent, the latter $O(kn)$ or $O(k \log n)$ shot-limited circuit evaluations—and not a complexity statement.

b. Losses estimated from model samples. When the expectation defining the loss is taken over something we do not have a model of, no propagation is available and the gradient must be estimated by drawing samples; the device then enters the training loop. Two distinct situations fall under this heading, and they differ in whether a variance guarantee survives.

The first is an external environment. In reinforcement learning the return depends on how the environment responds to the sampled action, and for the environments that motivate model-free methods in the first place no sufficiently accurate model is available to propagate through. Sampling from the policy is then unavoidable. The variance guarantee, however, is not lost: it requires the effective observable to be structurally of low body number, not to have known coefficients. A return of the form $R(s,a) = \sum_i r_i(s)\, a_i + \sum_{i<j} r_{ij}(s)\, a_i a_j$—the shape of a portfolio payoff with linear rewards and quadratic costs, and of shaped rewards generally—induces a two-body effective observable whose weights are fixed by the environment and unknown to us, which is exactly the hypothesis under which Proposition 12(iii) applies; the statement holds conditionally on each environment state s, which fixes the weights; the state-averaged loss is by linearity again a two-body head, with weights $\mathbb{E}_s[r_{ij}(s)]$, to which the same bound applies. Such pipelines are trained on hardware and carry the $\Theta(k^3/n^5)$ guarantee.

The second is a high-order functional of the model's own samples: a deep discriminator applied to the shots, an indicator of a rare configuration, or a sparse terminal return that does not decompose into few-body terms. Here the effective observable has body number growing with k, the analysis of Section IV does not apply, and gradients may concentrate; nothing in this paper rules that out. These pipelines require the device and carry no variance guarantee.

c. Composition. The two regimes compose. Writing $\mathcal{L} = \mathcal{L}_{\mathbf{q}} + \lambda\, \mathcal{L}_{\mathrm{sample}}$ with an anchored first term guarantees that the total gradient retains a non-vanishing contribution regardless of how flat the second term is, and separating the two contributions during training diagnoses whether the sample-based term carries signal above shot noise. In summary, gradients are provably non-vanishing whenever the effective observable is of low body number, which covers returns and heads of the usual form; such gradients are estimated on the device when no model of the distribution is available, as in model-free reinforcement learning, and can also be computed classically with

polynomial overhead when one is. Supervised pipelines fall on either side of this division according to the head: one reading only $\mathbf{q}$ is classical end to end, while a set-valued head consuming the samples themselves is not (Section VI D).

D. Machine Learning Task Families

All task families use the magic-block encoding of Section II at the operating point of Eq. (7) ($k = 60$, paired blocks, Section II A) and consume the same sampler; they differ only in loss and head, and each is placed in the taxonomy of Section VI C by that choice.

a. Generative modeling (distributional loss). Train the QNN as a Born machine: given data samples $\{\mathbf{x}^{(1)}_{\text{data}}, \ldots, \mathbf{x}^{(N)}_{\text{data}}\}$, minimize a sample-based distributional loss such as maximum mean discrepancy (MMD) or KL divergence against the empirical data distribution. The QNN's output distribution $p(\cdot \mid \mathbf{x})$ itself is the model. Related architectures have been explored in quantum Born machines [79] and quantum GANs [80].

b. Reinforcement learning policies (distributional loss). Use $p(\cdot \mid \mathbf{x})$ as a policy: given an environment state $\mathbf{x}$, the QNN outputs a distribution over discrete actions. Train via policy gradient. Policy-gradient training consumes model samples inside the loop: the return is an arbitrary function of the sampled string, so the gradient is estimated on hardware via the parallel parameter-shift rule, and this task family is quantum at training time as well as at inference. The efficient training cost of the parallel PSR combines naturally with episodic policy updates for both architectures.

c. Supervised classification (sample-set head). For a classification task with C classes, define logits

$$y_c(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\phi}, W, b) = \sum_{i<j} W_{c,ij} \langle n_i n_j \rangle(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\phi}) + b_c, \; c \in [C], \tag{63}$$

where $W \in \mathbb{R}^{C \times \binom{n}{2}}$ and $b \in \mathbb{R}^C$ are classical parameters, trained alongside the quantum parameters $(\boldsymbol{\theta}, \boldsymbol{\phi})$. Apply softmax to obtain class probabilities, and train against labels with cross-entropy. This is a hybrid architecture: the quantum layer supplies the two-body correlator vector and a classical head maps it to predictions. In practice one can restrict to a subset of n two-body operators (e.g., nearest-neighbor pairs together with one long-range pair) to obtain an $n \to n$ head. The linear head may equally be replaced by any permutation-invariant model consuming the shots directly, with the consequences set out in Section VI C. One point is specific to the head: it changes the gradient *rate* and not merely its constant, entering through the two scalars c_1 and c_2 of Proposition 12, which can differ in order and in sign—$\Theta(k^3/n^5)$ for the single-pair readout against $\Theta(k^2/n^4)$ for the uniform nearest-neighbour head—while the absence of an exponential barren plateau is head-independent (Proposition 12(iii)).

d. Supervised regression (sample-set head). For scalar regression, predict $\hat{y}(\mathbf{x}) = \sum_{i<j} w_{ij} \langle n_i n_j \rangle + b$ and train with mean-squared error. Vector-valued regression follows the classification template.

e. Kernel and similarity tasks (overlap readout). The encoding $\mathbf{x} \mapsto |\psi(\mathbf{x})\rangle$ defines an implicit quantum kernel $K(\mathbf{x}, \mathbf{x}') = |\langle \psi(\mathbf{x}) | \psi(\mathbf{x}') \rangle|^2$. Overlaps can be estimated via SWAP tests or destructive SWAP measurements [81], enabling kernel-based classification, clustering, or nearest-neighbor methods. The magic-block encoding supports this readout directly.

In every case the classical simulation cost of the end-to-end model is governed by what the head consumes: few-body sample statistics are estimable at additive precision, while sample-level structure is accessible to a classical simulator only through the sampling algorithms of Section VII, at best-known cost $2^{\Omega(k)}$.

E. Sample Complexity and Generalization

Although the QNN operates in a $\binom{n}{k}$-dimensional Hilbert space that is superpolynomial in n, it does not require superpolynomially many training samples. The generalization error depends on the number of trainable parameters T, not on the Hilbert space dimension.

Theorem 30 (Generalization bound)**.** *Let $\mathcal{F}$ be the function class implemented by the unitary brick-wall or butterfly QNN with T trainable parameters ($T = 3n^2/2$ for the brick-wall, $T = \frac{3}{2} n \log n$ for the butterfly), with the magic-block encoding of Section II. With m training samples, the generalization error satisfies*

$$\mathbb{E}\left[\sup_{f \in \mathcal{F}} |\mathcal{L}_{\text{test}}(f) - \mathcal{L}_{\text{train}}(f)|\right] \leq O\left(\sqrt{\frac{T \log m}{m}}\right). \tag{64}$$

For ε-generalization it suffices to take $m = \Omega\big(T \log(T/\varepsilon)/\varepsilon^2\big)$ samples.

Proof. We apply Theorem 1 of Caro et al. [82] to the QNN with T trainable parameters.

Per-gate Lipschitz constant. Each parameter θ_i appears in exactly one gate $U_i(\theta_i) = \exp(i\theta_i G_i)$, where G_i is a traceless Hermitian generator with $\|G_i\|_{\text{op}} \leq 1$. For any normalized input state $|\psi\rangle$ and observable O with $\|O\| \leq 1$, the parameter-shift rule gives

$$\left|\frac{\partial \mathcal{L}}{\partial \theta_i}\right| = \left|\langle \psi | \, i[G_i, \tilde{O}] \, |\psi\rangle\right| \leq 2\|G_i\|_{\text{op}} \, \|O\| = O(1),$$

since all circuit outputs are bounded in $[-1, 1]$. Hence the loss has per-gate Lipschitz constant $O(1)$.

Covering number. An ε-grid of spacing ε in each coordinate of the parameter cube $[0, 2\pi]^T$ induces an $O(\varepsilon)$-cover of $\mathcal{F}$ in L^∞. The number of grid points is $(2\pi/\varepsilon)^T$, giving $\log \mathcal{N}(\mathcal{F}, \varepsilon) = O(T \log(1/\varepsilon))$.

Rademacher complexity bound. Substituting into the standard covering-number Rademacher bound with $\varepsilon \sim$

$1/\sqrt{m}$:

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} |\mathcal{L}_{\text{test}}(f) - \mathcal{L}_{\text{train}}(f)|\right] \le O\left(\sqrt{\frac{T\log m}{m}}\right). \quad (65)$$

Setting the right-hand side to ε and solving gives $m = O(T\log(T/\varepsilon)/\varepsilon^2)$. □

VII. CLASSICAL SIMULATION HARDNESS

This section establishes the classical simulation hardness of the pipeline. The hardness results are expressivity statements: the function classes accessible to the unitary brick-wall and butterfly QNNs contain output distributions that cannot be efficiently sampled by classical algorithms, at a hardness level controlled by the particle number k. This expressivity is structural—it follows from the non-Gaussianity of the magic-state encoding, the number of magic blocks, and the passive FLO structure of the circuit—and does not depend on which specific parameter values are found by training.

Because the hardness depends on the particle number k, we organize the results as a ladder in k (cf. Section II D), instantiated at the operating point of Eq. (7). Four levels emerge:

- $k = O(\log n)$*: no hardness.* The FLO extent is polynomial and the entire pipeline—amplitudes and sampling included—is classically simulable in polynomial time.
- $k = \omega(\log n)$*: hardness under best-known algorithms.* Sampling costs $2^{\Omega(k)}\,\text{poly}(n)$ under the best-known classical algorithms (Theorem 33). At the operating point $k = 60$ this is $\chi^2 = 2^{60}$ cross terms; at $k = \log^2 n$ it is $n^{\Omega(\log n)}$.
- $k = n^\epsilon$*: worst-case #P-hardness.* Exact computation of output probabilities is #P-hard for the brick-wall (Theorem 34); for the butterfly the same statement holds under a reachability assumption (Remark 35).
- $k = \Theta(n)$*: average-case hardness.* The fully-packed limit inherits the complete Fermion Sampling machinery—anticoncentration and the worst-to-average-case Cayley-path reduction—for the Haar-initialized brick-wall (Theorem 37).

We note that average-case hardness holds as of now only at the top rung, because both of its ingredients—anticoncentration and the worst-to-average-case reduction—are currently proven only for fully-packed magic inputs. How far down the ladder they survive is precisely the "fermion sampling with less magic" open problem of [2, Sec. IV]; we state our sharpened form of it as Conjecture 38 in Section VII E.

Orthogonal to the ladder, *expectation-value readouts of fixed body number* are classically easy at any k, via efficient r-RDM propagation (Proposition 8); the ladder therefore concerns sampling alone.

All costs are governed by the FLO extent of the input state (Definition 31), a quantity that measures non-Gaussianity, which we analyze first. We close with a direct engagement with the Cerezo et al. [38] simulability framework (Section VII F).

A. The Fermion Sampling Structure

The full inference circuit (after training) is

$$V(\boldsymbol{\theta},\boldsymbol{\phi},\mathbf{x}) = U(\boldsymbol{\theta},\boldsymbol{\phi})\cdot D(\boldsymbol{\varphi}(\mathbf{x}))\cdot U^{\text{had}} \quad (66)$$

applied to $|\Psi_{\text{in}}\rangle$, where $U(\boldsymbol{\theta},\boldsymbol{\phi})$ is either the unitary brick-wall or butterfly, U^{had} is the corresponding fixed Hadamard spreading circuit, and $|\Psi_{\text{in}}\rangle$ is the paired magic state of Section II. Each component is a passive FLO transformation: $U(\boldsymbol{\theta},\boldsymbol{\phi})$ and U^{had} are products of RBS and R_z gates, both particle-number preserving; $D(\boldsymbol{\varphi}(\mathbf{x})) = \bigotimes_i R_z(\varphi_i)$ is a diagonal one-body unitary. Their product V is therefore a passive FLO transformation with single-particle unitary $W \in U(n)$. The output distribution is

$$p(\mathbf{x}' \mid \mathbf{x}) = |\,\langle \mathbf{x}'|\, V(\boldsymbol{\theta},\boldsymbol{\phi},\mathbf{x})\,|\Psi_{\text{in}}\rangle\,|^2, \quad (67)$$

where $|\Psi_{\text{in}}\rangle$ is the non-Gaussian magic state. This is exactly the Fermion Sampling structure of Oszmaniec et al. [2]: a passive FLO unitary acting on a non-Gaussian fermionic input. The best-known-algorithm hardness results below apply to both architectures since they depend only on this Fermion Sampling structure, not on the specific connectivity pattern of the passive FLO circuit; the #P-hardness and average-case statements additionally require that the hard single-particle unitaries be *reachable*, which holds unconditionally for the brick-wall (surjective onto $U(n)$) and under a reachability assumption for the fixed-depth butterfly (Remark 35).

B. FLO-Extent-Based Classical Simulation

For every readout family, the relevant classical simulation algorithms are controlled by a single quantity, the FLO extent of the input state.

Definition 31 (FLO extent)**.** *The FLO extent of a fermionic state $|\psi\rangle$ is the smallest integer χ such that $|\psi\rangle = \sum_{s=1}^{\chi} c_s\,|\phi_s^{\text{Gauss}}\rangle$ for some Gaussian states $|\phi_s^{\text{Gauss}}\rangle$ and complex coefficients c_s.*

For the paired magic state $|\Psi_4\rangle^{\otimes k/2}$, each block contributes FLO extent 2, so $\chi = 2^{k/2} = 2^B$ with B the number of magic blocks. Passive FLO transformations preserve FLO extent, so χ of the output state $V|\Psi_{\text{in}}\rangle$ matches that of the input.

Proposition 32 (Classical cost of simulating the pipeline)**.** *For a state with FLO extent χ prepared by a passive FLO circuit with single-particle unitary W, the following classical costs hold:*

- Fixed-body expectation values, exact*:* $\mathrm{poly}(n)$ *for every fixed body number r, via propagation of the r-RDM through $\Lambda^r(W)$ (Proposition 8); for $r = 1$ this is $O(n^2)$, with $\langle\psi|V^\dagger n_j V|\psi\rangle = [W\rho_1^{\mathrm{in}}W^\dagger]_{jj}$. The cost is independent of χ and of the form-rank of the input.*
- Exact amplitudes*:* $O(\chi \cdot \mathrm{poly}(n))$*, by direct evaluation of the Gaussian-sum expansion (for paired blocks, the determinant sum of Eq.* (2)*).*
- r-body expectation values, fixed r, form-rank 2*:* $O(\mathrm{poly}(n)\cdot\epsilon^{-2})$ *to additive error ϵ, via the mixed-Pfaffian estimator of Oh et al. [51].*
- Exact sampling from $p(\cdot \mid \mathbf{x})$*:* $O(\chi^2 \cdot \mathrm{poly}(n))$ *per sample, via the chain rule over marginals.*

Proof. Each case is the corresponding best-known algorithm from the cited literature, specialized to our input; we recall the arguments. Throughout, the Gaussian terms in the decomposition of $|\psi\rangle$ are mutually non-orthogonal Slater determinants, and all matrix elements between them are evaluated by the generalized (Löwdin) Wick rules, each in $\mathrm{poly}(n)$ time. The form-rank-2 case is quoted from [51] and not reproved here. *Fixed-body case.* An r-body expectation value depends only on the r-RDM of the output state, which is $\Lambda^r(W)\rho_r^{\mathrm{in}}\Lambda^r(W)^\dagger$; this is Proposition 8. Since ρ_r^{in} has dimension $\binom{n}{r}$ regardless of the non-Gaussianity or form-rank of the input, classical computation takes $\mathrm{poly}(n)$ time for fixed r, and $O(n^2)$ for $r = 1$.

Amplitudes. Each Gaussian term contributes a determinant (Slater-determinant overlap) computable in $\mathrm{poly}(n)$ time; summing the χ terms gives the amplitude exactly.

Case $r \geq 2$, form-rank 2. The mixed-Pfaffian reduction of Oh et al. [51] compresses the χ^2 cross-term sum into a single coefficient of a multivariate Pfaffian polynomial, extractable via Monte Carlo with $O(\epsilon^{-2}\log\delta^{-1})$ samples each costing $O(N^3)$ time (where N is the number of paired blocks), independent of χ.

Sampling. Sample the modes one at a time. The marginal probability of any partial occupation pattern on modes $1,\ldots,m$ is

$$p(x_1,\ldots,x_m) = \sum_{s,s'} c_s^* c_{s'}\, \langle\phi_s^{\mathrm{Gauss}}|\, V^\dagger\Big(\prod_{j\leq m} P_{x_j}\Big)V\,|\phi_{s'}^{\mathrm{Gauss}}\rangle\,,$$

where $P_{x_j} \in \{n_j, 1-n_j\}$. Each cross-term is a matrix element of a product of occupation projectors between two Gaussian states evolved by FLO, computable *exactly* in $\mathrm{poly}(n)$ time by generalized (Löwdin) Wick rules for non-orthogonal Slater determinants. Conditional sampling of mode $m+1$ given $x_1,\ldots,x_m$ therefore costs $O(\chi^2\mathrm{poly}(n))$, and a full sample costs n such steps. Note that this argument requires *exact* (or relative-error) marginals; additive-error estimates do not suffice, since the conditionals divide by marginals that are generically exponentially small. □

Applying this to the magic-state construction at the operating point of Eq. (7) ($B = 30$ blocks, $\chi = 2^{30}$, $\chi^2 = 2^{60}$): because the input has form-rank 2, the mixed-Pfaffian algorithm of [51] achieves $\mathrm{poly}(n)\cdot\epsilon^{-2}$ cost for any fixed-body observable, regardless of k: few-body expectation values are classically tractable at additive precision at *any* k. The classically hard task is *sampling* (Theorem 33), at best-known cost $\chi^2 = 2^{60}$.

C. Hardness under Best-Known Algorithms at $k = \omega(\log n)$

Once the block number exceeds logarithmic, every algorithm of Proposition 32 that touches the non-Gaussian structure costs $2^{\Omega(k)}$. We record the resulting hardness statement. It is *algorithm-relative* by design; the calibration of Section II D converts it into a concrete compute budget, and any future algorithmic improvement is absorbed by increasing the block number B, at training cost linear in k.

Theorem 33 (Sampling hardness for paired-block inputs under best-known algorithms)**.** *Let the input be the paired magic-block state $|\Psi_4\rangle^{\otimes k/2}$ with $k = \omega(\log n)$. Then every currently known classical algorithm for sampling from the output distribution $p(\cdot \mid \mathbf{x})$ of the trained unitary brick-wall or butterfly QNN requires $2^{\Omega(k)}\,\mathrm{poly}(n)$ time. At the sampling operating point $k = 60$ the best-known cost is $\chi^2 = 2^{60} \approx 10^{18}$ cross terms, each costing $\mathrm{poly}(n)$ operations; at $k = \log^2 n$ it is $n^{\Omega(\log n)}$. Consequently, the function classes of both architectures contain output distributions inaccessible to classical samplers under best-known algorithms.*

Proof. Three routes to sampling are known, and each costs $2^{\Omega(k)}$. (i) Exact strong simulation via the determinant sum of Eq. (2) costs $\Theta(2^{k/2})$ determinant evaluations per amplitude. (ii) Chain-rule sampling via extent-based marginals (Proposition 32) costs $\Theta(2^k)$ cross-term evaluations per sample [1]. (iii) The additive-error correlator estimator of [51] does not yield approximate sampling, because chain-rule conditionals require relative precision on marginals that are generically exponentially small in k. Hence no known algorithm samples in polynomial time once $k = \omega(\log n)$. □

D. Worst-Case #P-Hardness at Polynomial Particle Number

The statements of the previous subsection are relative to the best-known algorithms. Genuine

complexity-theoretic hardness—#P-hardness of exact computation—requires the block number itself to be polynomial in n, because the hard-instance size of the underlying reduction is the block count (Remark 1).

Theorem 34 (Worst-case #P-hardness for paired blocks at $k = n^\epsilon$)**.** *Fix a constant $\epsilon \in (0, 1)$ and let $k = n^\epsilon$ (with $k \le n/2$). For the paired magic-block input $|\Psi_4\rangle^{\otimes k/2}$, there exist parameter settings $(\boldsymbol{\theta}, \boldsymbol{\phi})$ of the unitary brick-wall and data encodings $\mathbf{x}$ for which computing the output probability $p(\mathbf{x}' \mid \mathbf{x})$ exactly is #P-hard.*

Proof. For any passive FLO circuit with single-particle unitary $W \in U(n)$, the output amplitude is the sum of 2^B determinants of Eq. (2), $B = k/2$—a mixed discriminant in the sense of Ivanov and Gurvits (cf. [2, Lemma 11]). Computing this mixed discriminant exactly for arbitrary W is #P-hard when the block count is polynomially related to the instance size: an $m \times m$ permanent instance (with nonnegative entries, so that hardness carries to the probability $|\cdot|^2$, cf. [2, Remark 21]) embeds with polynomial overhead into the mixed discriminant of $O(m)$ blocks. Choosing $n = (2m)^{1/\epsilon} = \mathrm{poly}(m)$ makes the reduction polynomial-time in m. The brick-wall is surjective onto $U(n)$ (Proposition 4), so the hard single-particle unitary W^* is exactly reachable: choose $\mathbf{x} = 0$ (so that the data layer is trivial up to fixed phases) and set the brick-wall angles, via the Givens decomposition, to realize $W_P = W^*(W_D W^{\mathrm{had}})^{-1}$, whence the total single-particle unitary is W^*. □

Remark 35 (Reachability caveat for the butterfly)**.** *The reduction realizes a specific hard unitary W^*, which the brick-wall reaches by surjectivity. A single depth-$2\log n$ butterfly covers only a submanifold of $U(n)$ of dimension at most $\frac{3}{2} n \log n$ (Proposition 7), and whether that submanifold meets the Ivanov–Gurvits hard instances is open; Theorem 34 therefore holds for the butterfly only under the assumption that it does. Stacking $O(n/\log n)$ butterfly blocks restores surjectivity, and with it the unconditional statement, at linear depth. The best-known-algorithm hardness of Theorem 33 is unaffected either way, since it depends only on the input state.*

Remark 36 (What worst-case #P-hardness does and does not buy)**.** *Two contrasts delimit Theorem 34. At $k = O(\log n)$, Eq. (2) is itself a polynomial-time algorithm, so no #P-hardness is possible; at $k = \log^2 n$ the same expression gives a quasi-polynomial upper bound, which likewise precludes it under standard assumptions. Worst-case #P-hardness therefore begins only at polynomial particle number, at a training cost of $O(n^{1+\epsilon})$ (Section II D).*

This is not, however, the guarantee the framework rests on. #P-hardness concerns exact *computation on a* worst-case *instance, whereas a deployment needs that no algorithm compute the readout of the* trained *circuit, to the precision the task requires, in feasible time; neither implies the other. Oh et al. [51] are the case in point: the mixed discriminant of a form-rank-2 input is #P-hard to evaluate exactly, and yet its few-body expectation values are estimable to additive error in polynomial time. Conversely, the operating point of Eq. (7) carries no #P-hardness whatsoever—$B = 30$ is a constant—but places best-known classical simulation beyond 10^{24} operations at every n, which is the operationally relevant statement. Theorem 34 is best read as evidence that the $2^{\Omega(k)}$ cost of Theorem 33 reflects genuine structure rather than a gap in the algorithmic literature.*

E. Average-Case Hardness: the Fully-Packed Limit and the Sparse Conjecture

The strongest expressivity claim available is average-case sampling hardness: a randomly initialized circuit produces a classically hard distribution with high probability. The two required ingredients, anticoncentration and the worst-to-average-case Cayley-path reduction [83], are currently proven only for fully-packed magic inputs, i.e. at extensive particle number [2]. Since the brick-wall realizes the Haar measure on $U(n)$ under Haar initialization (Section IV B), the full machinery transfers verbatim in that regime.

Theorem 37 (Average-case sampling hardness for the brick-wall, fully-packed regime)**.** *Let $k = n/2$ with the fully-packed paired input $|\Psi_4\rangle^{\otimes n/4}$. Under the complexity-theoretic assumptions of [2] (noncollapse of the polynomial hierarchy and average-case #P-hardness of relative-error computation of output probabilities, their Conjectures 1–2), approximately sampling from $p(\mathbf{x}' \mid \mathbf{x})$ is classically hard for a unitary brick-wall $U_P(\boldsymbol{\theta}, \boldsymbol{\phi})$ at Haar-random parameters. Consequently, the function class of the unitary brick-wall contains output distributions that cannot be efficiently sampled by any classical algorithm.*

Proof. Under Haar-random parameters, the brick-wall realizes the Haar measure on $U(n)$ in the single-particle sector (Section IV B). The full circuit $V = U_P \circ D \circ U^{\mathrm{had}}$ is therefore a Haar-random passive FLO transformation composed with the fixed deterministic operators D and U^{had}. $D(\boldsymbol{\varphi}(\mathbf{x}))$ and U^{had} are fixed deterministic passive FLO unitaries with single-particle matrices W_D and W^{had} in $U(n)$; they do not depend on the brick-wall parameters $(\boldsymbol{\theta}, \boldsymbol{\phi})$. Since $W_P \sim \mathrm{Haar}(U(n))$ and the Haar measure on $U(n)$ is bi-invariant, for any fixed $A \in U(n)$ the product $A \cdot W_P$ is again Haar-distributed: $\mathbb{P}[AW_P \in S] = \mathbb{P}[W_P \in A^{-1}S] = \mathrm{Haar}(A^{-1}S) = \mathrm{Haar}(S)$ for any measurable $S \subseteq U(n)$. Applying this twice, $W = W_P \cdot W_D \cdot W^{\mathrm{had}}$ is Haar-distributed on $U(n)$.

The Oszmaniec et al. [2] anticoncentration result (their Theorem 1) applies verbatim: for Haar-random passive FLO circuits on $U(n)$ initialized in the fully-packed input $|\Psi_4\rangle^{\otimes n/4}$, the output distribution anticoncentrates with constant $C_{\mathrm{pas}} = 5.7$. The worst-to-average-case reduction of [2] (their Theorem 7, via the Cayley-path

argument) and their Stockmeyer reduction (their Theorems 2–3) then establish hardness of approximate sampling under the stated assumptions. Since the brick-wall's function class contains all distributions reachable at Haar-random parameters, it contains distributions inaccessible to efficient classical samplers. □

Conjecture 38 (Sparse-input average-case hardness)**.** *At $k = n^\epsilon$ with paired input $|\Psi_4\rangle^{\otimes k/2}$ and $n - 2k$ idle modes, the output distribution of a Haar-random passive FLO circuit anticoncentrates, and approximate sampling remains hard on average under the assumptions of [2]. This is a sharpened form of the "fermion sampling with less magic" open problem of [2, Sec. IV] and its resolution requires, in particular, redoing the anticoncentration second-moment computation of [2, Lemma 13] for sparse inputs. Resolving it affirmatively would place average-case hardness at the same particle numbers where the gradient variance of Theorem 9 is largest.*

For the butterfly, the average-case question is doubly open: beyond the sparse-input issue, anticoncentration at logarithmic depth is not known even at full packing—whether the collision-probability constant satisfies $C_k(n) = O(\mathrm{poly}(n))$, and hence whether the worst-to-average-case reduction of Oszmaniec et al. [2] applies, is open for both fully-packed and sparse magic-block inputs.

Hardness summary.

- $k = O(\log n)$: the entire pipeline is classically simulable in polynomial time.
- *Operating point, $B = 30$ blocks* ($k = 60$, paired blocks): best-known classical sampling costs $\chi^2 = 2^{60}$ cross terms, i.e. $\gtrsim 10^{24}$ elementary operations (Theorem 33), calibrated in Section II D to exceed feasible classical compute; future algorithmic improvements are absorbed by increasing B.
- $k = n^\epsilon$: worst-case #P-hardness of exact output probabilities for the brick-wall (Theorem 34); for the butterfly under the reachability assumption (Remark 35); average-case hardness conjectured (Conjecture 38).
- $k = \Theta(n)$: the full average-case machinery of [2] for the Haar-initialized brick-wall at $k = n/2$ (Theorem 37); the parallel PSR no longer yields an asymptotic reduction over the naive parameter-shift rule (Section II D).
- *Fixed-body expectation-value readouts*: classically easy at any k for both architectures, via r-RDM propagation (Proposition 8).

The last item identifies precisely which observables are classically easy: those whose expectation values are determined by a reduced density matrix of fixed body number, regardless of the input's higher-body non-Gaussian structure.

Concrete parameter values for both architectures at representative problem sizes are collected in Table II of Section VIII.

F. On the Cerezo et al. Simulability Framework

Cerezo et al. [38] established that QNNs with a polynomial-dimensional dynamical Lie algebra admit efficient classical simulation under a condition on the input state and the observable: both must be $\mathfrak{g}$-*compatible*, in the sense that their projections onto $\mathfrak{g}$ can be computed efficiently. Under this condition the loss $\mathcal{L} = \langle\psi_{\mathrm{in}}| V^\dagger O V |\psi_{\mathrm{in}}\rangle$ is evaluated classically by propagating the $\mathfrak{g}$-projection of the input through the $\mathfrak{g}$-adjoint action of V and pairing it with the $\mathfrak{g}$-projection of the observable. The conclusion concerns the value of an expectation; the argument makes no statement about the complexity of sampling from the output distribution of the circuit.

Both architectures considered here have polynomial dynamical Lie algebra, $\mathfrak{g} = \mathfrak{u}(n)$ with $\dim \mathfrak{g} = n^2$. We therefore examine whether the framework precludes quantum advantage in the present setting. It does not; we treat in turn the sampling task itself and the few-body sample statistics of Section VI B.

a. The sampling task. Consider the sampling pipeline at $k = \Theta(n)$. Each hypothesis of the framework is satisfied. The dynamical Lie algebra is $\mathfrak{u}(n)$, of dimension n^2. The $\mathfrak{g}$-projection of the magic-state input is its 1-RDM, which is diagonal with entries $\frac{1}{2}$ on the $2k$ active modes and computable in $O(n^2)$ time (Lemma 2). The readouts n_j are one-body observables, so their $\mathfrak{g}$-projections coincide with themselves. The conclusion of the framework holds as well: these expectation values are classically computable in $O(n^2)$ time, in agreement with Proposition 8.

The output distribution of the same circuit acting on the same input is nevertheless not classically accessible: approximate sampling is hard on average at $k = n/2$ under the assumptions of [2] (Theorem 37); exact computation of individual output probabilities is #P-hard in the worst case at $k = n^\epsilon$ (Theorem 34); and every known sampling algorithm costs more than 10^{24} operations at the operating point $k = 60$ (Theorem 33).

These statements are consistent. An individual output probability $p(\mathbf{x}' \mid \mathbf{x}) = |\langle \mathbf{x}'| V |\Psi_{\mathrm{in}}\rangle|^2$ is the expectation value of the rank-one projector $|\mathbf{x}'\rangle\langle\mathbf{x}'|$, an observable of body-number k; such an observable is not $\mathfrak{g}$-compatible, and the framework does not apply to it. It follows that the implication frequently drawn from the framework—that an architecture with polynomial dynamical Lie algebra cannot support quantum advantage—does not hold in general. The present construction satisfies every hypothesis of the framework and nonetheless generates classically intractable samples; at $k = cn$ with $c < 1/2$ it does so—under the best-known-algorithm calibration of Section II D, the average-case statement of Theorem 37

being available at exactly half filling—while retaining unconditional trainability, with gradient variance $\Theta(1/n^2)$, its largest value anywhere in the range (Theorem 9 and Theorem 13).

This regime is of central rather than peripheral interest for machine learning: generative modelling and reinforcement-learning policies both consume the circuit directly as a sampler (Section VI), so the classically intractable object is the model output itself.

b. Few-body sample statistics. For statistics of fixed body number the framework is informative, and its conclusion is the one we reach independently. The $\mathfrak{g}$-projection of an r-body observable is determined by the r-particle sector, and by Proposition 8 the corresponding expectation value is computable in poly(n) time regardless of k or of form-rank; for the paired inputs used here the mixed-Pfaffian reduction of Oh et al. [51] additionally gives an additive-error estimator with the shot complexity of the quantum device itself. Few-body readouts are therefore classically tractable and cannot witness quantum advantage.

c. Summary. Polynomial dimension of the dynamical Lie algebra does not imply classical simulability of the learning task. Two properties jointly determine tractability:

- the body number r of the readout observable,
- the precision regime (multiplicative for sampling, additive for expectation values).

Every fixed r is resolved, for arbitrary input and either architecture, by r-RDM propagation (Proposition 8); at form-rank 2 the mixed-Pfaffian reduction of [51] additionally supplies an additive-error estimator whose shot complexity matches the quantum device. Sample-based readouts fall outside the scope of the framework, and constitute the setting in which an architecture of polynomial dynamical Lie algebra is provably not classically simulable.

VIII. SUMMARY AND DISCUSSION

We have presented a provably scalable end-to-end quantum neural network framework built around two architectures — the unitary brick-wall for nearest-neighbor hardware and the unitary butterfly for all-to-all connectivity — that together address five theoretical challenges simultaneously: hardware-matched circuit depth (Section III), efficient training cost via the multi-layer parallel parameter-shift rule (Section V), polynomial gradient variance and absence of barren plateaus for readouts of fixed body number (Section IV), superpolynomial classical simulation hardness for the sampling task under the magic-state encoding (Section VII), and polynomial sample complexity for generalization (Section VI). The third and fourth of these are established for different objects of the same circuit—the gradient channel and the deployed output respectively—and Section VI C sets out which pipelines admit a variance guarantee and which require the quantum device in the training loop, and Section VI D instantiates the standard machine-learning tasks—generative modelling, reinforcement-learning policies, supervised classification and regression, kernel methods—with the loss and head appropriate to each. Both architectures share the same encoding pipeline and the same parallel parameter-shift rule; they differ in circuit depth, parameter count, training cost, and the strength of the trainability guarantee. The key structural insight is that all properties follow from interleaving particle-number-preserving RBS gates with single-qubit R_z gates — in a brick-wall connectivity pattern for nearest-neighbor hardware, or a butterfly connectivity pattern for all-to-all hardware — combined with a magic-state encoding whose number of magic blocks is calibrated against classical compute: at the operating point of Eq. (7), paired magic blocks $|\Psi_4\rangle^{\otimes k/2}$ at $k = 60$, at hardness level $\chi^2 = 2^{60}$.

A. Complete Parameter Summary

Table II summarizes all relevant quantities for both architectures at the operating point, at four representative problem sizes. The table is organized in three blocks: shared encoding-dependent quantities (identical for both architectures), architecture-dependent quantities for the butterfly, and architecture-dependent quantities for the brick-wall.

Several features of Table II deserve highlighting.

a. Trainability. The gradient variance for two-body correlator readouts is $\Theta(k^3/n^5)$ for both architectures, independent of block type. As established in Section IV, this rate is *unconditional* for the brick-wall (Theorem 9) and *conditional* on Conjecture 20 for the butterfly (Theorem 19), which also carries an unconditional polynomial lower bound $\Omega(k^4/n^6)$ (Theorem 17). Two body number is the only order at which we establish a rate; losses that depend on the samples through statistics of this order inherit it by linearity (Proposition 29), while losses sensitive to higher-order sample structure retain the same parameter-shift cost but carry no variance guarantee (Section VI C).

b. Classical hardness. The hard task is sampling from the model's output distribution—the deployed output of every pipeline in Section VI, and not the loss used to train it. By construction its best-known classical simulation cost is the same at every problem size: $\chi^2 = 2^{60} \approx 1.2\times10^{18}$ cross terms, each costing poly$(n) \gtrsim 10^6$ elementary operations, i.e. $\gtrsim 10^{24}$ operations in total—about two weeks on a 10^{18}-flop/s machine. Hardness is set by the block number $B = 30$, not by n; growing n buys expressivity (Hilbert-space dimension, feature dimension) and training efficiency, while the hardness margin is adjusted, if ever needed, by increasing B at cost linear in k (Section II D). These costs apply to both architectures

TABLE II. Complete parameter summary for the unitary brick-wall and butterfly QNNs at four problem sizes and the operating point of Eq. (7): $B = 30$ paired magic blocks $|\Psi_4\rangle^{\otimes 30}$, $k = 60$, FLO extent $\chi = 2^{30}$, hardness level $\chi^2 = 2^{60} \approx 1.2 \times 10^{18}$, fixed across n; encoding depth $\log n + 5$. **Shared quantities** (top block) depend only on the magic-state encoding and are identical for both architectures. The simulation cost row reports χ^2, the leading factor in the best-known classical cost of sampling (Theorem 33); few-body expectation values are additively tractable via the mixed-Pfaffian estimator [51]. **Architecture-dependent quantities** (middle and bottom blocks) reflect the depth–connectivity tradeoff: the butterfly targets all-to-all hardware at $O(\log n)$ depth with $T = \frac{3}{2} n \log n$ parameters; the brick-wall targets nearest-neighbor (ring) hardware at $O(n)$ depth with $T = 3n^2/2$ parameters. Naive PSR cost is $4n \log n$ for the butterfly and $4n^2$ for the brick-wall (4 circuit evaluations per RBS parameter and 2 per R_z parameter); parallel PSR cost is $4k \log n$ for the butterfly and $4kn$ for the brick-wall; the reduction factor n/k, identical for the two architectures, grows linearly with n at fixed hardness and exceeds 1 throughout the admissible range $2k \leq n$. Gradient variance for two-body correlator readouts is $\Theta(k^3/n^5)$ *unconditionally* for the brick-wall (Theorem 9, Corollary 11 and Theorem 13) and $\Theta(k^3/n^5)$ *conditional* on Conjecture 20 for the butterfly (Theorem 19).

	$n = 256$	$n = 512$	$n = 1024$	$n = 4096$
Shared quantities (encoding-dependent, identical for both architectures)				
Magic blocks B	30	30	30	30
Active qubits $2k$	120	120	120	120
Hilbert space $\binom{n}{k}$	2.0×10^{59}	1.2×10^{79}	8.6×10^{97}	4.3×10^{134}
FLO extent $\chi = 2^{30}$	1.1×10^{9}	1.1×10^{9}	1.1×10^{9}	1.1×10^{9}
Simulation cost $\chi^2 = 2^{60}$	1.2×10^{18}	1.2×10^{18}	1.2×10^{18}	1.2×10^{18}
Gradient variance (2-body)		$\Theta(k^3/n^5)$		
Sample complexity		$\tilde{O}(T)$		
Unitary butterfly (all-to-all connectivity, $O(\log n)$ depth)				
Total circuit depth	30	33	36	42
Parameters $T = \frac{3}{2} n \log n$	3,072	6,912	15,360	73,728
Naive PSR cost $4n \log n$	8,192	18,432	40,960	196,608
Parallel PSR cost $4k \log n$	1,920	2,160	2,400	2,880
Reduction factor n/k	4.3×	8.5×	17.1×	68.3×
Unitary brick-wall (nearest-neighbor connectivity, $O(n)$ depth)				
Total circuit depth	773	1,541	3,077	12,293
Parameters $T = 3n^2/2$	98,304	393,216	1,572,864	2.5×10^{7}
Naive PSR cost $4n^2$	262,144	1,048,576	4,194,304	6.7×10^{7}
Parallel PSR cost $4kn$	61,440	122,880	245,760	983,040
Reduction factor n/k	4.3×	8.5×	17.1×	68.3×

identically, since they depend only on the magic-state encoding and not on the connectivity pattern of the passive FLO circuit.

c. Training cost. At fixed hardness level (fixed k), the parallel-PSR reduction factor n/k grows linearly with n: it reaches 17.1× at $n = 1024$ and 68.3× at $n = 4096$. The butterfly's absolute cost, $4k \log n$, is nearly independent of n: between 1,920 and 2,880 circuit evaluations per gradient step across the entire table. Training a hardness-calibrated model therefore becomes *relatively* cheaper as the machine grows. The parallel rule is cheaper than the naive one whenever $k < n$, so the binding constraint on the smallest useful machine is the encoding requirement $2k \leq n$—$n \geq 120$ at $k = 60$—rather than the parameter-shift rule; already at $n = 256$ the reduction is 4.3×. The butterfly's polylogarithmic per-step cost makes it the architecture of choice when circuit evaluations are the bottleneck; the brick-wall's $O(n)$ depth makes it the natural target for nearest-neighbor hardware regardless of evaluation cost.

B. Outlook

The main open question is empirical: whether the proposed framework delivers practical advantages on real machine learning tasks. The theoretical guarantees established here — trainability, efficient gradients, classical hardness, polynomial generalization — are necessary but not sufficient for useful quantum advantage. What remains to be demonstrated is that quantum models built on either architecture solve problems classical models cannot solve as well, whether standalone or as components in ensemble methods. This requires both mature quantum hardware and careful empirical study. For the butterfly, $n = 512$ or $n = 1024$ with all-to-all or reconfigurable connectivity appears to us to be the scale at which meaningful comparisons become possible—the operating point of Eq. (7) fits comfortably ($2k \leq n$) and the parallel PSR delivers a material advantage (Table II); for the brick-wall, the same n on nearest-neighbor hardware—a ring, or a chain via the open-boundary variant of Remark 6. The verification regime $k = O(\log n)$ provides the natural on-ramp: the identical architecture can be

validated end to end against exact classical simulation at small block number before B is scaled up.

The analysis of Section VI C narrows where such demonstrations should be attempted. The criterion is the *deployed output*: it must be the sample set or a functional of it beyond fixed body number, since a model whose output is a few-body statistic is classically simulable end to end regardless of k. Whether the device is also needed during training varies, and either mode is viable: a loss anchored on estimable statistics can be trained classically, with the device reserved for sampling at inference, while a loss that is an expectation over something we have no model of requires the device in the loop—which can happen even for a low-body effective observable, when its weights are supplied by an environment rather than known to us. Two families meet the criterion. The first is joint multivariate generative modelling, where the object consumed downstream is the dependence structure of a high-dimensional distribution— joint tail probabilities, scenario ensembles, risk functionals—rather than a collection of marginals; sampling is then the deployed operation and the two-body moments that the trainability analysis controls are exactly the pairwise co-occurrence statistics such applications already use. The second is model-free reinforcement learning with a structurally low-body return, which is the one setting identified in Section VI C in which a proven gradient rate, classical hardness of the deployed policy, and the need for the device inside the training loop hold simultaneously. Financial decision problems—portfolio construction, execution—are a natural instance of the second family: the return is linear in the positions with quadratic costs and risk, hence two-body in the action, while no sufficiently accurate model of the environment is available to propagate through, which is why model-free methods are used in the first place. A further architectural fit is that particle-number preservation makes every shot a valid Hamming-weight-k configuration, so constrained selection problems—allocate across k of n assets, route k of n child orders—are sampled feasibly by construction, without masking or repair. We note that these are candidates identified on structural grounds; whether the resulting models are *useful* is the empirical question above, and the $k = O(\log n)$ regime is where it should first be asked.

On the theory side, several questions remain. The first concerns the separation of objects described in Section VI C. The hardness guarantees of Section VII are sampling statements: they operate at multiplicative precision on individual output probabilities, and the tasks that consume the QNN as a sampler—generative modeling and reinforcement-learning policies—inherit them directly; Oh et al. [51] explicitly disclaim any implications of additive-error estimation for this regime, and any future algorithmic advance is absorbed by raising the block number B. The trainability guarantees, by contrast, are established for readouts of fixed body number, and precisely that property makes them classically computable whenever a model of the distribution is available (Proposition 8). Whether the two can be carried by a single object—whether a *variationally trained* circuit can have an expectation-value readout that is at once provably free of barren plateaus and classically hard—is open, and it is not clear how to achieve it: few-body expectation values of bounded observables sit naturally in the additive-precision regime, where estimators such as the mixed-Pfaffian reduction of Oh et al. [51] already cover broad input classes, while observables of growing body order face vanishing signals and gradients. The question is not whether *some* expectation-value model can be classically hard—fixed, cryptographically structured feature maps achieve this, as in the discrete-logarithm construction of Liu et al. [84], where no variational training is involved and the hardness is imported from the planted problem rather than arising from the architecture—but whether hardness of this kind can coexist with a trained circuit and a proven gradient rate. A related question is how much of the output distribution a low-body loss actually shapes: training against few-body statistics controls those statistics, while the higher-order structure that carries the hardness is inherited from the architecture rather than learned, and the practical value of that unshaped remainder is an empirical matter. We stress that this is a question about proofs rather than about capability: the sampler supports the full range of machine-learning tasks catalogued in Section VI D, and none of them requires an expectation-value readout to be classically hard. What an affirmative answer would add is a second, independent route to advantage, not a task the present framework cannot address.

A distinct contribution of the present framework is a sharp statement of where the magic-count threshold for hardness sits, updating the "fermion sampling with less magic" open problem of [2] in light of the extent-based simulation algorithms of [1]: at $B = O(\log n)$ blocks the pipeline is classically simulable outright, including sampling; at $B = \omega(\log n)$ all known algorithms are superpolynomial (Theorem 33); at $B = n^{\Omega(1)}$ worst-case #P-hardness is restored (Theorem 34); and the average-case machinery is currently available only at extensive filling. Closing the remaining gap—average-case hardness and anticoncentration for sparsely packed magic inputs (Conjecture 38)—is, in our view, the most consequential open problem our construction inherits from the Fermion Sampling literature.

Other theoretical directions remain open. Resolving Conjecture 20 — extending the transfer-matrix spectral-gap proof of Appendix B to the $\Lambda^2\mathbb{C}^n$ representation — would promote the butterfly's $\Theta(k^3/n^5)$ gradient variance from conditional to unconditional, placing both architectures on equal theoretical footing. A second, more technical question concerns the brick-wall's per-parameter statement. The gradient-variance rate is exact and unconditional for the invariant derivative at a Haar point, for the n phases of the front R_z layer, and for every gate of the circuit in the independent-halves ensemble.

What is not settled is the pointwise statement for an interior gate of the *literal* single mesh, which reduces to a single second-moment question: writing $\Pi^{(\ell)} = W_1 \Pi W_1^\dagger$ for the spread projector propagated through the first $\ell-1$ columns, does

$$\mathbb{E}\big[\, \|[g, \Pi^{(\ell)}]\|_F^2 \,\big] \;=\; \Theta\!\left(\frac{k(n-2k)}{n^2}\right) \tag{68}$$

hold at the intermediate depths $1 < \ell < n$—equivalently, can a shallow partial mesh re-localize the Walsh–Hadamard-spread projector at the level of second moments? Proposition 14 answers yes at both endpoints of the prefix, with matching constants, and establishes the corresponding first-moment statement at every intermediate depth. Settling the second moment requires a transfer analysis of the partial mesh in the Hurwitz angle densities, which are position-dependent with $\mathbb{E}[\cos^2\vartheta] \neq \frac{1}{2}$, in the sector carrying $|\Pi^{(\ell)}_{jj}|^2$ and $|\Pi^{(\ell)}_{ij}|^2$; the Clements-mesh version of the Hurwitz factorization [74, 85] is the natural starting point. We regard the question as well posed and self-contained. The anticoncentration question for the butterfly at logarithmic depth remains open; the single-particle 2-design does not by itself control the k-particle collision probability, and a direct argument in the k-particle sector may be possible. Understanding the optimization landscape — whether layer-wise block coordinate descent converges faster than full gradient descent and under what conditions — is an empirical question we leave open; polynomial gradient variance rules out exponentially flat landscapes but not other obstructions, such as the proliferation of poor local minima identified in [86]. The relationship between body number and precision regime suggests a refined hierarchy of QNN expressivity worth mapping systematically: our paper identifies two structural distinctions in the classical-tractability landscape—observables of fixed body number are always easy, by reduced-density-matrix propagation (Proposition 8) and, at additive precision on paired inputs, by the estimator of Oh et al. [51]; while sampling at multiplicative precision is hard at sufficient block number via Fermion Sampling—and the intermediate regime of observables whose body number grows with n is unmapped in both directions.

A separate question is whether the alternative Dicke-state encoding (Remark 3) offers practical or theoretical advantages that justify its higher preparation depth. Its permutation-symmetric structure provides natural inductive biases for set-valued data, and its 1-RDM is proportional to the identity, so the trainability analysis of Section IV—which is driven by $\|[g, \rho_1]\|_F^2$—does not apply and the gradient variance is governed instead by connected two-body correlations. Characterizing that rate, and determining whether these advantages outweigh the depth cost in practice, is an interesting direction for future work.

Despite these open directions, the framework established here provides a complete answer to the theoretical challenges that have dominated quantum machine learning for the past several years, at two levels. As a matter of theory, both the unitary brick-wall and the unitary butterfly are trainable throughout the regime $n - 2k = \Omega(n)$—polynomial gradient variance from $k = O(1)$ to any constant filling fraction $c < 1/2$—while at extensive particle number the same architecture carries the classical-hardness guarantees of Section VII, up to the full average-case Fermion Sampling machinery at half filling. As a matter of practice, the operating point of Eq. (7) places best-known classical simulation beyond 10^{24} operations while the multi-layer parallel parameter-shift rule reduces the training cost by a factor n/k that grows linearly with n. The brick-wall achieves this with unconditional trainability guarantees on nearest-neighbor hardware; the butterfly achieves this at polylogarithmic depth on all-to-all hardware, with the $\Theta(k^3/n^5)$ trainability rate as an open question based on a well-formed conjecture. Whether either architecture is useful is now a question for experiment.

NOTE ADDED

Version 1 of this manuscript additionally introduced a second, triplet (form-rank-3) magic-block encoding and claimed best-known-algorithm classical hardness for the associated two-body expectation-value readout. That claim was refuted by Erfan Amidi [87], who observed that two-body expectation values after passive fermionic linear optics are determined by the input two-particle reduced density matrix—explicitly computable for the block-product inputs—and are therefore classically computable in $O(n^4)$ time, independently of the particle number and the form-rank. The corresponding statement for observables of arbitrary fixed body number is Proposition 8. We thank him for the correction. Version 2 removes the triplet encoding and all expectation-value hardness claims, and presents the framework in unified sample-based form; the trainability, parameter-shift, and sampling-hardness results are unaffected, and the existence of a trainable, classically hard expectation-value pipeline is now stated as an open question (Section VIII).

AUTHOR CONTRIBUTIONS

The author conceived the framework, proved the results, and wrote the manuscript. Large language model assistants were used for literature search, for checking algebraic manipulations and numerical evaluations, and for copy-editing; all definitions, statements and proofs were verified by the author.

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Appendix A: Proofs for Section IV

This appendix collects the supporting lemmas and the full proofs of the trainability results of Section IV.

1. Structural lemmas

Lemma 39 (Locality of coordinate-local gate generators on Λ^2)**.** *Let G be a trainable gate generator—either an RBS generator G_{ij} (acting on the mode pair $T = \{i,j\}$) or an R_z generator $\hat{n}_j$ (acting on $T = \{j\}$)—with $\|G^{(2)}\|_{\mathrm{op}} = O(1)$, so that G is supported on the $O(1)$ coordinates T. Let $\sigma = \Lambda^2(V')\,\sigma^{\mathrm{in}}\,\Lambda^2(V')^\dagger$, where $V' = D_1 H$ is the encoding (Walsh–Hadamard spreading followed by the diagonal data layer) and σ^{in} is supported on a set of r basis pairs with $\|\sigma^{\mathrm{in}}\|_{\mathrm{op}} = O(1)$ and $\|\sigma^{\mathrm{in}}\|_F^2 = \Theta(r)$. Then*

$$\big\|[G^{(2)},\sigma]\big\|_F^2 \;=\; O\big(\tfrac{1}{n}\big)\;\|\sigma\|_F^2, \tag{A1}$$

uniformly in the data $\mathbf{x}$. Both the disconnected part $\rho_1 \wedge \rho_1$ (with $r = \binom{2k}{2}$) and the connected part ρ_2^{conn} (with $r = \Theta(k)$) of the encoded 2-RDM satisfy the hypotheses. The hypothesis that G be supported on $O(1)$ coordinates *is used through the alignment of $\mathcal{S}_T$ with the computational basis in which V' delocalizes; for a general unitary conjugate $V^\dagger G V$ the statement is false pointwise, and its correct averaged replacement is Lemma 42.*

Proof. The single-particle generator g satisfies $g\,|a\rangle = 0$ for $a \notin T$; since $G^{(2)}\,|\{a,b\}\rangle = (g\,|a\rangle)\wedge|b\rangle + |a\rangle\wedge(g\,|b\rangle)$, the lift annihilates every basis pair disjoint from T, so $G^{(2)} = P_{\mathcal{S}_T} G^{(2)} P_{\mathcal{S}_T}$ with $\mathcal{S} := \mathcal{S}_T = \mathrm{span}\{|\{a,b\}\rangle : \{a,b\}\cap T \neq \emptyset\}$ and $\dim\mathcal{S} = \binom{n}{2} - \binom{n-|T|}{2} = \Theta(n)$, a subspace spanned by basis pairs, and

$$\big\|[G^{(2)},\sigma]\big\|_F \le 2\|G^{(2)}\|_{\mathrm{op}}\,\|P_{\mathcal{S}}\,\sigma\|_F. \tag{A2}$$

It remains to bound $\|P_{\mathcal{S}}\sigma\|_F^2 = \sum_{P\in\mathcal{S}}(\sigma^\dagger\sigma)_{PP}$. Writing $v_P := \Lambda^2(V')^\dagger\,|P\rangle$, a unit vector, we have $(\sigma^\dagger\sigma)_{PP} = \langle v_P, (\sigma^{\mathrm{in}})^\dagger\sigma^{\mathrm{in}} v_P\rangle$. Since $|V'_{ab}| = n^{-1/2}$ we have $|\Lambda^2(V')_{PQ}| \le 2/n$, so the restriction of v_P to the r basis pairs carrying σ^{in} has squared norm at most $4r/n^2$, and therefore

$$(\sigma^\dagger\sigma)_{PP} \le \|\sigma^{\mathrm{in}}\|_{\mathrm{op}}^2\cdot\frac{4r}{n^2} = O\Big(\frac{r}{n^2}\Big) = O\bigg(\frac{\|\sigma\|_F^2}{d}\bigg), \tag{A3}$$

uniformly in P and in $\mathbf{x}$, using $\|\sigma\|_F^2 = \|\sigma^{\mathrm{in}}\|_F^2 = \Theta(r)$ and $d = \Theta(n^2)$. Summing over the $\dim\mathcal{S} = \Theta(n)$ pairs in $\mathcal{S}$ gives $\|P_{\mathcal{S}}\sigma\|_F^2 = O(\|\sigma\|_F^2/n)$, which proves (A1).

For the disconnected part, $\rho_1^{\mathrm{in}} \wedge \rho_1^{\mathrm{in}}$ is diagonal on Λ^2 with entry $\frac{1}{4}$ on each of the $\binom{2k}{2}$ active–active pairs, so $r = \binom{2k}{2}$, $\|\cdot\|_{\mathrm{op}} = \frac{1}{4}$ and $\|\cdot\|_F^2 = \Theta(r)$. For the connected part, the input is a product over the $k/2$ (resp. $k/3$) magic blocks, so ρ_2^{conn} is block-diagonal with $\Theta(k)$ blocks of $O(1)$ size and constant norm, giving $r = \Theta(k)$, $\|\cdot\|_{\mathrm{op}} = O(1)$ and $\|\cdot\|_F^2 = \Theta(k)$. □

Lemma 40 (Λ^4 trace formula)**.** *Let P_{Λ^4} denote the orthogonal projector of $\Lambda^2\mathbb{C}^n \otimes \Lambda^2\mathbb{C}^n$ onto its $\Lambda^4\mathbb{C}^n$ component. Then for all $M, N \in \mathrm{End}(\Lambda^2\mathbb{C}^n)$,*

$$\mathrm{tr}\big(P_{\Lambda^4}\,(M\otimes N)\big) = \tfrac{1}{6}\Big[\mathrm{tr}M\;\mathrm{tr}N + \mathrm{tr}(MN) - \mathrm{tr}\big(\Gamma(M)\Gamma(N)\big)\Big]. \tag{A4}$$

Proof. Let $\mathcal{A}_4 = \frac{1}{24}\sum_{\pi\in S_4}\mathrm{sgn}(\pi)\,\pi$ be the total antisymmetrizer on $(\mathbb{C}^n)^{\otimes 4}$. The subspace $\Lambda^2\otimes\Lambda^2$ is $\mathcal{A}_4$-invariant, $\mathcal{A}_4$ is self-adjoint and idempotent, and its image inside $\Lambda^2\otimes\Lambda^2$ is Λ^4; hence P_{Λ^4} is the restriction

of $\mathcal{A}_4$. Extending M and N to $(\mathbb{C}^n)^{\otimes 4}$ by composing with the antisymmetrizer on each index pair contributes a factor $\frac{1}{2}$ per operator, so

$$\mathrm{tr}\big(P_{\Lambda^4}(M\otimes N)\big) = \frac{1}{24}\cdot\frac{1}{4}\sum_{\pi\in S_4}\mathrm{sgn}(\pi)\, M^{i_1 i_2}_{i_{\pi(1)}i_{\pi(2)}}\, N^{i_3 i_4}_{i_{\pi(3)}i_{\pi(4)}}. \tag{A5}$$

The 24 permutations fall into three orbits.

(i) π preserves $\{1,2\}$ and $\{3,4\}$ (4 permutations). For $\pi = \mathrm{id}$ the contraction is $(M^{i_1 i_2}_{i_1 i_2})(N^{i_3 i_4}_{i_3 i_4}) = 4\mathrm{tr}M\,\mathrm{tr}N$. Each transposition inside a pair flips the sign of the corresponding kernel by antisymmetry and carries $\mathrm{sgn} = -1$, so all four terms contribute $+4\mathrm{tr}M\mathrm{tr}N$, totalling $16\,\mathrm{tr}M\,\mathrm{tr}N$.

(ii) π exchanges $\{1,2\}$ with $\{3,4\}$ (4 permutations). For $\pi = (13)(24)$ the contraction is $M^{i_1 i_2}_{i_3 i_4}N^{i_3 i_4}_{i_1 i_2} = 4\mathrm{tr}(MN)$, with $\mathrm{sgn} = +1$; the other three differ by transpositions inside a pair and, by the same sign cancellation, also contribute $+4\mathrm{tr}(MN)$, totalling $16\,\mathrm{tr}(MN)$.

(iii) mixed π (16 permutations). Each such permutation sends exactly one element of $\{1,2\}$ into $\{1,2\}$ and the other into $\{3,4\}$, so the contraction closes one index of M against one of N twice. For $\pi = (13)$, for instance,

$$M^{i_1 i_2}_{i_3 i_2}\, N^{i_3 i_4}_{i_1 i_4} = \sum_{i_1,i_3}\Gamma(M)^{i_1}{}_{i_3}\,\Gamma(N)^{i_3}{}_{i_1} = \mathrm{tr}\big(\Gamma(M)\Gamma(N)\big), \tag{A6}$$

with $\mathrm{sgn}(\pi) = -1$; every mixed permutation reduces to the same expression with the same sign after using the antisymmetry of the kernels, totalling $-16\,\mathrm{tr}(\Gamma(M)\Gamma(N))$.

Collecting, $\frac{1}{96}\big[16\mathrm{tr}M\mathrm{tr}N + 16\mathrm{tr}(MN) - 16\mathrm{tr}(\Gamma(M)\Gamma(N))\big]$, which is (A4). As a check, at $M = N = I_{\Lambda^2}$ the right-hand side equals $\frac{1}{6}[d^2 + d - n(n-1)^2] = \frac{1}{24}n(n-1)(n-2)(n-3) = \binom{n}{4} = \dim\Lambda^4 = \mathrm{tr}P_{\Lambda^4}$, for every n. □

Lemma 41 (The contraction intertwines the adjoint action)**.** *For every single-particle operator g and every $Y \in \mathrm{End}(\Lambda^2\mathbb{C}^n)$,*

$$\Gamma\big([\,d\Lambda^2(g),\, Y\,]\big) \;=\; [\,g,\,\Gamma(Y)\,]. \tag{A7}$$

Proof. In kernel form, $\big(d\Lambda^2(g)\,Y\big)^{ij}_{kl} = g^i{}_a Y^{aj}_{kl} + g^j{}_a Y^{ia}_{kl}$ and $\big(Y\,d\Lambda^2(g)\big)^{ij}_{kl} = Y^{ij}_{al}\,g^a{}_k + Y^{ij}_{kb}\,g^b{}_l$. Contracting j against l and summing, the first term of each expression yields $g\,\Gamma(Y)$ and $\Gamma(Y)\,g$ respectively. The two remaining terms are $\sum_{j,a} g^j{}_a Y^{ia}_{kj}$ and $\sum_{j,b} g^b{}_j Y^{ij}_{kb}$, which coincide after the relabelling $a \leftrightarrow j$ and cancel in the difference. □

Lemma 42 (Averaged locality of conjugated gate generators)**.** *Let g be a Hermitian single-particle operator, let σ be any Hermitian operator on $\Lambda^2\mathbb{C}^n$, and for $V \in U(n)$ write $\tilde{g} = V^\dagger g V$, so that $\tilde{G}^{(2)} = d\Lambda^2(\tilde{g}) = \Lambda^2(V)^\dagger\, d\Lambda^2(g)\,\Lambda^2(V)$. Then, for $V \sim \mathrm{Haar}(U(n))$,*

$$\mathbb{E}_V\big\|[\tilde{G}^{(2)},\sigma]\big\|_F^2 \;\le\; \frac{4\,\big[(n-2)\mathrm{tr}(g^2) + (\mathrm{tr}g)^2\big]}{\binom{n}{2}}\,\|\sigma\|_F^2. \tag{A8}$$

For an RBS generator the right-hand side is $16(n-2)\|\sigma\|_F^2/(n(n-1))$ and for a phase generator $g = \hat{n}_j$ it is $8\|\sigma\|_F^2/n$: in both cases $O(\|\sigma\|_F^2/n)$, the same rate as the pointwise bound of Lemma 39 for coordinate-local generators.

Proof. Write $A = \tilde{G}^{(2)}$, which is Hermitian. Expanding the commutator, $\|[A,\sigma]\|_F^2 = 2\big[\mathrm{tr}(A^2\sigma^2) - \mathrm{tr}(A\sigma A\sigma)\big]$, and by Cauchy–Schwarz $|\mathrm{tr}(A\sigma A\sigma)| = |\langle (A\sigma)^\dagger, A\sigma\rangle| \le \|A\sigma\|_F^2 = \mathrm{tr}(A^2\sigma^2)$, so $\|[A,\sigma]\|_F^2 \le 4\,\mathrm{tr}(A^2\sigma^2)$. Now $A^2 = \Lambda^2(V)^\dagger(G^{(2)})^2\Lambda^2(V)$, and $\Lambda^2\mathbb{C}^n$ is an irreducible $U(n)$-module, so by Schur's lemma $\mathbb{E}_V[A^2] = \big(\mathrm{tr}((G^{(2)})^2)/\binom{n}{2}\big)I$. Finally, in kernel form $\big(d\Lambda^2(g)\big)^{ij}_{kl} = g^i{}_k\delta^j_l + \delta^i_k g^j{}_l - g^i{}_l\delta^j_k - \delta^i_l g^j{}_k$, and $\mathrm{tr}(MN) = \frac{1}{4}M^{ij}_{kl}N^{kl}_{ij}$ gives

$$\mathrm{tr}\big((d\Lambda^2(g))^2\big) = (n-2)\mathrm{tr}(g^2) + (\mathrm{tr}g)^2 \tag{A9}$$

(checked at $g = I_n$, where $d\Lambda^2(I) = 2I_{\Lambda^2}$ and both sides equal $2n(n-1)$). Combining the three displays gives (A8); the two special cases use $\mathrm{tr}(g^2) = 2$, $\mathrm{tr}g = 0$ for an RBS generator and $\mathrm{tr}(g^2) = \mathrm{tr}g = 1$ for a phase generator. □

Lemma 43 (Conjugation neutrality of the one-body commutator)**.** *Let Π be a rank-m orthogonal projector on $\mathbb{C}^n$ and let g be Hermitian. For $V \sim \mathrm{Haar}(U(n))$,*

$$\mathbb{E}_V\big\|[\,g,\,V\Pi V^\dagger\,]\big\|_F^2 \;=\; \frac{2m(n-m)}{n^2-1}\,\Big[\mathrm{tr}(g^2) - \tfrac{(\mathrm{tr}g)^2}{n}\Big]. \tag{A10}$$

At $m = 2k$ this gives $8k(n-2k)/(n^2-1)$ for an RBS generator and $4k(n-2k)/\big(n(n+1)\big)$ for a phase generator, to be compared with the unconjugated values $8k(n-2k)/n^2$ and $4k(n-2k)/n^2$ of Step 4 of Theorem 9: the two agree to leading order and with the same constants, the ratios being $n^2/(n^2-1)$ and $n/(n+1)$. Haar conjugation is thus neutral for τ, constants included.

Proof. Write $Q = V\Pi V^\dagger$, so $Q^2 = Q$ and, expanding the commutator, $\|[g,Q]\|_F^2 = 2\big[\mathrm{tr}(g^2Q) - \mathrm{tr}(gQgQ)\big]$. Since $\mathbb{E}_V[Q] = (m/n)I_n$, the first term averages to $(m/n)\mathrm{tr}(g^2)$. For the second, $\mathrm{tr}(gQgQ) = \mathrm{tr}\big[(g\otimes g)(Q\otimes Q)S\big]$ with S the swap on $\mathbb{C}^n\otimes\mathbb{C}^n$, and $\mathbb{E}_V[Q\otimes Q]$ is invariant under $U\otimes U$ for every $U \in U(n)$, hence equals $aI + bS$ by Schur–Weyl duality. Pairing with I and with S gives $an^2 + bn = (\mathrm{tr}Q)^2 = m^2$ and $an + bn^2 = \mathrm{tr}(Q^2) = m$, whence

$$a = \frac{m(mn-1)}{n(n^2-1)}, \qquad b = \frac{m(n-m)}{n(n^2-1)}. \tag{A11}$$

Since $\mathrm{tr}[(g\otimes g)S] = \mathrm{tr}(g^2)$ and $\mathrm{tr}(g\otimes g) = (\mathrm{tr}g)^2$, we get $\mathbb{E}_V\,\mathrm{tr}(gQgQ) = a\,\mathrm{tr}(g^2) + b\,(\mathrm{tr}g)^2$, and therefore

$$\mathbb{E}_V\|[g,Q]\|_F^2 = 2\Big[\big(\tfrac{m}{n} - a\big)\mathrm{tr}(g^2) - b\,(\mathrm{tr}g)^2\Big]. \tag{A12}$$

Finally $\frac{m}{n} - a = m(n-m)/(n^2-1)$ and $b = \frac{1}{n}\,m(n-m)/(n^2-1)$, which is (A10). As a check, at $m = n$ (so $Q = I$) both sides vanish. □

2. Proof of Theorem 9

Proof. Fix a direction $g \in \mathfrak{u}(n)$ and let $W \sim \mathrm{Haar}(U(n))$. Right invariance of the Haar measure makes We^{-isg} Haar-distributed for every s, so $s \mapsto \mathbb{E}[\mathcal{L}(We^{-isg})]$ is constant and $\mathbb{E}[D_g\mathcal{L}] = 0$. We do *not* split W into independent factors—the partial products of Givens rotations before and after a fixed gate are not individually Haar-distributed—and instead use that the Haar measure on $U(n)$ is an exact unitary 2-design on every polynomial irreducible representation, applying the resulting second-moment identity directly on the two-particle module $\Lambda^2\mathbb{C}^n$. This is the dynamical-Lie-algebra variance computation of Fontana et al. [36] and Larocca et al. [37] specialized to the antisymmetric module.

Since $M = n_i n_j$ is a two-body observable, its parameter-dependent expectation value is governed entirely by the action of W on $\Lambda^2\mathbb{C}^n$. Writing $M^{(2)} = |\{i,j\}\rangle\langle\{i,j\}|$ for the image of M on $\Lambda^2\mathbb{C}^n$ and $\rho_2(\mathbf{x})$ for the two-particle reduced density matrix of $|\psi(\mathbf{x})\rangle$, the invariant derivative is

$$D_g\mathcal{L} \;=\; i\,\mathrm{tr}_{\Lambda^2}\big([G^{(2)},\, \Lambda^2(W)^\dagger M^{(2)}\Lambda^2(W)]\,\rho_2(\mathbf{x})\big). \tag{A13}$$

By cyclicity of the trace, $D_g\mathcal{L} = -i\,\mathrm{tr}_{\Lambda^2}(\Lambda^2(W)^\dagger M^{(2)}\Lambda^2(W)\,B)$ with $B = [G^{(2)}, \rho_{2,\mathfrak{g}_2}(\mathbf{x})]$: the identity component of ρ_2 commutes with $G^{(2)}$ and drops out. This moves the commutator onto the state factor and lets the second-moment identity act directly on $M^{(2)\otimes 2}$. The variance over $W \sim \mathrm{Haar}(U(n))$ is then

$$\mathrm{Var}[D_g\mathcal{L}] \;= \\ -\,\mathbb{E}_{W\sim\mathrm{Haar}(U(n))}\Big[\mathrm{tr}\Big(\big(\Lambda^2(W)^\dagger M^{(2)}\Lambda^2(W)\big)^{\otimes 2} B^{\otimes 2}\Big)\Big]. \tag{A14}$$

Step 1: $U(n)$-Weingarten formula on Λ^2. Since W is exactly Haar-distributed on $U(n)$, the $U(n)$-Weingarten formula applies to its action on $\Lambda^2\mathbb{C}^n$. As a $U(n)$-representation,

$$\Lambda^2\mathbb{C}^n \otimes \Lambda^2\mathbb{C}^n \;=\; V_{(2,2)} \oplus V_{(2,1,1)} \oplus V_{(1,1,1,1)}, \tag{A15}$$

with dimensions (hook length formula)

$$D_{(2,2)} := \dim V_{(2,2)} = \tfrac{n^2(n^2-1)}{12}, \tag{A16}$$

$$D_{(2,1,1)} := \dim V_{(2,1,1)} = \tfrac{n(n+1)(n-1)(n-2)}{8}, \tag{A17}$$

$$D_{\Lambda^4} := \dim V_{(1,1,1,1)} = \tbinom{n}{4}, \tag{A18}$$

all $\Theta(n^4)$ (at $n=4$: $20+15+1 = 36 = \binom{4}{2}^2$). Let P_λ be the orthogonal projectors onto these irreducibles. By Schur's lemma the Haar second-moment operator is $\mathbb{E}_V[\Lambda^2(V)^{\otimes 2} Y \Lambda^2(V)^{\dagger\otimes 2}] = \sum_\lambda \frac{\mathrm{tr}(P_\lambda Y)}{\dim V_\lambda} P_\lambda$, exact with no approximation error. Applying this to the $M^{(2)\otimes 2}$ factor in (A14) and pairing with $B^{\otimes 2}$ via $\mathrm{tr}(P_\lambda P_\mu) = \delta_{\lambda\mu}\dim V_\lambda$ yields the exact sum

$$\mathrm{Var}[D_g\mathcal{L}] \;=\; -\sum_\lambda \frac{\mathrm{tr}\big(P_\lambda\, M^{(2)\otimes 2}\big)\,\mathrm{tr}\big(P_\lambda\, B^{\otimes 2}\big)}{\dim V_\lambda}. \tag{A19}$$

The generator enters *only* through B; there is no separate scalar prefactor.

We record once the two facts used to evaluate the three weights. Let S be the exchange of the two Λ^2 factors and $P_{\mathrm{sym}} = \frac{1}{2}(\mathbf{1}+S)$, $P_{\mathrm{anti}} = \frac{1}{2}(\mathbf{1}-S)$. By the plethysm $\mathrm{Sym}^2(\Lambda^2\mathbb{C}^n) = V_{(2,2)} \oplus \Lambda^4\mathbb{C}^n$ and $\Lambda^2(\Lambda^2\mathbb{C}^n) = V_{(2,1,1)}$ (dimension check at $n=4$: $20+1 = 21 = \binom{6+1}{2}$ and 15), so

$$P_{(2,1,1)} = P_{\mathrm{anti}}, \qquad P_{(2,2)} = P_{\mathrm{sym}} - P_{\Lambda^4}. \tag{A20}$$

Consequently, for *any* $X \in \mathrm{End}(\Lambda^2\mathbb{C}^n)$,

$$\begin{aligned} \mathrm{tr}\big(P_{(2,1,1)}X^{\otimes 2}\big) &= \tfrac{1}{2}\big[(\mathrm{tr}X)^2 - \mathrm{tr}(X^2)\big], \\ \mathrm{tr}\big(P_{\Lambda^4}X^{\otimes 2}\big) &= \tfrac{1}{6}\big[(\mathrm{tr}X)^2 + \mathrm{tr}(X^2) - \mathrm{tr}\big(\Gamma(X)^2\big)\big], \\ \mathrm{tr}\big(P_{(2,2)}X^{\otimes 2}\big) &= \tfrac{1}{3}\big[(\mathrm{tr}X)^2 + \mathrm{tr}(X^2)\big] + \tfrac{1}{6}\mathrm{tr}\big(\Gamma(X)^2\big), \end{aligned} \tag{A21}$$

the first from $\mathrm{tr}(SX^{\otimes 2}) = \mathrm{tr}(X^2)$, the second from Lemma 40 with $M = N = X$, and the third from (A20). As a check, at $X = I_{\Lambda^2}$ the three right-hand sides return $D_{(2,1,1)}$, D_{Λ^4} and $D_{(2,2)}$ respectively, for every n. We stress that the $V_{(2,1,1)}$ weight in (A21) does *not* vanish merely because $X^{\otimes 2}$ commutes with S; it vanishes precisely when $(\mathrm{tr}X)^2 = \mathrm{tr}(X^2)$.

Step 2: the observable weight sits on $V_{(2,2)}$. $M^{(2)} = |\{i,j\}\rangle\langle\{i,j\}|$ is a rank-one projector, so $\mathrm{tr}M^{(2)} = \mathrm{tr}(M^{(2)2}) = 1$ and, by Lemma 41's contraction, $\Gamma(M^{(2)}) = \hat{n}_i + \hat{n}_j$, whence $\mathrm{tr}(\Gamma(M^{(2)})^2) = 2$. Substituting into (A21),

$$\mathrm{tr}\big(P_{(2,1,1)}M^{(2)\otimes 2}\big) \;=\; \tfrac{1}{2}[1-1] = 0, \tag{A22}$$

$$\mathrm{tr}\big(P_{\Lambda^4}M^{(2)\otimes 2}\big) \;=\; \tfrac{1}{6}[1+1-2] = 0, \tag{A23}$$

and $\mathrm{tr}(P_{(2,2)}M^{(2)\otimes 2}) = \frac{1}{3}[1+1] + \frac{1}{6}\cdot 2 = 1$. It is the rank-one *projector* property $(\mathrm{tr}M^{(2)})^2 = \mathrm{tr}(M^{(2)2})$, and not any exchange symmetry of $M^{(2)\otimes 2}$, that annihilates the $V_{(2,1,1)}$ channel; the general two-body readout of Proposition 12 does not have this property. Therefore (A19) collapses to the single term

$$\mathrm{Var}[D_g\mathcal{L}] = -\frac{\mathrm{tr}\big(P_{(2,2)}B^{\otimes 2}\big)}{D_{(2,2)}} = -\frac{12\,\mathrm{tr}\big(P_{(2,2)}B^{\otimes 2}\big)}{n^2(n^2-1)}. \tag{A24}$$

Step 3: evaluation of the $(2,2)$ weight of $B^{\otimes 2}$. B is anti-Hermitian and traceless, so $\mathrm{tr}B = 0$, $\mathrm{tr}(B^2) = -\|B\|_F^2$, and, $\Gamma(B)$ being anti-Hermitian as well (Lemma 41), $\mathrm{tr}(\Gamma(B)^2) = -\|\Gamma(B)\|_F^2$. Substituting $X = B$ into (A21) gives all three weights at once:

$$\begin{aligned} \mathrm{tr}\big(P_{(2,1,1)}B^{\otimes 2}\big) &= \tfrac{1}{2}\|B\|_F^2, \\ \mathrm{tr}\big(P_{\Lambda^4}B^{\otimes 2}\big) &= \tfrac{1}{6}\big[\|\Gamma(B)\|_F^2 - \|B\|_F^2\big], \\ \mathrm{tr}\big(P_{(2,2)}B^{\otimes 2}\big) &= -\tfrac{1}{3}\|B\|_F^2 - \tfrac{1}{6}\|\Gamma(B)\|_F^2. \end{aligned} \tag{A25}$$

The $V_{(2,1,1)}$ weight of $B^{\otimes 2}$ is thus *nonzero*; it is eliminated from (A19) by the observable, through the first identity

of (A22), and not by any property of B. Substituting the last line of (A25) into (A24) gives the exact identity (22).

Step 4: the one-body term. By Lemma 41 and the contraction identity $\Gamma(\rho_2) = (k-1)\rho_1$ of Eq. (19)—the identity component of ρ_2 contributing only a multiple of I_n, which commutes with g—we obtain $\Gamma(B) = [g, \Gamma(\rho_{2,\mathfrak{g}_2})] = (k-1)[g, \rho_1(\mathbf{x})]$. With $\rho_1(\mathbf{x}) = \frac{1}{2}\Pi$ this is $\frac{k-1}{2}[g,\Pi]$, so $\|\Gamma(B)\|_F^2 = \frac{(k-1)^2}{4}\tau$ with $\tau = \|[g,\Pi]\|_F^2$. For a rank-one phase generator $g = \hat{n}_j$ (so $g^2 = g$),

$$\begin{aligned}\tau &= 2\big[\mathrm{tr}(g\Pi) - \mathrm{tr}(g\Pi g\Pi)\big] = 2\,\Pi_{jj}(1-\Pi_{jj}) \\ &= \frac{2\cdot 2k(n-2k)}{n^2},\end{aligned} \tag{A26}$$

using $\Pi_{jj} = 2k/n$ (Lemma 2). For an RBS generator $g = G_{ij}$ on any mode pair $\{i,j\}$ (so g^2 is the projector onto $\mathrm{span}\{e_i, e_j\}$),

$$\begin{aligned}\tau &= 2\big[\Pi_{ii} + \Pi_{jj} - 2\Pi_{ii}\Pi_{jj} + 2\,\mathrm{Re}\,\Pi_{ij}^2\big] \\ &= \frac{4\cdot 2k(n-2k)}{n^2} + 4\,\mathrm{Re}\,\Pi_{ij}^2.\end{aligned} \tag{A27}$$

The off-diagonal entry is a Walsh–Hadamard character sum: before the diagonal data layer $\Pi = H\Pi_A H$ with Π_A the projector onto the $2k$ active modes, so $\Pi_{ij} = \frac{1}{n}\sum_{c<2k}(-1)^{\langle \mu,\, c\rangle}$ with $\mu := i \oplus j$, and the data layer only multiplies it by a phase. Let 2^b be the lowest set bit of μ and write $2k = q\,2^{b+1} + r$, $0 \le r < 2^{b+1}$. The involution $c \leftrightarrow c \oplus 2^b$ flips $(-1)^{\langle \mu, c\rangle}$ and preserves each full block of 2^{b+1} consecutive values of c, so those blocks cancel term by term and only the residue survives, giving

$$|\Pi_{ij}| \;=\; \tfrac{1}{n}\,\big|\min(r,\, 2^{b+1} - r)\big| \;\le\; \frac{\min(2^b,\, 2k)}{n}, \tag{A28}$$

$$\Pi_{ij} \;=\; 0 \quad \text{whenever } 2^{b+1} \mid 2k. \tag{A29}$$

For the brick-wall every trainable RBS gate acts on a nearest-neighbour pair of the ring: for a chain pair, $\mu = i \oplus (i{+}1)$ has $b = 0$; for the boundary pair $(n,1)$, the binary labels $n-1$ and 0 give $\mu = 2^K - 1$, again with $b = 0$. Since $2 \mid 2k$ always, $\Pi_{ij} = 0$ identically and (A27) reduces to $\tau = 8k(n-2k)/n^2$ exactly, for every RBS gate of the circuit. (For the long-stride pairs of the butterfly, $\mu = 2^b$ and (A28) gives $\Pi_{ij} = 0$ whenever $2^{b+1} \mid 2k$, and $|\Pi_{ij}| \le 2^b/n \le 2k/n$ otherwise; this is used in Claim 2 of Section IV C.) In every case $\tau \ge 8k(n-2k)/n^2 - 4|\Pi_{ij}|^2$, so $\tau = \Theta\big(k(n-2k)/n^2\big)$ whenever $n - 2k = \Omega(n)$ and $|\Pi_{ij}|^2 \le c\,k(n-2k)/n^2$ for a sufficiently small constant c; for the brick-wall $\Pi_{ij} = 0$ and the evaluation is exact and unconditional.

Step 5: the two-body term. Decompose the 2-RDM into its disconnected (Hartree–Fock) and connected (cumulant) parts, $\rho_2 = \rho_1 \wedge \rho_1 + \rho_2^{\mathrm{conn}}$, and correspondingly $B = B^{\mathrm{disc}} + B^{\mathrm{conn}}$. Since $\rho_1 \wedge \rho_1 = \frac{1}{4}\Lambda^2(\Pi)$ and, by functoriality of the exterior square, $[\,d\Lambda^2(g), \Lambda^2(\Pi)\,] = \partial_u \Lambda^2(\Pi + uC)|_{u=0}$ with $C := [g,\Pi]$, we may evaluate $\mathrm{tr}\big((B^{\mathrm{disc}})^2\big)$ by polarizing the identity $\mathrm{tr}\big(\Lambda^2(A)\Lambda^2(A')\big) = e_2(AA') = \frac{1}{2}[(\mathrm{tr}AA')^2 - \mathrm{tr}((AA')^2)]$ at $A = \Pi + uC$, $A' = \Pi + vC$. Using $\Pi^2 = \Pi$, $\mathrm{tr}C = \mathrm{tr}(\Pi C) = 0$ and $\mathrm{tr}(\Pi C^2) = \frac{1}{2}\mathrm{tr}(C^2) = -\frac{\tau}{2}$, one finds $\partial_u\partial_v(\mathrm{tr}AA')^2 = -2m\tau$ and $\partial_u\partial_v \mathrm{tr}((AA')^2) = 4\mathrm{tr}(\Pi C^2) = -2\tau$ with $m = 2k$ (the second using $\mathrm{tr}(C\Pi C) = \mathrm{tr}(\Pi C^2)$ by cyclicity), whence

$$\big\|B^{\mathrm{disc}}\big\|_F^2 = -\mathrm{tr}\big((B^{\mathrm{disc}})^2\big) = \frac{(m-1)\,\tau}{16} = \frac{(2k-1)\,\tau}{16}, \tag{A30}$$

exactly and independently of $\mathbf{x}$. For the connected part, Lemma 39 with $r = \Theta(k)$ gives $\|B^{\mathrm{conn}}\|_F^2 = O(k/n)$; the same lemma applied with $r = \binom{2k}{2}$ confirms $\|B^{\mathrm{disc}}\|_F^2 = O(k^2/n)$, consistent with the exact value. The error term in (24) is dominated by the cross term: by Cauchy–Schwarz, $2|\langle B^{\mathrm{disc}}, B^{\mathrm{conn}}\rangle| \le 2\|B^{\mathrm{disc}}\|_F \|B^{\mathrm{conn}}\|_F = O\big(k/\sqrt{n}\big)\cdot O\big(\sqrt{k/n}\big) = O(k^{3/2}/n)$, so that

$$\Big|\, \|B\|_F^2 - \tfrac{(2k-1)\tau}{16} \,\Big| \;=\; O\!\left(\frac{k^{3/2}}{n}\right), \tag{A31}$$

which is the order recorded in (24) and is negligible against the leading $\Theta(k^2/n)$ whenever $k = \omega(1)$ (and against the one-body term $\Theta(k^3/n)$ throughout).

Step 6: the rate. Substituting (24) into (22) and using $\tau = \Theta(k(n-2k)/n^2)$,

$$\begin{aligned}4\|B\|_F^2 + 2\|\Gamma(B)\|_F^2 &= \frac{\tau}{4}\Big[(2k-1) + 2(k-1)^2\Big] + O\Big(\tfrac{k^{3/2}}{n}\Big) \\ &= \Theta\!\left(\frac{k^3\,(n-2k)}{n^2}\right),\end{aligned} \tag{A32}$$

so that for $n - 2k = \Omega(n)$ the numerator is $\Theta(k^3/n)$ and $\mathrm{Var}[D_g\mathcal{L}] = \Theta(k^3/n^5)$, which is (25). We note that the two-sided bound holds for every $k \ge 2$, including $k = O(1)$, where the error term in (A31) is of the same order as the two-body signal: the lower bound uses no error control, since $4\|B\|_F^2 \ge 0$ and the one-body term is exact,

$$4\|B\|_F^2 + 2\|\Gamma(B)\|_F^2 \ge 2\,\|\Gamma(B)\|_F^2 = \frac{(k-1)^2}{2}\,\tau \tag{A33}$$

$$= \Theta\!\left(\frac{k^3(n-2k)}{n^2}\right), \tag{A34}$$

while for the upper bound the error term $O(k^{3/2}/n)$ of (A31) is itself $O(k^3/n)$ for every $k \ge 2$ and is absorbed. The quantities τ, $\|B^{\mathrm{disc}}\|_F^2$ and $\|\Gamma(B)\|_F^2$ depend on the encoding only through $\rho_1 = \frac{1}{2}\Pi$, with Π the rank-$2k$ projector of Lemma 2.

□

3. Proof of Proposition 12

Proof. Equation (A19) and the three weights (A21) of Step 1 hold for arbitrary $M^{(2)}$; only Step 2 used the rank-one structure. Adding a multiple of I_{Λ^2} to $M^{(2)}$

shifts $\mathcal{L}$ by a constant and leaves $D_g\mathcal{L}$ unchanged, so we may replace $M^{(2)}$ by $\tilde{M}$ throughout. With $\mathrm{tr}\tilde{M}=0$, $\mathrm{tr}(\tilde{M}^2)=p$ and $\mathrm{tr}(\Gamma(\tilde{M})^2)=q$ (both $\tilde{M}$ and $\Gamma(\tilde{M})$ being Hermitian), (A21) gives

$$t_{(2,1,1)}=-\tfrac{p}{2},\qquad t_{\Lambda^4}=\tfrac{1}{6}(p-q),\qquad t_{(2,2)}=\tfrac{p}{3}+\tfrac{q}{6},$$

while (A25) gives $s_{(2,1,1)}=\frac{1}{2}\|B\|_F^2$, $s_{\Lambda^4}=\frac{1}{6}(\|\Gamma(B)\|_F^2-\|B\|_F^2)$ and $s_{(2,2)}=-\frac{1}{3}\|B\|_F^2-\frac{1}{6}\|\Gamma(B)\|_F^2$. Substituting into $\mathrm{Var}=-\sum_\lambda t_\lambda s_\lambda/D_\lambda$ and collecting the coefficients of $\|B\|_F^2$ and $\|\Gamma(B)\|_F^2$ gives

$$c_2 \;=\; \frac{t_{(2,2)}}{3D_{(2,2)}}-\frac{t_{(2,1,1)}}{2D_{(2,1,1)}}+\frac{t_{\Lambda^4}}{6D_{\Lambda^4}}, \tag{A35}$$

$$c_1 \;=\; \frac{1}{6}\left(\frac{t_{(2,2)}}{D_{(2,2)}}-\frac{t_{\Lambda^4}}{D_{\Lambda^4}}\right), \tag{A36}$$

and clearing denominators with $D_{(2,2)}=\frac{n^2(n^2-1)}{12}$, $D_{(2,1,1)}=\frac{n(n+1)(n-1)(n-2)}{8}$, $D_{\Lambda^4}=\frac{n(n-1)(n-2)(n-3)}{24}$ yields (29): the numerator of c_1 over the common denominator $n^2(n^2-1)(n-2)(n-3)$ is $\frac{1}{6}\big[(4p+2q)(n-2)(n-3)-4(p-q)n(n+1)\big]=q(n^2-n+2)-4p(n-1)$, and that of c_2 over $n^2(n+1)(n-2)(n-3)$ is $4[(n-2)p-q]$.

(i) is a direct substitution (for $M=n_in_j$ use $\mathrm{tr}M^{(2)}=1$, $\mathrm{tr}(M^{(2)2})=1$, $\mathrm{tr}(\Gamma(M^{(2)})^2)=2$ and subtract the trace part).

(ii) follows by inserting $\|\Gamma(B)\|_F^2=\frac{(k-1)^2}{4}\tau$ and $\|B\|_F^2=\frac{(2k-1)\tau}{16}+O(k^{3/2}/n)$ from Steps 4–5 into (28); the quoted values for the two example heads are the substitution of p,q into (29), and the resulting rates follow from $\tau=\Theta(k/n)$ together with $2k\le n$.

(iii) Writing the bracket over the common denominator $n^2(n^2-1)(n-2)(n-3)$, its numerator equals

$$\begin{aligned}&4p(n-1)\big[(2k-1)(n-2)-4(k-1)^2\big]\\&+4q\big[(k-1)^2(n^2-n+2)-(2k-1)(n-1)\big].\end{aligned} \tag{A37}$$

For $k\ge 2$ and $n\ge 2k+2$ both square brackets are positive: the first because $\frac{4(k-1)^2}{2k-1}<2k\le n-2$ (equivalently $6k>4$), and the second because $(k-1)^2(n^2-n+2)-(2k-1)(n-1)\ge(n-1)[(k-1)^2n-(2k-1)]>0$ using $(k-1)^2n\ge n\ge 2k+2$. Since $q\ge 0$, dropping the q term gives (31). For $n-2k=\Omega(n)$ we have $(2k-1)(n-2)-4(k-1)^2=\Omega(kn)$ and $\tau=\Theta(k/n)$, so the leading term of (30) is $\Omega(p\,k^2/n^5)$. Finally $q\le 4np$ (Cauchy–Schwarz on $\Gamma(\tilde{M})^i{}_k=\sum_j\tilde{M}^{ij}_{kj}$), so $|c_2|=O(p/n^4)$ and the correction term in (30) is $O(p\,k^{3/2}/n^5)$, smaller than the leading term by a factor $O(k^{-1/2})$; it is therefore absorbed for k above a constant. □

4. Interior gates

Lemma 44 (The dominant terms depend on the generator only through τ)**.** *Let h be* any *Hermitian single-particle operator, put $H^{(2)}=d\Lambda^2(h)$, $B_h=[H^{(2)},\rho_{2,\mathfrak{g}_2}(\mathbf{x})]$ and $\tau_h:=\|[h,\Pi]\|_F^2$. Then, exactly and for every $\mathbf{x}$,*

$$\|\Gamma(B_h)\|_F^2=\frac{(k-1)^2}{4}\,\tau_h,\qquad \|B_h^{\mathrm{disc}}\|_F^2=\frac{(2k-1)\,\tau_h}{16}, \tag{A38}$$

where $B_h^{\mathrm{disc}}=[H^{(2)},\rho_1\wedge\rho_1]$.

Proof. Steps 4 and 5 of the proof of Theorem 9 use exactly three properties of $C:=[h,\Pi]$, namely $\mathrm{tr}C=0$, $\mathrm{tr}(\Pi C)=0$ and $\mathrm{tr}(\Pi C^2)=\frac{1}{2}\mathrm{tr}(C^2)$. All three follow from $\Pi^2=\Pi$ and cyclicity of the trace for an arbitrary Hermitian h: the first two are immediate, and expanding C^2 gives $\mathrm{tr}(\Pi C^2)=\mathrm{tr}(\Pi h\Pi h)-\mathrm{tr}(\Pi h^2)=\frac{1}{2}\mathrm{tr}(C^2)$. No property of h beyond Hermiticity—in particular no locality in the computational basis—enters either computation. □

Proof. Since $U_\theta=e^{-i\theta g}$ commutes with g, we have $\partial_\theta W=-i\,W\tilde{g}$ with $\tilde{g}=W_1^\dagger gW_1$; that is, $\partial_\theta\mathcal{L}=D_{\tilde{g}}\mathcal{L}$ in the notation of (20). Condition on W_1. Writing $W=W_2\,(U_\theta W_1)$ with W_2 Haar and independent of W_1, the total single-particle unitary W is, conditionally on W_1, Haar-distributed on $U(n)$ and independent of the (now fixed) direction $\tilde{g}$. Hence Theorem 9 applies conditionally and yields $\mathbb{E}[\partial_\theta\mathcal{L}\mid W_1]=0$ together with $\mathrm{Var}[\partial_\theta\mathcal{L}\mid W_1]=\big(4\|B_{\tilde{g}}\|_F^2+2\|\Gamma(B_{\tilde{g}})\|_F^2\big)/(n^2(n^2-1))$. Because the conditional mean vanishes identically, the law of total variance reduces to $\mathrm{Var}=\mathbb{E}_{W_1}\big[\mathrm{Var}[\,\cdot\mid W_1]\big]$, which is (32).

It remains to average the two norms. By Lemma 44 with $h=\tilde{g}$, $\|\Gamma(B_{\tilde{g}})\|_F^2=\frac{(k-1)^2}{4}\tau_{\tilde{g}}$ and $\|B_{\tilde{g}}^{\mathrm{disc}}\|_F^2=(2k-1)\tau_{\tilde{g}}/16$ with $\tau_{\tilde{g}}=\|[\tilde{g},\Pi]\|_F^2=\|[g,W_1\Pi W_1^\dagger]\|_F^2$. Lemma 43 with $m=2k$ then gives $\mathbb{E}_{W_1}[\tau_{\tilde{g}}]=\Theta\big(k(n-2k)/n^2\big)$ for both generator families, with the same constants as the unconjugated values of Step 4, so $\mathbb{E}\|\Gamma(B_{\tilde{g}})\|_F^2=\Theta(k^3/n)$ and $\mathbb{E}\|B_{\tilde{g}}^{\mathrm{disc}}\|_F^2=\Theta(k^2/n)$. For the connected part, Lemma 42 applied to $\sigma=\rho_2^{\mathrm{conn}}$, with $\|\rho_2^{\mathrm{conn}}\|_F^2=\Theta(k)$, gives $\mathbb{E}\|B_{\tilde{g}}^{\mathrm{conn}}\|_F^2=O(k/n)$, and by Cauchy–Schwarz the cross term is at most $2\big(\mathbb{E}\|B^{\mathrm{disc}}\|_F^2\big)^{1/2}\big(\mathbb{E}\|B^{\mathrm{conn}}\|_F^2\big)^{1/2}=O(k^{3/2}/n)$. Collecting, $4\mathbb{E}\|B_{\tilde{g}}\|_F^2+2\mathbb{E}\|\Gamma(B_{\tilde{g}})\|_F^2=\Theta(k^3/n)$ whenever $n-2k=\Omega(n)$, and dividing by $n^2(n^2-1)$ gives the stated rate. □

Proof. (a) and (b) restate Corollary 11 and Theorem 13. For (c) we induct on the column index, the base case $\Pi^{(1)}=\Pi$ having uniform diagonal $2k/n$ by Lemma 2. Assume $\mathbb{E}[\Pi_{jj}^{(\ell)}]=2k/n$ for all j. Column ℓ consists of phases followed by RBS rotations on disjoint mode pairs. Conjugation by a diagonal phase matrix leaves the diagonal unchanged and multiplies $\Pi_{ab}^{(\ell)}$ by $e^{i(\varphi_a-\varphi_b)}$; since each Givens rotation of the column carries its own phase, uniform and independent of everything preceding it, the off-diagonal entry of each rotated pair has zero conditional expectation. An RBS rotation of angle ϑ on the pair $\{a,b\}$ sends (Π_{aa},Π_{bb}) to $(\cos^2\vartheta\,\Pi_{aa}+\sin^2\vartheta\,\Pi_{bb}\mp$

$\sin 2\vartheta\,\mathrm{Re}\,\Pi_{ab},\ \sin^2\vartheta\,\Pi_{aa}+\cos^2\vartheta\,\Pi_{bb}\pm\sin 2\vartheta\,\mathrm{Re}\,\Pi_{ab}$) and leaves every other diagonal entry untouched. Taking expectations, the cross terms drop out and the pair map becomes a doubly stochastic average of (Π_{aa},Π_{bb}), whatever the marginal law of ϑ. A constant vector is a fixed point of every doubly stochastic map, so $\mathbb{E}[\Pi_{jj}^{(\ell+1)}] = 2k/n$ for all j. □

5. Proof of Theorem 17

Proof. Identify the modes with $(\mathbb{Z}_2)^K$ through their binary labels, so that layer ℓ pairs $b \leftrightarrow b\oplus e_{\ell-1}$, and write $s := e_{K-1}$ for the stride of the last layer.

Step 1: Parseval. $\mathcal{L}$ is a trigonometric polynomial in the full parameter vector $\Theta = (\boldsymbol{\theta},\boldsymbol{\phi}) \in \mathbb{T}^{\tilde{T}}$; write $\mathcal{L} = \sum_{\boldsymbol{\delta}} c_{\boldsymbol{\delta}}\, e^{i\boldsymbol{\delta}\cdot\Theta}$. Since $\mathcal{L}$ is periodic in each coordinate, $\mathbb{E}_\Theta[\partial_\theta\mathcal{L}] = 0$, and by Parseval

$$\mathrm{Var}\left[\frac{\partial\mathcal{L}}{\partial\theta}\right] \;=\; \sum_{\boldsymbol{\delta}} \delta_\theta^2\,|c_{\boldsymbol{\delta}}|^2 \;\geq\; 2\,(\delta_\theta^*)^2\,|c_{\boldsymbol{\delta}^*}|^2 \tag{A39}$$

for any single frequency vector $\boldsymbol{\delta}^* \neq 0$ with $\delta_\theta^* \neq 0$, the factor 2 accounting for the conjugate pair $\pm\boldsymbol{\delta}^*$ ($\mathcal{L}$ being real). The variance of the gradient is thereby reduced to a *first* moment of $\mathcal{L}$; no second-moment property of the circuit enters.

Step 2: choice of the frequency vector. Fix $i \neq j$ with $j \notin \{i, i\oplus s\}$, put $M = n_i n_j$, and write $i' = i\oplus s$, $j' = j\oplus s$. Let θ be the RBS angle of the layer-K gate acting on $\{i,i'\}$, let θ' be the RBS angle of the layer-K gate acting on $\{j,j'\}$, and let $\boldsymbol{\delta}^*$ carry frequency 2 on each of θ,θ' and 0 elsewhere.

These are the only gates of layer K whose modes meet $\{i,j\}$, so with $c_A = \cos\theta$, $s_A = \sin\theta$, $c_B = \cos\theta'$, $s_B = \sin\theta'$,

$$\begin{aligned}\Lambda^2\big(U^{(K)}\big)^\dagger\,|i\wedge j\rangle = c_A c_B\,|i\wedge j\rangle + c_A s_B\,|i\wedge j'\rangle \\ + s_A c_B\,|i'\wedge j\rangle + s_A s_B\,|i'\wedge j'\rangle\,.\end{aligned} \tag{A40}$$

Let σ denote the two-particle reduced density matrix entering the RBS sublayer of layer K, i.e. after layers $1,\dots,K-1$ and after the full-width phase layer of layer K, and let $\bar\sigma = \mathbb{E}[\sigma]$, the expectation over all parameters other than θ,θ'. The phase layer averages every off-diagonal entry of σ to zero, so $\bar\sigma$ is diagonal in the pair basis and, writing $\bar\sigma_{ab} := \bar\sigma_{\{a,b\},\{a,b\}}$,

$$\begin{aligned}\mathbb{E}\big[\mathcal{L}\big] = c_A^2 c_B^2\,\bar\sigma_{ij} + c_A^2 s_B^2\,\bar\sigma_{ij'} \\ + s_A^2 c_B^2\,\bar\sigma_{i'j} + s_A^2 s_B^2\,\bar\sigma_{i'j'}.\end{aligned} \tag{A41}$$

Using $\mathbb{E}_\theta[\cos^2\theta\,e^{-2i\theta}] = \frac14$ and $\mathbb{E}_\theta[\sin^2\theta\,e^{-2i\theta}] = -\frac14$,

$$c_{\boldsymbol{\delta}^*} = \tfrac{1}{16}\,D, \qquad D := \bar\sigma_{ij} - \bar\sigma_{ij'} - \bar\sigma_{i'j} + \bar\sigma_{i'j'}. \tag{A42}$$

The certificate is a *second* difference of the first moment $\bar\sigma$ across the stride of the last layer; the single differences, which correspond to a frequency on one gate only, cancel.

Step 3: only the $y = 0$ Fourier block contributes. In the notation of Appendix B, parametrize a pair by base point and difference, $P_{(b,\delta)} = P_{\{b,\,b\oplus\delta\}}$, and decompose along the characters $\hat P_{\delta,y}$ of the translation group $(\mathbb{Z}_2)^K$. The four terms of (A42) carry the base points i and $i\oplus s$ with equal signs, so only characters with $y\cdot s = 0$ survive the base sum. Layers $1,\dots,K-1$ annihilate $\mathcal{B}_y$ for every y with $y\cdot e_{\ell-1} = 1$, $\ell \leq K-1$; together with $y_{K-1} = 0$ this leaves only $y = 0$. Consequently, for the purposes of D, the weight $\bar\sigma_{(b,\delta)}$ is the uniform average $\frac{2}{n}F^{(K-1)}(\delta)$ of the base-summed weights

$$F^{(m)}(\delta) := \sum_b \mathbb{E}\big[\sigma^{(m)}_{\{b,\,b\oplus\delta\}}\big], \qquad \sum_{\delta\neq0} F^{(0)}(\delta) = \tbinom{k}{2}, \tag{A43}$$

where for each $\delta \neq 0$ the sum over b runs over a transversal of the involution $b \leftrightarrow b\oplus\delta$, i.e. over one representative per unordered pair $\{b, b\oplus\delta\}$, so that each pair is counted once; and, with $d := i\oplus j$,

$$D = \frac{4}{n}\Big[F^{(K-1)}(d) - F^{(K-1)}(d\oplus s)\Big]. \tag{A44}$$

On the $y = 0$ block the layer action is the scalar recursion of Appendix B(iii): layer ℓ fixes $F(e_{\ell-1})$ and replaces every other $F(\delta)$ by $\frac12[F(\delta) + F(\delta\oplus e_{\ell-1})]$.

Step 4: the two top-bit sectors carry equal weight. Split the differences by their top bit and set

$$T_0 := \sum_{\delta\neq0,\ \delta_{K-1}=0} F^{(0)}(\delta), \qquad T_1 := \sum_{\delta_{K-1}=1} F^{(0)}(\delta), \tag{A45}$$

so that $T_0 + T_1 = \binom{k}{2}$. Writing N_0 for the number of particles of the encoded state in the half $\{a : a_{K-1} = 0\}$, one has $T_1 = \mathbb{E}[N_0(k-N_0)]$ and $T_0 = \binom{k}{2} - T_1$. Let P be the single-particle projector onto that half. Since $H_{aj}H_{al} = \frac1n(-1)^{\langle a,\,j\oplus l\rangle}$, and the sum of $(-1)^{\langle a,\mu\rangle}$ over the $n/2$ labels a with $a_{K-1} = 0$ equals $\frac{n}{2}$ when $\mu \in \{0,s\}$ and 0 otherwise,

$$H^T P H = \tfrac12\big(I + \Sigma\big), \qquad \Sigma : |j\rangle \mapsto |j\oplus s\rangle\,, \tag{A46}$$

so that on the k-particle sector $N_0 = \frac12(k + \hat\Sigma)$, with $\hat\Sigma$ the one-body operator of Σ evaluated on the *pre-spread* magic state. That state occupies only the modes $0,\dots,2k-1$, and $4k \leq n$ gives $2k \leq n/2$, so Σ maps every occupied mode to an unoccupied one. Hence $\langle\Psi_{\mathrm{in}}|\,\hat\Sigma\,|\Psi_{\mathrm{in}}\rangle = 0$, while $\hat\Sigma\,|\Psi_{\mathrm{in}}\rangle$ is, within each Fock component, a superposition of k mutually orthogonal states of unit norm, giving $\langle\Psi_{\mathrm{in}}|\,\hat\Sigma^2\,|\Psi_{\mathrm{in}}\rangle = k$. Therefore $\mathbb{E}[N_0] = k/2$, $\mathbb{E}[N_0^2] = \frac14 k(k+1)$, and

$$T_0 \;=\; T_1 \;=\; \frac{k(k-1)}{4}. \tag{A47}$$

Step 5: the recursion cannot equalize both sectors. The strides $e_0, \dots, e_{K-2}$ used by layers $1, \dots, K-1$ never flip the top bit, so the two sectors evolve separately and T_0, T_1 are conserved. In the top-bit-1 sector no difference equals a stride $e_{\ell-1}$, so no term is ever frozen and the recursion averages its $n/2$ values to the uniform value $2T_1/n$ after $K-1$ layers. The top-bit-0 sector contains only $n/2-1$ admissible differences. Hence, using (A47),

$$\sum_{\delta\neq 0, \delta_{K-1}=0} \Big[F^{(K-1)}(\delta) - F^{(K-1)}(\delta \oplus s)\Big] \quad \text{(A48)}$$
$$= \quad T_0 - \Big(\tfrac{n}{2} - 1\Big)\tfrac{2T_1}{n} = \tfrac{k(k-1)}{2n},$$

which is nonzero for every $k \geq 2$. The sum runs over $n/2-1$ terms, so some difference d with $d_{K-1} = 0$ satisfies

$$\big|F^{(K-1)}(d) - F^{(K-1)}(d\oplus s)\big| \;\geq\; \frac{k(k-1)}{2n\,(n/2-1)} \;>\; \frac{k(k-1)}{n^2}.$$

Choosing i, j with $i \oplus j = d$ and combining with (A44) and (A42),

$$|c_{\boldsymbol{\delta}^*}| \;=\; \frac{|D|}{16} \;\geq\; \frac{1}{16}\cdot\frac{4}{n}\cdot\frac{k(k-1)}{n^2} \;=\; \frac{k(k-1)}{4n^3}. \quad \text{(A49)}$$

Inserting this into (A39) with $\delta^*_\theta = 2$ gives $\mathrm{Var}[\partial_\theta \mathcal{L}] \geq 8|c_{\boldsymbol{\delta}^*}|^2 \geq k^2(k-1)^2/(2n^6)$. □

Appendix B: Spectral Gap of the Butterfly's Second-Moment Operator

This appendix proves Theorem 16 of Section IV: the second-moment operator of the unitary butterfly, restricted to the antisymmetric sector of the tensor square of the single-particle representation, has spectral gap $1 - 1/n$, from which the ε-approximate 2-design property with $\varepsilon = O(1/n)$ on that sector follows as a corollary. The antisymmetric sector is the one that carries every quantity appearing in this paper: k-particle states and particle-number-preserving observables evolve in exterior powers of the single-particle action, and the variance computations of Section IV engage the second moment only through antisymmetric-sector operators.

1. Setup

Let $W_n = U_K \cdots U_1$ be the unitary butterfly on $n = 2^K$ modes, where layer ℓ applies independent Haar-$U(2)$ gates on $n/2$ disjoint pairs at stride $2^{\ell-1}$. This is the *proof model* for the spectral gap: each two-qubit gate is treated as a Haar-random $U(2)$ element restricted to the single-excitation subspace, which is a sufficient condition for the fixed-point structure and spectral gap established below. The actual parametrized circuit uses $\mathrm{RBS}(\theta)\cdot(R_z(\phi_a)\otimes R_z(\phi_b))$ per pair with (θ, ϕ_a, ϕ_b) sampled uniformly over their domains. What the recursion below actually requires of each gate, *on the antisymmetric sector*, is *not* Lie closure to $\mathfrak{u}(2)$, but only the two moments invoked in Facts 51 and 52 and Lemmas 47–49: the deterministic antisymmetric phase $|\det V|^2 = 1$ and the first-moment identity $\mathbb{E}[V|i\rangle\langle i|V^\dagger] = I/2$ on each pair. The uniform-angle gate supplies both. Its single-particle action on a pair is $R(\theta)\,\mathrm{diag}(e^{-i\phi_a/2}, e^{-i\phi_b/2})$ with $R(\theta) = \left(\begin{smallmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{smallmatrix}\right)$. For a computational-basis input $|i\rangle$ the diagonal R_z factor cancels in the outer product $V|i\rangle\langle i|V^\dagger$ (its phase has unit modulus and does not appear), so this moment is controlled by θ alone: the off-diagonal entries are proportional to $\mathbb{E}[\sin\theta\cos\theta]$, which vanishes when θ is sampled over a full period, while $\mathbb{E}[\cos^2\theta] = \mathbb{E}[\sin^2\theta] = \frac{1}{2}$ equalizes the diagonal, giving $\mathbb{E}[V|i\rangle\langle i|V^\dagger] = I/2$. Hence, on the antisymmetric sector, the Haar-$U(2)$ model and the uniform-angle $\mathrm{RBS}\cdot R_z$ gate have identical moments and the recursion below applies to both. We emphasize that this equivalence is a sector statement: on the *symmetric* sector the two gate distributions differ—the RBS rotation is real, and the phases cancel between ket and bra precisely on the pair-diagonal components $|ab\rangle\langle ab|$ and $|ab\rangle\langle ba|$, which carry the additional symmetric-sector eigenvectors.

The *second-moment operator* on $\mathrm{End}(\mathbb{C}^n \otimes \mathbb{C}^n)$ is

$$\Phi_2[Y] = \mathbb{E}_{W_n}\left[(W_n \otimes W_n)\, Y\, (W_n^\dagger \otimes W_n^\dagger)\right].$$

Since $W_n \otimes W_n$ commutes with SWAP, the antisymmetric sector $\mathrm{End}(\Lambda^2\mathbb{C}^n) = \{P_A Y P_A\}$, with $P_A = (I_{n^2} - \mathrm{SWAP})/2$, is Φ_2-invariant; all statements below concern the restriction Φ_2^A of Φ_2 to this sector.

Theorem 45 (Spectral gap on the antisymmetric sector)**.** *Φ_2^A has fixed-point space* $\mathrm{span}\{P_A\}$ *(eigenvalue 1) and all remaining eigenvalues equal to 0 or $1/n$ (the eigenvalue $1/n$ having multiplicity $K-1$). Hence the spectral gap on the antisymmetric sector equals $1 - 1/n$.*

Corollary 46 (Approximate 2-design on the antisymmetric sector)**.** *The butterfly W_n is an ε-approximate unitary 2-design on the antisymmetric sector of the single-particle tensor square, with*

$$\varepsilon_{2\mathrm{d}}(n) := \|\Phi_2^{W_n,A} - \Phi_2^{\mathrm{Haar},A}\|_{\mathrm{op}} = O(1/n).$$

This is Theorem 16 of the main text.

The fixed-point statement of Theorem 45 follows from successive layer projections as in the main text (note $P_A = \frac{1}{2}(I\otimes I - \mathrm{SWAP})$ is fixed deterministically). We prove the eigenvalue statement: all non-trivial eigenvalues on the sector are 0 or $1/n$.

Throughout the proof, W_n is an exact single-particle 1-design ($\mathbb{E}[W_{ij}W^*_{kl}] = \delta_{ik}\delta_{jl}/n$), which holds for the $U(n)$ butterfly via the parity-decoupling recursion applied at the 1-design level (each $U(2)$ layer gate is itself a 1-design on its pair).

2. Step A: Partial trace identity

Lemma 47. *For any exact 1-design ν and any $Y \in \mathrm{End}(\mathbb{C}^n \otimes \mathbb{C}^n)$:*

$$\mathrm{Tr}_2(\Phi_2^\nu[Y]) = \mathrm{Tr}_1(\Phi_2^\nu[Y]) = (\mathrm{tr}(Y)/n)\, I_n.$$

Proof. Tracing over the second factor sets $d = b$ and sums over b:

$$[\mathrm{Tr}_2\, \Phi_2[Y]]_{a,c} = \sum_{b,e,f,g,h} \mathbb{E}[W_{ae} W_{bf} W_{cg}^* W_{bh}^*]\, Y_{(e,f),(g,h)}.$$

Unitarity gives $\sum_b W_{bf} W_{bh}^* = \delta_{fh}$ *deterministically*, so the expectation factors: $\sum_b \mathbb{E}[\cdots] = \mathbb{E}[W_{ae} W_{cg}^*]\delta_{fh}$. The 1-design gives $\mathbb{E}[W_{ae} W_{cg}^*] = \delta_{ac}\delta_{eg}/n$, yielding $[\mathrm{Tr}_2\Phi_2[Y]]_{a,c} = (\delta_{ac}/n)\mathrm{tr}(Y)$. The case Tr_1 is symmetric. □

Corollary 48. *The subspace $\mathcal{T} = \{Y : \mathrm{Tr}_1 Y = \mathrm{Tr}_2 Y = 0\}$ is Φ_2-invariant; all non-trivial eigenvalues lie in $\mathcal{T}$.*

3. Step B: Wick pairings vanish on traceless inputs

Lemma 49. *For any exact 1-design ν and traceless $A, B \in \mathrm{End}(\mathbb{C}^n)$: the factored-1-design contribution to $\Phi_2[A \otimes B]$ is zero.*

Proof. The 1-design second-moment formula $\mathbb{E}[W_{ae} W_{cg}^*] = \delta_{ac}\delta_{eg}/n$, applied independently to each factor, yields the single contraction

$$(\delta_{ac}/n)(\delta_{bd}/n)\,\mathrm{tr}(A)\,\mathrm{tr}(B) = 0$$

since $\mathrm{tr}(A) = \mathrm{tr}(B) = 0$. □

4. Step C: Transfer matrix in the antisymmetric projector basis

For modes $i \neq j$ define $P_{ij} = |\psi_{ij}\rangle\langle\psi_{ij}|$ with $|\psi_{ij}\rangle = |ij\rangle - |ji\rangle$.

Lemma 50 (Closed basis)**.** $\Phi_2^{W_n}[P_{ij}] \in \mathrm{span}\{P_{kl} : k < l\}$ *for all $i < j$.*

This follows from the two layer formulas below by induction over layers: each layer maps $\mathrm{span}\{P_{kl} : k < l\}$ into itself (Facts 51 and 52), so the composition W_n does as well, for every $n = 2^K$. (We have also checked it explicitly, with zero remainder, for all pairs at $n = 4$ and $n = 8$.)

Two consequences of Steps A–B fix the scope of the eigenvalue problem solved below. First, by Corollary 48 every non-trivial eigenvector lies in the traceless subspace $\mathcal{T}_n$, so it suffices to bound $\Phi_2^{W_n}|_{\mathcal{T}_n}$. Second, Lemma 49 shows that the factored-1-design contribution vanishes on the traceless inputs P_{ij} − (trace part), so the entire action on $\mathcal{T}_n$ is carried by the connected (cross-pair) part captured by the transfer matrix in the P_{ij} basis below.

Fact 51 (Same-pair)**.** *If $\{i, j\}$ is a pair in layer ℓ:*

$$\Phi_2^{(\ell)}[P_{ij}] = P_{ij}.$$

Proof. For $V \in U(2)$, $(V \otimes V)|\psi_{ij}\rangle = (\det V)|\psi_{ij}\rangle$, since $|\psi_{ij}\rangle$ is antisymmetric and $V \otimes V$ acts on antisymmetric vectors by multiplication by $\det V$. Hence $(V \otimes V)P_{ij}(V \otimes V)^\dagger = |\det V|^2 P_{ij} = P_{ij}$, and the expectation is P_{ij} regardless of the sampling distribution on V. □

Fact 52 (Cross-pair)**.** *If $i \in \alpha$, $j \in \beta$ are in different pairs α, β of layer ℓ (independent gates V_α, V_β):*

$$\Phi_2^{(\ell)}[P_{ij}] = \frac{1}{|\alpha||\beta|} \sum_{k\in\alpha,\, l\in\beta} P_{kl} =: Z_{\alpha\beta}.$$

This output is the same *for all $i \in \alpha$, $j \in \beta$.*

Proof. Independence and the 1-design property $\mathbb{E}[V_\alpha|i\rangle\langle i|V_\alpha^\dagger] = I_\alpha/|\alpha|$ (which holds for both $U(2)$ and $SU(2)$ Haar, and is the 1-design moment formula specialized to the single-particle sector) give $\Phi_2^{(\ell)}[|ij\rangle\langle ij|] = (1/|\alpha||\beta|)I_\alpha \otimes I_\beta$ and $\Phi_2^{(\ell)}[|ij\rangle\langle ji|] = (1/|\alpha||\beta|)S_{\alpha\beta}$, so $\Phi_2^{(\ell)}[P_{ij}] = (1/|\alpha||\beta|)(I_\alpha \otimes I_\beta + I_\beta \otimes I_\alpha - S_{\alpha\beta} - S_{\alpha\beta}^\dagger) = (1/|\alpha||\beta|)\sum_{k,l} P_{kl}$. □

a. Explicit computation for $n = 4$ ($W_4 = L_2 \circ L_1$). Set L_1 pairs $A = \{0, 1\}$, $B = \{2, 3\}$; L_2 pairs $C = \{0, 2\}$, $D = \{1, 3\}$.

After L_1: $\Phi_2^{L_1}[P_{01}] = P_{01}$, $\Phi_2^{L_1}[P_{23}] = P_{23}$ (Fact 51). All four cross-pairs ($i \in A, j \in B$) give the same output: $\Phi_2^{L_1}[P_{ij}] = \frac{1}{4}(P_{02} + P_{03} + P_{12} + P_{13}) =: Z$.

After L_2: P_{01}, P_{23} are cross-pairs of L_2 (Fact 52): $\Phi_2^{L_2}[P_{01}] = \Phi_2^{L_2}[P_{23}] = \frac{1}{4}(P_{01} + P_{03} + P_{12} + P_{23}) =: v_1$. Applying L_2 to Z (using Facts 51 and 52 for $P_{02}, P_{13} \in$ same pairs; $P_{03}, P_{12} \in$ cross pairs of L_2):

$$\begin{aligned}\Phi_2^{L_2}[Z] &= \tfrac{1}{4}\big[P_{02} + \tfrac{1}{4}(P_{01}{+}P_{03}{+}P_{12}{+}P_{23}) \\ &\quad + \tfrac{1}{4}(P_{01}{+}P_{03}{+}P_{12}{+}P_{23}) + P_{13}\big] \\ &= \tfrac{1}{4}(P_{02} + P_{13}) + \tfrac{1}{8}(P_{01} + P_{03} + P_{12} + P_{23}) =: v_2.\end{aligned} \tag{B1}$$

Transfer matrix:

$$\Phi_2^{W_4}[P_{ij}] = \begin{cases} v_1 = \frac{1}{4}(P_{01} + P_{03} + P_{12} + P_{23}) \\ \qquad \text{if } (i,j) \in \{(0,1),(2,3)\}, \\ v_2 = \frac{1}{4}(P_{02} + P_{13}) \\ \quad +\frac{1}{8}(P_{01} + P_{03} + P_{12} + P_{23}) \text{ otherwise.} \end{cases} \tag{B2}$$

The matrix has rank 2; image $= \mathrm{span}\{v_1, v_2\}$.

Eigenvector and eigenvalue: Let $Y^* = c_1(P_{01} + P_{03} + P_{12} + P_{23}) + c_2(P_{02} + P_{13})$. The constraint $\mathrm{Tr}_1(Y^*) = 0$ (mode 0 check: $c_1 + c_2 + c_1 = 0$) gives $c_2 = -2c_1$. Computing:

$$\begin{aligned}\Phi_2^{W_4}[Y^*] &= 2c_1 v_1 + 2(c_1 + c_2)v_2 \\ &= \tfrac{c_1}{4}(P_{01}{+}P_{03}{+}P_{12}{+}P_{23}) - \tfrac{c_1}{2}(P_{02}{+}P_{13}) \\ &= \tfrac{1}{4}\, Y^*.\end{aligned} \tag{B3}$$

The maximum eigenvalue in $\mathcal{T}$ is $1/4 = 1/n$, completing the $n = 4$ base case.

b. Step D: General $n = 2^K$ via the doubly-stochastic structure.

Lemma 53 (Doubly stochastic transfer matrix)**.** *For any $n = 2^K$ ($K \geq 2$), the transfer matrix M defined by*

$$\Phi_2^{W_n}[P_{ij}] = \sum_{k<l} M_{(ij),(kl)}\, P_{kl}$$

is doubly stochastic: every row sum and every column sum equals 1.

Proof. Row sums. Φ_2 is trace-preserving, so

$$2\sum_{k<l} M_{(ij),(kl)} = \mathrm{Tr}(\Phi_2[P_{ij}]) = \mathrm{Tr}(P_{ij}) = 2.$$

Column sums.

$$\sum_{i<j} M_{(ij),(kl)} = \tfrac{1}{4}\,\mathrm{Tr}\Big(P_{kl}\cdot \Phi_2\Big[\textstyle\sum_{i<j} P_{ij}\Big]\Big).$$

We verify that $\sum_{i<j} P_{ij} = \mathbf{I}\otimes\mathbf{I} - \mathrm{SWAP}$: writing $\Pi = \sum_i |ii\rangle\langle ii|$,

$$\sum_{i<j} P_{ij} = (\mathbf{I}\otimes\mathbf{I} - \Pi) - (\mathrm{SWAP} - \Pi) = \mathbf{I}\otimes\mathbf{I} - \mathrm{SWAP}.$$

Both $\mathbf{I}\otimes\mathbf{I}$ and SWAP are fixed points of $\Phi_2^{W_n}$ (deterministically, for every unitary: $WW^\dagger = I$ and SWAP commutes with $W\otimes W$), so $\Phi_2[\mathbf{I}\otimes\mathbf{I}-\mathrm{SWAP}] = \mathbf{I}\otimes\mathbf{I}-\mathrm{SWAP}$. Finally, $\mathrm{Tr}(P_{kl}(\mathbf{I}\otimes\mathbf{I}-\mathrm{SWAP})) = 2-(-2) = 4 = \mathrm{Tr}(P_{kl}^2)$, giving column sum $4/4 = 1$. □

Proposition 54 (General spectral gap)**.** *For $n = 2^K$ ($K \geq 2$), every non-zero eigenvalue of $\Phi_2^{W_n}|_{\mathcal{T}_n}$ equals exactly $1/n$. Hence $\lambda_{\max}(\Phi_2^{W_n}|_{\mathcal{T}_n}) = 1/n$.*

Proof. Throughout, $F := \mathbf{I}\otimes\mathbf{I}-\mathrm{SWAP} = \sum_{i<j} P_{ij}$ is the fixed point; we write $\sigma(Y) := \sum_{i<j} c_{ij}$ for the *coefficient sum* of $Y = \sum c_{ij}P_{ij}$.

c. Step 1 (Image basis). By Facts 51 and 52, $\Phi_2^{W_n}[P_{ij}]$ depends on the pair (i,j) only through its *type*—the position of the highest set bit of $i\oplus j$ (equivalently, the butterfly layer at which i and j are first paired)—and therefore takes at most K distinct values as (i,j) ranges over all $\binom{n}{2}$ pairs. Denote them $f_1,\dots,f_K$, one per butterfly layer. The image satisfies $\mathrm{Im}(\Phi_2) = \mathrm{span}\{f_1,\dots,f_K\}$.

d. Step 2 (T-constraint in the image basis). We show: $v = \sum_t a_t f_t \in \mathcal{T}_n$ if and only if $\sum_t a_t = 0$.

Row sums of M equal 1 (Lemma 53), so $\sigma(f_t) = 1$ for every t. Lemma 47 gives $\mathrm{Tr}_1(f_t) = (2/n)I_n$ for every t. Hence for any mode m:

$$\sum_{l\neq m}[v \text{ coeff at } (m,l)] = \sum_t a_t\cdot\frac{2}{n}.$$

This is zero (T-constraint) if and only if $\sum_t a_t = 0$.

e. Step 3 (Key Formula). We claim: for every type $t = 1,\dots,K$,

$$\Phi_2^{W_n}[f_t] = \frac{1}{n}\,f_t + \frac{2}{n^2}\,F. \tag{B4}$$

Verification at $n = 4$. From the explicit transfer matrix (B2), the two image vectors are $v_1 = \frac{1}{4}(P_{01}+P_{03}+P_{12}+P_{23})$ and $v_2 = \frac{1}{4}(P_{02}+P_{13})+\frac{1}{8}(P_{01}+P_{03}+P_{12}+P_{23})$. Apply $\Phi_2^{W_4}$ to each using (B2):

$$\begin{aligned}\Phi_2[v_1] &= \tfrac{1}{4}[\Phi_2[P_{01}]+\Phi_2[P_{03}]+\Phi_2[P_{12}]+\Phi_2[P_{23}]]\\ &= \tfrac{1}{4}[v_1+v_2+v_2+v_1] = \tfrac{1}{2}v_1+\tfrac{1}{2}v_2.\end{aligned}$$

One checks directly that $\frac{1}{2}v_1+\frac{1}{2}v_2 = \frac{3}{16}(P_{01}+P_{03}+P_{12}+P_{23})+\frac{1}{8}(P_{02}+P_{13})$, while $\frac{1}{4}v_1+\frac{1}{8}F$ equals the same expression (expanding v_1 and F). Similarly $\Phi_2[v_2] = \frac{1}{2}v_1+\frac{1}{2}v_2$ and $\Phi_2[v_2] = \frac{1}{4}v_2+\frac{1}{8}F$. Hence (B4) holds for both types at $n = 4$.

General $n = 2^K$. We prove (B4) for all K by diagonalizing $\Phi_2^{W_n}$ on $\mathcal{A} := \mathrm{span}\{P_{ij} : i<j\}$ in closed form, exploiting the translation symmetry of the butterfly. Write $g^{(\ell)} := \Phi_2^{(\ell)}$ for the single-layer operator.

(i) Translation symmetry and Fourier blocks. Identify the modes with $(\mathbb{Z}_2)^K$ via their K-bit labels, so layer ℓ pairs $i \leftrightarrow i\oplus e_{\ell-1}$, with e_b the b-th coordinate vector ($b = 0,\dots,K-1$). For $c\in(\mathbb{Z}_2)^K$, the relabeling $\pi_c : i\mapsto i\oplus c$ maps each layer's pair partition to itself; since the gates are i.i.d. across pairs, the law of W_n is invariant under π_c, so $\Phi_2^{W_n}$ commutes with conjugation by $\pi_c\otimes\pi_c$ for every c. Parametrize a pair by base point and difference, $P_{(i,\delta)} := P_{\{i,\,i\oplus\delta\}}$ with $\delta\in(\mathbb{Z}_2)^K\setminus\{0\}$; then π_c sends $P_{(i,\delta)}\mapsto P_{(i\oplus c,\,\delta)}$. Diagonalizing the commuting $(\mathbb{Z}_2)^K$-action, set

$$\hat{P}_{\delta,y} := \sum_i (-1)^{y\cdot i}\,P_{(i,\delta)}, \qquad y\in(\mathbb{Z}_2)^K,\quad y\cdot\delta = 0$$

(the constraint makes the summand well defined under $i\sim i\oplus\delta$). Since π_c scales $\hat{P}_{\delta,y}$ by $(-1)^{y\cdot c}$, the operator $\Phi_2^{W_n}$ is block-diagonal in y, preserving each $\mathcal{B}_y := \mathrm{span}\{\hat{P}_{\delta,y} : \delta\neq 0,\ y\cdot\delta = 0\}$.

(ii) Blocks with $y\neq 0$ vanish. Fix layer ℓ and write $e = e_{\ell-1}$, $\eta = y\cdot e$. In base-difference form, Facts 51–52 read

$$g^{(\ell)}[P_{(i,\delta)}] = \begin{cases} P_{(i,e)}, & \delta = e,\\ \frac{P_{(i,\delta)}+P_{(i\oplus e,\delta)}+P_{(i,\delta\oplus e)}+P_{(i\oplus e,\delta\oplus e)}}{4}, & \delta\neq e.\end{cases}$$

Summing against $(-1)^{y\cdot i}$ and shifting the summation index gives, for $\delta\neq e$,

$$g^{(\ell)}[\hat{P}_{\delta,y}] = \tfrac{1}{4}(1+(-1)^\eta)\,\hat{P}_{\delta,y} + \tfrac{1}{4}(1+(-1)^\eta)\,\hat{P}_{\delta\oplus e,\,y},$$

and $g^{(\ell)}[\hat{P}_{e,y}] = \hat{P}_{e,y}$. Hence if $\eta = y\cdot e_{\ell-1} = 1$, layer ℓ annihilates all of $\mathcal{B}_y$. Every $y\neq 0$ has a coordinate with $y\cdot e_{\ell-1} = 1$, and every layer preserves $\mathcal{B}_y$; therefore

$\Phi_2^{W_n}|_{\mathcal{B}_y} = 0$ for all $y \neq 0$. All non-zero eigenvalues of $\Phi_2^{W_n}$ thus lie in $\mathcal{B}_0 = \mathrm{span}\{S_\delta : \delta \neq 0\}$, where $S_\delta := \hat{P}_{\delta,0} = \sum_i P_{(i,\delta)}$.

(iii) The block $y = 0$. On $\mathcal{B}_0$ every layer has $\eta = 0$, so (with $S_0 := 0$)

$$g^{(\ell)}[S_\delta] = \begin{cases} S_{e_{\ell-1}}, & \delta = e_{\ell-1}, \\ \frac{1}{2}S_\delta + \frac{1}{2}S_{\delta\oplus e_{\ell-1}}, & \delta \neq e_{\ell-1}. \end{cases}$$

We prove the following by induction on K.

Claim $H(K)$**.** *On $\mathcal{B}_0$ ($\dim = n-1$), $\Phi_2^{W_n}$ has rank K; its non-zero spectrum consists of a simple eigenvalue 1 with eigenvector $F_n = \sum_{\delta\neq 0} S_\delta$ (the unique image vector of non-zero coefficient sum), together with the eigenvalue $1/n$ of multiplicity $K-1$ (all of whose eigenvectors have coefficient sum zero).*

Write $\tilde{\sigma}(\sum_\delta c_\delta S_\delta) := \sum_\delta c_\delta$ for the coefficient sum in the S_δ basis (distinct from the P_{ij}-basis coefficient sum σ defined above: since each $S_\delta = \sum_i P_{(i,\delta)}$ collects $n/2$ pairs, one has $\sigma = (n/2)\,\tilde{\sigma}$ on $\mathcal{B}_0$). Each $g^{(\ell)}$ preserves $\tilde{\sigma}$ (this is the row-sum statement of Lemma 53), so $\tilde{\sigma} \circ \Phi_2^{W_n} = \tilde{\sigma}$, and every eigenvector with eigenvalue $\neq 1$ has $\tilde{\sigma} = 0$.

The base case $H(1)$ ($n = 2$) is immediate: $\mathcal{B}_0 = \mathbb{C}\,S_{e_0}$ and $g^{(1)}[S_{e_0}] = S_{e_0} = F_2$. For the inductive step, split $(\mathbb{Z}_2)^K$ by the top bit and write $W_n = U_K \circ (W_A \oplus W_B)$, so $\Phi_2^{W_n} = g^{(K)} \circ \Psi$ with $\Psi := g^{(K-1)} \cdots g^{(1)}$. Let ι_b embed a lower difference $\delta' \in (\mathbb{Z}_2)^{K-1}$ as (δ', b). The inner layers act only on the low $K-1$ bits, and their same-pair exception requires top bit 0; hence $\Psi[\iota_0 x] = \iota_0\,\Phi_2^{W_{n/2}}[x]$ (the lower butterfly), while on the top-bit-1 sector no exception fires and the inner layers average the low bits to uniform, giving $\Psi[S_{(\delta',1)}] = \frac{2}{n}\,u_1$ for every δ', where $u_1 := \sum_{\delta'} S_{(\delta',1)}$. Applying $g^{(K)}$ (which flips the top bit and fixes only $S_{e_{K-1}} = S_{(0,1)}$) yields, for $x \in \mathcal{B}_0^{(n/2)}$,

$$\Phi_2^{W_n}[\iota_0 x] = \tfrac{1}{2}(\iota_0 + \iota_1)\,\Phi_2^{W_{n/2}}[x], \tag{B5}$$

$$\Phi_2^{W_n}[S_{(\delta',1)}] = w := \tfrac{1}{n}\big(F_n + S_{e_{K-1}}\big) \quad (\forall \delta'), \tag{B6}$$

the second line using $g^{(K)}[u_1] = \frac{1}{2}S_{e_{K-1}} + \frac{1}{2}u_1 + \frac{1}{2}\iota_0 F_{n/2}$ and $u_1 + \iota_0 F_{n/2} = F_n$.

Each lower eigenvector x with eigenvalue $\mu \neq 1$ has $\tilde{\sigma}(x) = 0$, so $\Phi_2^{W_n}[\iota_1 x] = \tilde{\sigma}(x)\,w = 0$ by (B6), and then (B5) makes $(\iota_0 + \iota_1)x$ an eigenvector with eigenvalue $\mu/2$. By $H(K-1)$ the lower eigenvalues $\mu \neq 1$ are $2/n$ (multiplicity $K-2$) and 0; these lift to eigenvalue $1/n$ (multiplicity $K-2$) and 0. On the remaining two-dimensional space $\mathrm{span}\{p, w\}$, with $p := (\iota_0 + \iota_1)F_{n/2}$, equations (B5)–(B6) together with $\Phi_2^{W_{n/2}}[F_{n/2}] = F_{n/2}$, $\tilde{\sigma}(F_{n/2}) = n/2 - 1$, $\Phi_2^{W_n}[S_{e_{K-1}}] = w$, and $w = \frac{1}{n}(p + 2S_{e_{K-1}})$ give

$$\Phi_2^{W_n}[p] = \tfrac{1}{2}p + \big(\tfrac{n}{2} - 1\big)w, \qquad \Phi_2^{W_n}[w] = \tfrac{1}{2n}p + \big(\tfrac{1}{2} + \tfrac{1}{n}\big)w.$$

This 2×2 matrix has trace $1 + 1/n$ and determinant $1/n$, hence eigenvalues 1 and $1/n$; the eigenvalue-1 vector is $\frac{1}{2}p + \frac{n}{2}w = F_n$. Since $\mathrm{Im}(\Phi_2^{W_n}) = (\iota_0 + \iota_1)\,\mathrm{Im}(\Phi_2^{W_{n/2}}) + \mathbb{C}\,w$ is spanned by p, the $K-2$ lifts of the lower $2/n$-eigenvectors, and w (dimension K), and decomposes into the $\Phi_2^{W_n}$-invariant pieces $\mathrm{span}\{p, w\}$ and the $(\iota_0 + \iota_1)$-lifts, collecting eigenvalues proves $H(K)$: eigenvalue 1 once, $1/n$ with multiplicity $(K-2) + 1 = K - 1$, and 0 elsewhere.

(iv) The Key Formula. By $H(K)$ the image $\mathcal{I} = \mathrm{span}\{f_1, \ldots, f_K\}$ splits as $\mathbb{C}F_n \oplus \mathcal{I}_0$, where $\mathcal{I}_0 = \{v \in \mathcal{I} : \sigma(v) = 0\}$ is the eigenvalue-$1/n$ eigenspace. For $v \in \mathcal{I}$, writing $v = \frac{\sigma(v)}{\sigma(F_n)}F_n + v_0$ with $v_0 \in \mathcal{I}_0$ and using $\sigma(F_n) = \binom{n}{2}$,

$$\begin{aligned}\Phi_2^{W_n}[v] &= \tfrac{\sigma(v)}{\sigma(F_n)}F_n + \tfrac{1}{n}v_0 = \tfrac{1}{n}v + \Big(1 - \tfrac{1}{n}\Big)\frac{\sigma(v)}{\binom{n}{2}}\,F_n \\ &= \tfrac{1}{n}v + \tfrac{2}{n^2}\,\sigma(v)\,F_n.\end{aligned}$$

Each $f_t = \Phi_2^{W_n}[P_{ij}]$ for a type-t pair has $\sigma(f_t) = \sigma(P_{ij}) = 1$, so $\Phi_2^{W_n}[f_t] = \frac{1}{n}f_t + \frac{2}{n^2}F_n$, which is (B4), now established in closed form for all $n = 2^K$.

(v) Conclusion. By $H(K)$ the only eigenvalue of $\Phi_2^{W_n}$ on $\mathcal{A}$ exceeding $1/n$ is the simple eigenvalue 1 at $F_n = \mathbf{I}\otimes\mathbf{I} - \mathrm{SWAP}$; the blocks $y \neq 0$ contribute only 0. Each eigenvalue-$1/n$ eigenvector v has $\sigma(v) = 0$, hence $\mathrm{tr}(v) = 2\sigma(v) = 0$, and from $v = n\,\Phi_2^{W_n}[v]$ with Lemma 47 we get $\mathrm{Tr}_1(v) = \mathrm{tr}(v)\,I_n = 0$ (and likewise Tr_2); thus $v \in \mathcal{T}_n$. Since $F_n \notin \mathcal{T}_n$ (indeed $\mathrm{Tr}_1 F_n = (n-1)I_n$), every non-trivial eigenvalue in $\mathcal{T}_n$ equals $1/n$, i.e. $\lambda_{\max}(\Phi_2^{W_n}|_{\mathcal{T}_n}) = 1/n$. □

This completes the proof of Theorem 45.

Proof of Corollary 46. The spectral gap $1 - 1/n$ between the fixed-point space $\mathrm{span}\{P_A\}$ and all remaining eigenvalues on the antisymmetric sector implies directly that $\|\Phi_2^{W_n,A} - \Phi_2^{\mathrm{Haar},A}\|_{\mathrm{op}} \leq 1/n$, since $\Phi_2^{\mathrm{Haar},A}$ is the projector onto the same fixed-point space (on the antisymmetric sector the Haar second moment fixes only P_A, as $\mathrm{span}\{I_{n^2}, \mathrm{SWAP}\} \cap \{P_A Y P_A\} = \mathbb{C}P_A$) and the two operators differ only on the traceless part of the sector, where the butterfly's operator has spectral norm $1/n$. This is the definition of an ε-approximate 2-design with $\varepsilon = O(1/n)$ on the sector. □